APACHE POINT OBSERVATORY FOLLOW-UP OF ACCELERATING CANDIDATE EXOPLANET HOST STARS & CHARACTERIZATION OF THE EXOPLANET HOST HD 143811 AB

by

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For Mom, Dad, Adrian, Avery, Ginny, Basil, and Max; and in honor of Grandma Lynn, a true academic.

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Peck, A., Roberson, W., Nielsen, E., De Rosa, R., Jones, N., Wang, J., Macintosh, B., Lewis, B., Duchêne, G., Metchev, S., Abbas, A., Xuan, J., Sanghi, A., Patience, J., Barman, T., Bulger, J., Chilcote, J., Esposito, T., Fitzgerald, M., Follette, K., Gallamore, H., Goodsell, S., Graham, J., Greenbaum, A., Hibon, P., Ingraham, P., Kalas, P., Konopacky, Q., Marchis, F., Maire, J., Marois, C., Matthews, B., Mawet, D., Millar-Blanchaer, M., Oppenheimer, R., Palmer, D., Perrin, M., Poyneer, L., Pueyo, L., Rajan, A., Rameau, J., Rantakyrö, F., Ren, B., Ruffio, JB., Savransky, D., Schneider, A., Sivaramakrishnan, A., Smith, A., Song, I., Soummer, R., Thomas, A., Ward-Duong, K., Wolff, S., *Characterization of the Host Binary of the Directly Imaged Exoplanet HD 143811 AB b.* ApJ Letters, 995, 40, doi: 10.3847/2041-8213/ae1a67

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Contents

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS

Instruments, Telescopes & Observatories

Acronyms and Jargon

LIST OF SYMBOLS

Symbol	Meaning
$q = M_2/M_1,\ M_1 > M_2$	mass ratio
M_1	mass of more massive binary component
M_2	mass of less massive binary component

1. APACHE POINT OBSERVATORY FOLLOW-UP OF ACCELERATING CANDIDATE EXOPLANET HOST STARS (APO ACCELS): AGES FOR 166 ACCELERATING STARS IN THE NORTHERN HEMISPHERE

Directly imaged substellar companions with well-constrained ages and masses serve as vital empirical benchmarks for planet formation and evolution models. Potential benchmark companions can be identified from astrometric accelerations of their host stars. We use *Gaia* DR3 and *Hipparcos* astrometry to identify 166 northern hemisphere stars with astrometric accelerations consistent with a substellar companion between 0.5" and 1". For this accelerating sample we identify young stars using APO/ARCES spectra and $TESS$ light curves. From spectroscopic screening of the sample, we measure ages for 24 stars with detectable amounts of lithium, place lower age limits on 135 stars with lithium non-detections, and measure ages from R'_{HK} for 34 stars. 129 stars have $TESS$ light curves from which we measure ages for 20 stars with rotation rates < 15 days, and we identify 3 eclipsing binaries. We present median ages and confidence intervals of age posteriors for the entire sample and discuss how the overall age distribution of our sample compares to a uniform star formation rate in the solar neighborhood. We identify 47 stars with median ages < 2 Gyr, 31 stars with median ages < 1 Gyr, and 14 stars with median ages < 0.5 Gyr, making them high-priority targets for direct imaging follow-up.

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1.1. Introduction

Direct imaging of exoplanets allows for demographic studies of wide-separation giant planets and for analysis of these planets' atmospheres through spectra and photometry (e.g. Nielsen et al. 2019; Vigan et al. 2021). Each direct imaging detection allows for detailed characterization of the companion (e.g. Marois et al. 2008, 2010; Lagrange et al. 2009; Macintosh et al. 2015; Matthews et al. 2024). While wide-separation giant planets have relatively low occurrence rates (e.g. Nielsen et al. 2019; Vigan et al. 2021), additional companions can be identified from astrometric accelerations of their host stars (e.g. Brandt 2021). A targeted survey informed by astrometric accelerations, radial velocities, and stellar ages can complement previous blind surveys by increasing the number of directly imaged substellar companions. Planets detected via astrometry have the added benefit of measurable dynamical masses making them key benchmarks for planet formation and evolution models (e.g. Allard et al. 2003; Marley et al. 2021; Morley et al. 2024), especially if their ages are known (e.g. Nielsen et al. 2020; De Rosa et al. 2023). The mass of a companion can be determined by combining imaging and astrometry, and the companion's age can be measured from observations of the host star. A broader sample of benchmark exoplanets will help constrain planet formation and evolution models.

Gaia astrometry is expected to be sensitive to planets between $\sim 1-15\ M_{\rm Jup}$ with periods between $\sim 0.5-10$ yr around a wide range of stellar host masses (Perryman et al. 2014). *Gaia* Data Release 4 (DR4) is scheduled for mid-2026 and is set to include a list of astrometrically detected exoplanets between $\sim 1-5$ AU detected using only *Gaia* (Gaia Collaboration et al. 2016). Prior to the release of DR4, substellar companions can be detected by combining positions, parallaxes, and proper motions from the *Hipparcos* Catalog (Perryman et al. 1997; van Leeuwen et al. 1997) and *Gaia* Data Release 3 (DR3) (Gaia Collaboration et al. 2023). Using astrometry the influence of a massive planet on the path of a star can be observed, and longer-period companions can be detected by linking these two missions (e.g. Perryman et al. 2014; Brandt 2021; Kervella et al. 2024). Velocity is measured as proper motion for stars in both *Hipparcos* (1991.25) and *Gaia* DR3 (2016.0) as well as from the difference between the two catalog positions. An acceleration of the photocenter can be identified from discrepancies between the three velocity vectors. Combined *Hipparcos* and *Gaia* astrometry yield typical acceleration precision as low as $\sim$1 $\mathrm{m\,s^{-1}\,yr^{-1}}$ for nearby stars, and is sensitive to longer periods than *Gaia* alone (Brandt 2021).

Several substellar companions have been detected using this method. The

$3-7$ $M_{\rm Jup}$ planet AF Lep b was identified by the acceleration of its young host star, a member of the ~ 24 Myr β Pictoris moving group (Bell et al. 2015), before being detected by direct imaging (De Rosa et al. 2023; Mesa et al. 2023; Franson et al. 2023). Higher mass substellar companions have been imaged around older accelerating stars including the $13.9-16.1$ $M_{\rm Jup}$ companion to the ≤ 500 Myr HIP 99770 (Currie et al. 2023b), the ~ 28 $M_{\rm Jup}$ companion to the ~ 750 Myr HIP 21152 (Kuzuhara et al. 2022), and the ~ 31 $M_{\rm Jup}$ companion to the $\lesssim 1.5$ Gyr HIP 5319 (Swimmer et al. 2022). Since these accelerations are from astrometric measurements of position and velocity with a limited time baseline, there is a degeneracy between the companion mass and orbital parameters, especially the period. For example, the acceleration of the ~ 500 Myr HR 1645 appeared to be consistent with a brown dwarf companion on a ~ 100 yr orbit but was instead shown to be caused by a stellar binary with a < 1 yr orbit (De Rosa et al. 2019).

Planets cool monotonically over time, with younger, more massive planets being brighter. Given current ground-based direct imaging sensitivity, substellar companions with a contrast ratio brighter than 10^{-6} can generally be detected to within $\sim 0.5''$ of their host stars (e.g. Macintosh et al. 2014; Currie et al. 2023a). Depending on the luminosity and distance of the host star, a wide separation ($\sim 10-30$ AU) giant planet can be detected around stars younger than a few hundred Myr, and brown dwarfs around stars younger than ~ 1 Gyr. Thus, measuring accurate ages of host stars and identifying young systems is crucial when selecting high-priority direct imaging targets. We measure the ages of 166 northern hemisphere accelerating stars so that we can identify the youngest systems and prioritize them for direct imaging follow-up.

We use the following methods to derive the ages of our sample stars.

1. **Ca H and K emission (R'_{HK}):** The calcium H and K emission lines are observed as cores within deep absorption lines. The absorption lines form in the optically thick photosphere and the emission cores are generated in the optically thin chromosphere via magnetic heating. Thus, these emission cores trace magnetic activity in stars. As a star spins down with age, its magnetic and chromospheric activity decreases (e.g. Skumanich 1971). The youngest stars have the fastest rotation and, thus, the strongest magnetic activity, which causes the strongest calcium emission. R'_{HK} is calibrated as an age tracer for F5-K2 stars (e.g. Soderblom et al. 1991; Mamajek & Hillenbrand 2008; Stanford-Moore et al. 2020).

2. **Lithium:** Throughout the main sequence lifetime of solar-type and lower mass stars, convection transports surface lithium to deeper, hotter layers of the star where it is destroyed via fusion. Lithium depletion is most efficient in low-mass stars that are either fully convective or have thick convective envelopes (Soderblom et al. 1993; Llorente de Andrés et al. 2021; Soderblom 2010) making it a useful age tracer for G, K, and M stars. Lithium equivalent

width is a powerful age tracer for very young late-type (later than $\sim$K5) stars due to their rapid lithium depletion, but after about 150 Myr only equivalent width upper limits can be established for these stars.

3. **Gyrochronology:** Stars later than $\sim$F7 spin down as they age due to angular momentum loss via magnetic braking (Kraft 1967). Rotation periods can be measured from flux variations due to star spots rotating on and off the disk of the star. Longer rotation periods correspond to older ages.

4. **Color-Magnitude Diagram Positions:** After reaching the zero-age main sequence, the position of a star on the CMD does not change significantly during the first third of its main sequence lifetime. During the final two-thirds of its main sequence lifetime, the core becomes more helium-rich, increasing the star's luminosity. This results in detectable movement across the CMD. Age posteriors for these stars can be derived by comparing their photometry to theoretical isochrones (e.g. Nielsen et al. 2013, 2019). This method is especially powerful for A and F-type stars which have shorter main-sequence lifetimes between $\sim 1-5$ Gyr.

An approximate schematic of which tracer can produce the most precise age for different stars is shown in Figure 1.1. Overlap allows for robust age measurements from multiple tracers for some stars. Significant scatter is observed in the age indicators for a given age and mass, which can be quantized to produce a posterior probability distribution on age. The precision of a given age tracer depends on the mass and age of the star. Several tools have been developed to derive age posteriors from these tracers such as `BAFFLES` (Stanford-Moore et al. 2020), `EAGLES` (Jeffries et al. 2023), `stardate` (Angus et al. 2019), and `gyro-interp` (Bouma et al. 2023). We present median ages and 68% and 95% confidence intervals for 166 northern hemisphere accelerating stars based on these four age tracers below.

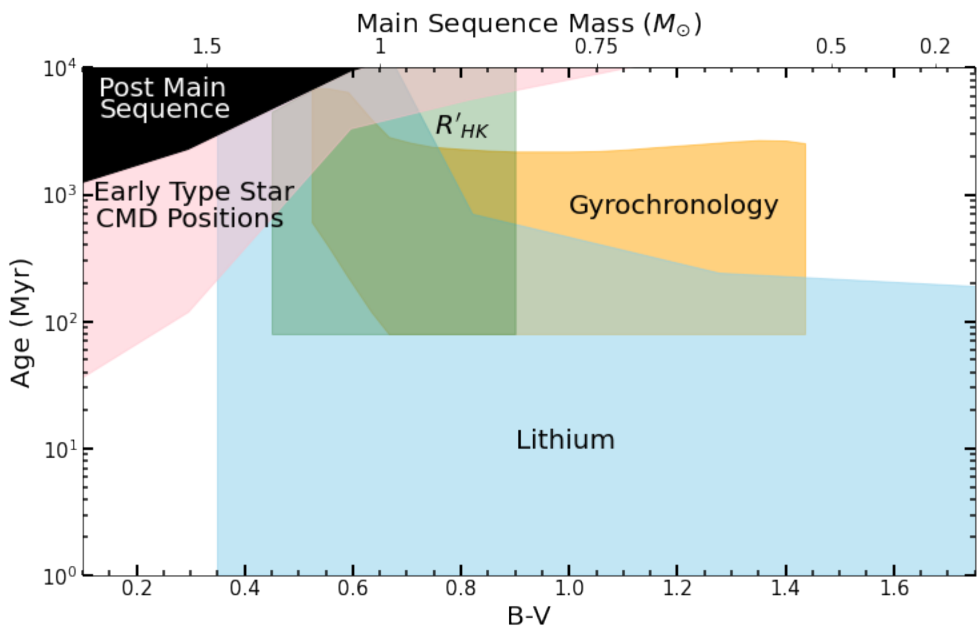

Fig. 1.1.— Age tracers used in this work for young, low-mass stars. This approximate schematic shows which combinations of age and mass are robustly covered by different age tracers. Each shaded region indicates where each tracer is most robust but does not include the regions where limits can be derived from each tracer (e.g. above the blue shaded region lithium is not expected to be detectable, but upper limits on lithium abundance can be used to derive lower limits on the age).

1.2. Sample, observations, and data reduction

We identify candidate exoplanet hosts by examining the *Hipparcos-Gaia* Catalog of Accelerations presented in Brandt (2021) created using the *Gaia* DR3 catalog. We select stars with a $> 4\sigma$ proper motion anomaly, combining the significance of the three pairs of proper motions. We exclude known close binaries reported in the Washington Double Star Catalog (WDS) (Mason et al. 2001) or the Ninth Catalog of Spectroscopic Binary Orbits (SB9) (Pourbaix et al. 2004) and stars with obvious companions in archival imaging data. Of the remaining stars, we reject those whose astrometric signal is consistent with a stellar companion between $0.1 - 1.0''$. We determine the allowed combinations of companion mass and orbital period using the rejection sampling algorithm described in detail in De Rosa et al. (2023). We compare simulated astrometric measurements from the simulated photocenter motion caused by an orbiting companion to the observed accelerations. We use the scan timings and scan angles of the *Hipparcos* and *Gaia* satellites to forward model the position and proper motion of the photocenter measured by the two missions. The photocenter motion is modeled as a combination of the linear motion of the system through space and the orbit of the primary star about the system barycenter. Figure 1.2 shows an example of the allowed mass-period parameter space for a star in our sample (HIP 669) based on the observed astrometric acceleration.

We limit our sample to stars observable from Apache Point Observatory (APO), i.e. stars with declinations $> -30°$. We also limit the sample to stars within 50 pc to maximize the angular separation between the star and the potential companion to maximize the chance of the companion being directly imaged. We do not include stars that have archival radial velocities (e.g. from ESO/HARPS or Keck/HIRES). Figure 1.3 shows the distance, color, and mag-

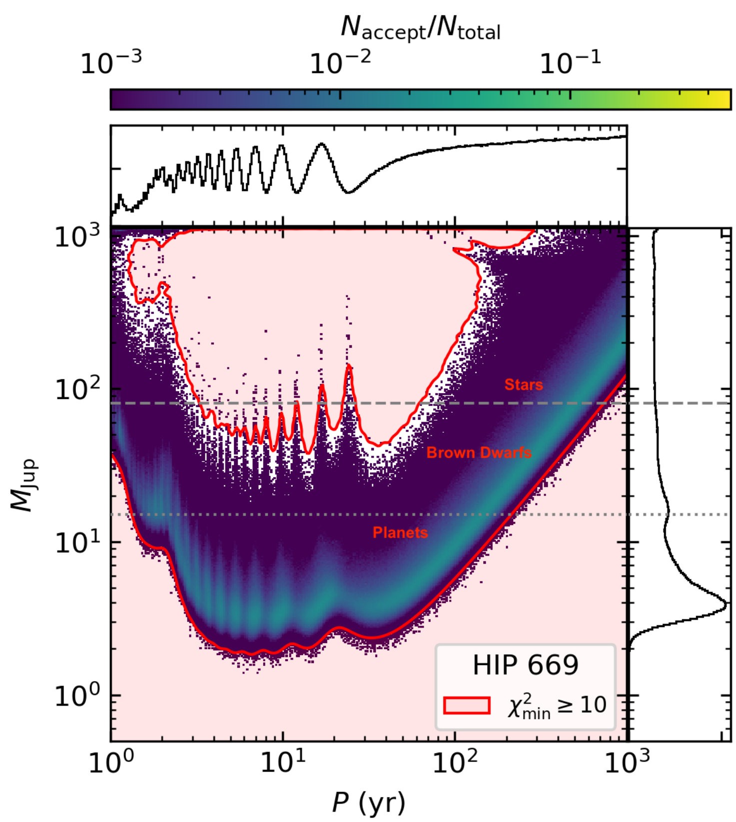

Fig. 1.2.— Results from rejection sampling of combined *Hipparcos* and *Gaia* DR3 astrometry show the possible companion mass-period combinations responsible for the observed acceleration of HIP 669. Candidate companions include stellar binaries, brown dwarfs, and wide-separation giant planets. The near-infrared contrast of these substellar companions depends on the age of the star.

nitudes of our sample. The majority of our sample is between 30-50 pc with apparent G magnitudes between 6 and 11.

Age posteriors for lithium and R'_{HK} from `BAFFLES` (Stanford-Moore et al. 2020) are calculated as a function of $B-V$. We adopt the $B-V$ values and associated errors that are reported in the *Hipparcos* catalog (van Leeuwen et al. 1997). Five stars in our sample (HIP 33368, HIP 46385, HIP 63661, HIP 69860, and HIP 81655) do not have a $B-V$ reported in the *Hipparcos* Catalog. An additional five stars (HIP 9067, HIP 111976, HIP 59953, HIP 1068, and HIP 1771) have very high uncertainties (∼0.5 magnitudes) on their reported B-V colors in the *Hipparcos* catalog. We derive $B-V$ for these targets by fitting a relation to the $B-V$ / $G_{BP}-G_{RP}$ color-color plot (Figure A.0.1 in Appendix). We use the $B-V$ values from the *Hipparcos* Catalog (van Leeuwen et al. 1997) and the $G_{BP}-G_{RP}$ values from *Gaia* DR3 (Gaia Collaboration et al. 2023). We filter the *Gaia* catalog for all stellar entries that were cross-matched to *Hipparcos* by the *Gaia* team, merged the *Hipparcos*/*Tycho* data, we exclude variable stars, and then apply the following filters: $RUWE \leq 1.6$, $Gmag < 13$, $Gmag_{err} < 0.01$, $Hpmag_{err} < 0.1$, $(B-V)_{err} < 0.1$, $B-V \neq 0.0$, and $B-V < 2.0$.

We find the best-fitting relationship is

$$B-V = -0.102(G_{BP}-G_{RP})^4 + 0.226(G_{BP}-G_{RP})^3$$

$$+0.017(G_{BP}-G_{RP})^2 + 0.688(G_{BP}-G_{RP}) + 0.002$$

Based on the scatter in the relationship, we adopt a 0.1 magnitude uncertainty on the derived $B-V$ colors.

Gyrochronological age posteriors from `gyro-interp` depend on effective temperature (Bouma et al. 2023). We determine effective temperatures for our sample by interpolating the grid from Pecaut & Mamajek (2013) to the measured *Gaia* $G_{BP}-G_{RP}$ colors. To estimate the uncertainty on the effective temperature we find the range of effective temperatures corresponding to the range of $G_{BP}-G_{RP} \pm 0.1$.

To measure lithium equivalent widths and R'_{HK}, we obtain spectra using the Astrophysical Research Consortium Echelle Spectrograph (ARCES) on the APO 3.5 meter telescope. ARCES is a multi-use spectrograph with an average spectral resolution $R = \frac{\Delta\lambda}{\lambda} = 35,000$ covering 3,200 Å to 10,000 Å (Wang et al. 2003). We achieve SNRs between ~ 20 for our faintest targets and ~ 200 for our brightest stars. The observing campaign began in October 2020 and is ongoing. We aim to observe each star at least three times over several months to years so that we can vet for binaries, which we will discuss in detail in future work. As of writing, we have observed each star at least once and 98 out of the 166 stars at least three times. We reduce the data using the python package, `pyvista`[1]. Each spectrum is bias-corrected, extracted, flat-fielded, and continuum-corrected. Wavelength

[1] https://pyvista.readthedocs.io/

solutions are extracted from the ThAr taken nearest in time to the science frame and are then applied to the science spectrum.

We measure rotation periods from $TESS$ light curves reduced with the SPOC pipeline (Jenkins et al. 2016). All light curves with cadences between 20-1800s are downloaded from the Mikulski Archive for Space Telescopes (MAST). We found light curves for 129 of our sample stars and identify periodic signals in the light curves of 34 of these 129 stars.

1.3. Lithium

We measure lithium 6707.79 Å equivalent widths from ARCES spectra so that we can use them and the python package `BAFFLES` (Stanford-Moore et al. 2020) to derive age posteriors. First, we bring the spectrum to the approximate rest frame using the *Gaia* DR3 radial velocity measurement (Gaia Collaboration et al. 2023) for the star and correct for the Earth's motion by applying a barycentric correction. To find the lithium 6707.79 Å feature and measure its equivalent width, we fit a double Gaussian with a slope to the spectrum with the functional form

$$f_{mod} = (m(\lambda - 6707.0) + b)(C_1 e^{-0.5(6707.79 - RV_s - \lambda)^2/\sigma^2} + C_2 e^{-0.5(\lambda_{ref} - RV_s - \lambda)^2/\sigma^2}) \tag{1.1}$$

where λ is the wavelength array, m and b are the slope and intercept of the continuum; λ_{ref} is the central wavelength of some reference feature; C_1 is the depth of the lithium feature; C_2 is the depth of the reference line; σ represents the standard deviation of both Gaussians; RV_s represents an additional radial velocity shift to the *Gaia* DR 3 measurement. We include this additional RV offset term in our fit to account for potential orbiting companions or any radial velocity drift in ARCES. For most stars, we adopt $\lambda_{ref} = 6705.1$ Å, a prominent Fe line. For lower mass stars where the 6705.1 Å Fe line is not present, we adopt $\lambda_{ref} = 6717.8$ Å, a prominent Ca line. The reference line prevents the fit from selecting a value for the radial velocity, RV_s that shifts another line (or noise trough) to the location of the lithium line. This also prevents the fit from adjusting the width of the lithium feature to match a noise trough. This is especially useful in cases where the lithium line is weak or not detectable. When $\lambda_{ref} = 6705.1$ Å we use the region of spectra from 6704 Å to 6710 Å. When $\lambda_{ref} = 6717.8$ Å we use the region of the spectrum from 6706 Å to 6720 Å. Both of these regions fall on a single ARCES order.

Since the 6707.79 Å feature is a blend of the ^6Li and ^7Li doublets, we find the effective center of the entire feature by fitting the double Gaussian to a narrow region of spectrum from 6704 Å to 6710 Å for several stars where $\lambda_{ref} = 6705.1$ Å such that

$$f_{mod} = (m(\lambda - 6707.0) + b)(C_1 e^{-0.5(6707.79+s-RV_s-\lambda)^2/\sigma^2} + C_2 e^{-0.5(6705.105-RV_s-\lambda)^2/\sigma^2}) \tag{1.2}$$

Here, s is the deviation of the center of the lithium feature from 6707.79 Å.

We use the python package *emcee* (Foreman-Mackey et al. 2013) to run Markov Chain Monte Carlo (MCMC) fits on several spectra and found that s for the best fit was consistently 0.0119 Å regardless of spectral type, so we adopt 6707.819 Å as the effective center of the lithium feature.

With Equation 3 and *emcee*, we run MCMC to find the best-fitting parameters and estimates of their uncertainties for each spectrum. The uncertainties in each pixel of the spectrum are assumed to follow a Poisson distribution and include estimates of the gain and read noise.

First, we run a preliminary MCMC with 100 walkers and 10,000 steps to find good starting positions. Then, a second round of MCMC is run from these starting positions, with 100 walkers, for 50,000 steps with a 2,000 step burn-in, which was sufficient to achieve convergence for most stars. 16 stars with non-detections required 100,000 steps with a 10,000 step burn-in, and 2 additional stars required 400,000 steps with a 200,000 step burn-in. We assessed convergence both by visual inspection and by comparing the posteriors at the start and end of each chain.

We also compute the integrated autocorrelation time (IAT) of our chains, which has a maximum value of 1.2; we therefore estimate that there are at least 35,000 independent samples of the posterior, consistent with the start and end of each chain being independent. Comparing the median of each parameter at the first 10% of each chain to the final 10% we find the largest fractional difference to be 9%. A similar test comparing the standard deviation from the first and last 10% yields a maximum fractional difference of 6%. These results are consistent with convergence. To calculate the lithium equivalent width, we integrate a Gaussian with parameters from the MCMC chains

$$\Sigma_{Li} = \sqrt{2\pi} C_1 \sigma \tag{1.3}$$

which we converted to an equivalent width by scaling the area by the height of the nearby continuum, H

$$EW_{Li} = \frac{\sqrt{2\pi} C_1 \sigma}{H} \tag{1.4}$$

To find the height of the nearby continuum, we evaluate a line defined by the slope (m) and intercept (b) of the fit at 6707.819 Å (the center of the lithium feature). We use our chains to compute the posterior on equivalent width and report the median. We take the standard deviation of the equivalent width posterior from the converged chain as our uncertainty.

For stars that do not have a clear lithium feature, we use the same MCMC fitting method to calculate upper limits on the lithium equivalent width. For upper limits, C_1 in Equation 3 approaches zero, and σ and RV_s are set almost entirely by the reference line. We identify upper limits by measuring the skew of the equivalent widths' distribution. If the skew exceeds 0.225 or the mean of the distribution is below 5 mÅ, we flag the spectrum as a candidate upper limit. We confirm our upper limit selection by visually inspecting the spectrum. We calculate the 1, 2, and 3σ upper limits by calculating the 68th, 95th, and 99.7th percentiles of the cumulative distribution function. We over-plot Gaussians with σ determined by the reference line and depths set by the 1, 2, and 3σ upper limits to visually assess the corresponding upper limit on the equivalent width. We report the 2σ upper limits.

We calculate the equivalent width for each star using both spectral orders that include the lithium 6707.8 Å feature. We adopt the equivalent width measurement or upper limit from the order/observation combination that yields the best fit to the data. This is generally the segment with the highest signal-to-noise ratio and no cosmic rays. We confirm our selection by visually assessing the fits. Each fit typically returns similar measurements of the lithium equivalent width, with error bars scaling with SNR as we would expect. Figure 1.4 shows examples of a lithium detection and a lithium upper limit determined from this process.

For stars with large amounts of lithium or high rotational velocities, the lithium 6707.8 Å feature can be blended with the 6707.4 Å Fe line. Past work has introduced a correction factor to equivalent widths to subtract the contribution of the iron line from the blended equivalent width. For example Soderblom et al. (1993) calculate iron 6707.4 Å equivalent widths as a function of $B - V$, and subtract this from their blended equivalent widths. Similar methods are used by Favata et al. (1993); Wichmann et al. (2003); Takeda & Kawanomoto (2005); White et al. (2007); Guillout et al. (2009). To assess the degree to which including an iron blend correction improves our lithium equivalent width measurements, we compiled literature values where available for our target stars and S-value calibration targets (discussed below) from Pace (2013). In Figure 1.5, we compare our measured lithium equivalent widths with and without the Soderblom et al. (1993) correction to these literature values for 12 stars. We find that our method with no correction applied yields more consistent equivalent widths compared to these literature values. Applying the Soderblom et al. (1993) correction systematically under-predicts the literature values. We attribute this to our method of fitting a Gaussian to the full absorption profile and taking the integral of the Gaussian, rather than summing fluxes in a specific wavelength range directly from the spectrum. As a result, since the asymmetric wing from the iron line introduces a negligible effect on the final equivalent width, we do not apply a correction to our measured equivalent widths.

To ensure there are no systematic offsets in our lithium fitting technique, we inject a Gaussian of a given equivalent width into an ARCES spectrum with no detectable lithium. Then we feed the injected spectrum through our fitting routine and recover the equivalent width. We conduct this test on five spectra with moderate to high signal-to-noise corresponding to stars at a range of spectral types. For each star, we repeat the injection/recovery test for 13 different values of injected lithium equivalent widths from 5-100 mÅ. Figure 1.6 compares the recovered equivalent widths to the injected lines, which agree to $\lesssim$1 mÅ, well within the reported error bars for each measurement. The main source of this disagreement is noise in each spectrum with the noise being on average slightly above or slightly below the continuum at the location of the lithium line, resulting in a slight underestimate or overestimate of the equivalent width, respectively.

We calculate age posteriors from lithium equivalent widths and uncertainties for our target stars using Bayesian Ages For Field LowEr-mass Stars (`BAFFLES`) (Stanford-Moore et al. 2020). `BAFFLES` utilizes a Bayesian framework that is calibrated on stars in clusters and associations of known ages. Functional forms were fit both to lithium equivalent width as a function of $B-V$ for each calibration cluster, and the intrinsic scatter in equivalent width. Final age posteriors represent a combination of the astrophysical scatter in the age/equivalent width relation and the measurement error (Stanford-Moore et al. 2020). `BAFFLES` uses a uniform star formation history as the age prior. The uncertainty on age is largely driven by the observed astrophysical scatter in lithium equivalent width at a constant age, with a much smaller contribution from the measurement uncertainty for a single star. `BAFFLES` is calibrated to calculate age posteriors from lithium for stars with $B-V$ between 0.35 and 1.9 from both measurements and upper limits. Figure 1.7 shows an example `BAFFLES` posterior for HIP 41277, where an equivalent width of 125 ± 5 mÅ translates to an age posterior of 174^{+79}_{-77} Myr. We detect lithium in 26 stars, and determine upper limits for 140 stars; of the full sample, 159 stars are in the `BAFFLES` $B-V$ range. We report $B-V$, lithium equivalent width measurements (or upper limits), and median ages and confidence intervals from `BAFFLES` (where applicable) for our sample in Table A.0.1.

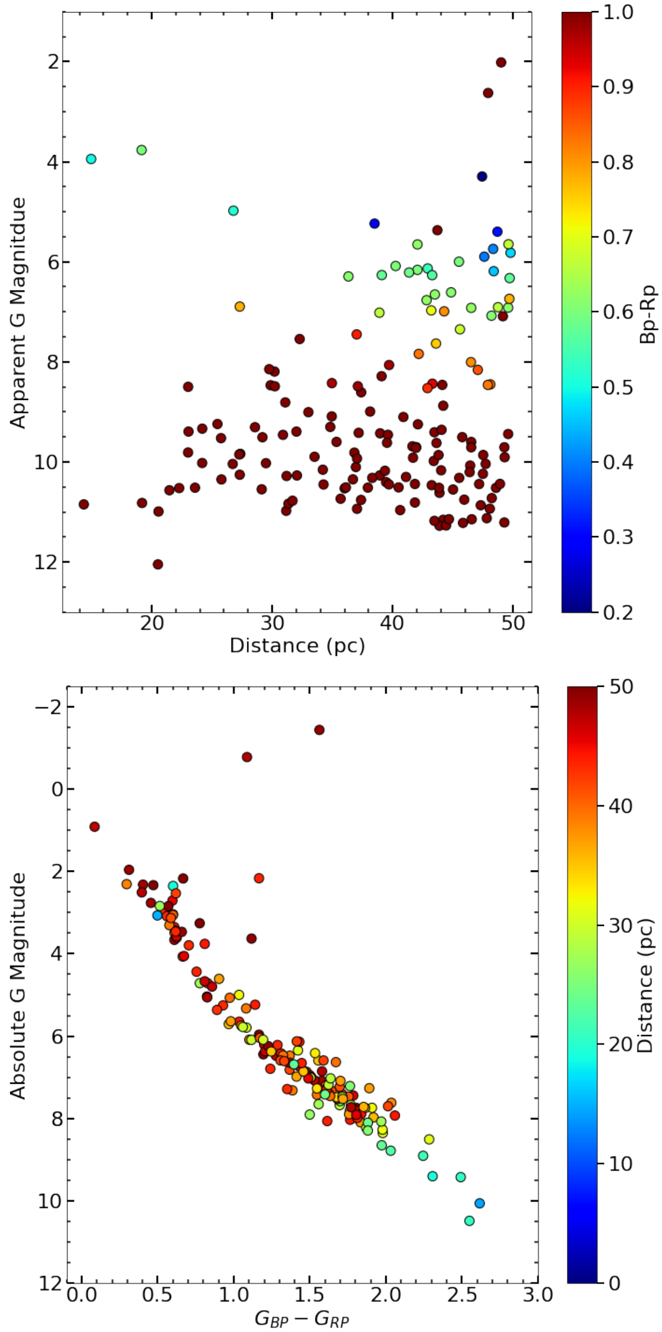

Fig. 1.3.— **Left:** Apparent *Gaia* G magnitude of the accelerating stars as a function of distance. **Right:** Color-magnitude diagram of the sample. The median of our sample is $G_{BP} - G_{RP} = 1.4$ (∼K5), 41.5 pc, and $G = 9.6$. Our sample covers a wide range of spectral types, so we require a variety of age tracers to derive ages for each star.

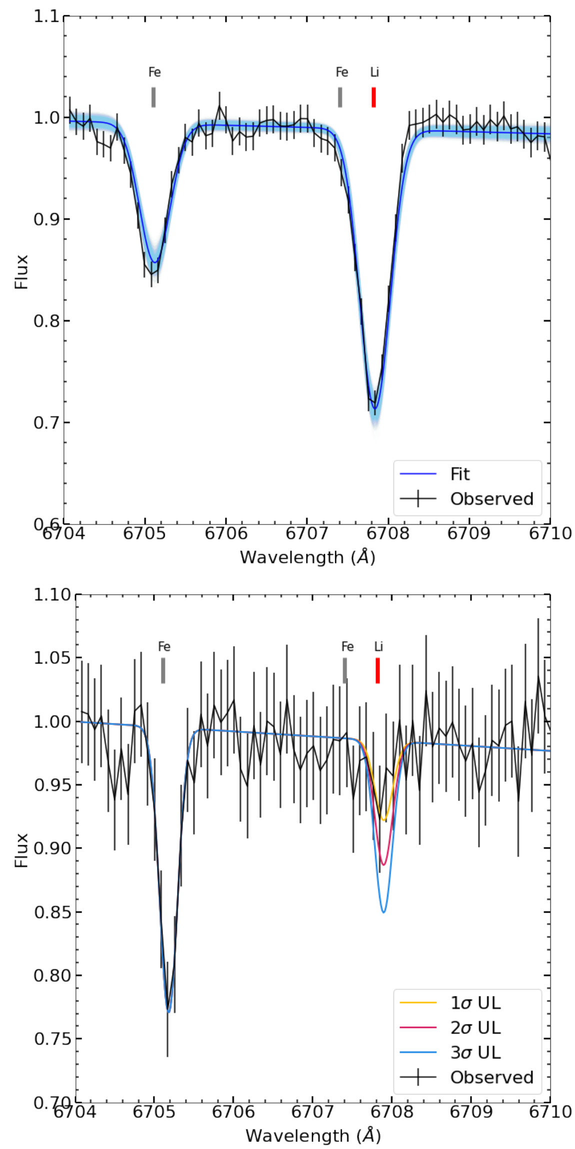

Fig. 1.4.— Our double Gaussian fitting technique allows us to robustly measure lithium equivalent widths and, in cases where lithium is not detected, place upper limits on the lithium equivalent widths. **Left:** A spectrum of HIP 41277 ($SNR \sim 75$) covers both the Fe 6705.1 Å line (used as a reference line) and the Li 6707.79 Å feature. Black points with error bars are the APO/ARCES data, the dark blue line represents the highest likelihood fit from the MCMC, and the light blue lines are random draws from the posterior. We find a lithium equivalent width for HIP 41277 of 125 ± 5 mÅ. **Right:** An example spectrum ($SNR \sim 30$) of an upper limit for lithium, from HIP 110663. Data are again black points with error bars, while yellow, red, and blue represent 1, 2, and 3σ upper limits. We find a 2σ upper limit of 30 mÅ for HIP 110663, which we use to set a lower limit on the age of the star.

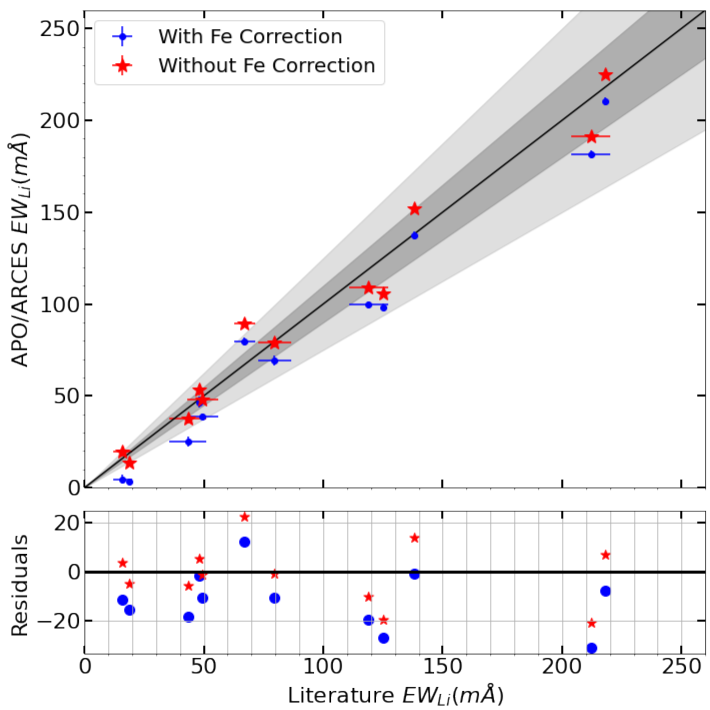

Fig. 1.5.— Comparing our lithium equivalent width measurements to results from the literature for 12 stars (red stars) shows good agreement. The black line shows the 1-to-1 relation, with ±10% in dark gray and ±25% in light gray. We find that applying a correction for the 6707.4 Å iron line (Soderblom et al. 1993) results in a systematic underestimate of the literature abundances (blue points). We conclude that our Gaussian fitting technique is robust to contamination from the 6707.4 Å iron line, so we do not include an additional correction to our reported equivalent width values. We take literature lithium equivalent widths for HIP 104075, HIP 105232, HIP 113905, HIP 35628, HIP 97779 from Guillout et al. (2009), HIP 115147, HIP 43410, HIP 52462, HIP 71631 from Wichmann et al. (2003), HIP 42438 from White et al. (2007), and HIP 43726 from Takeda & Kawanomoto (2005).

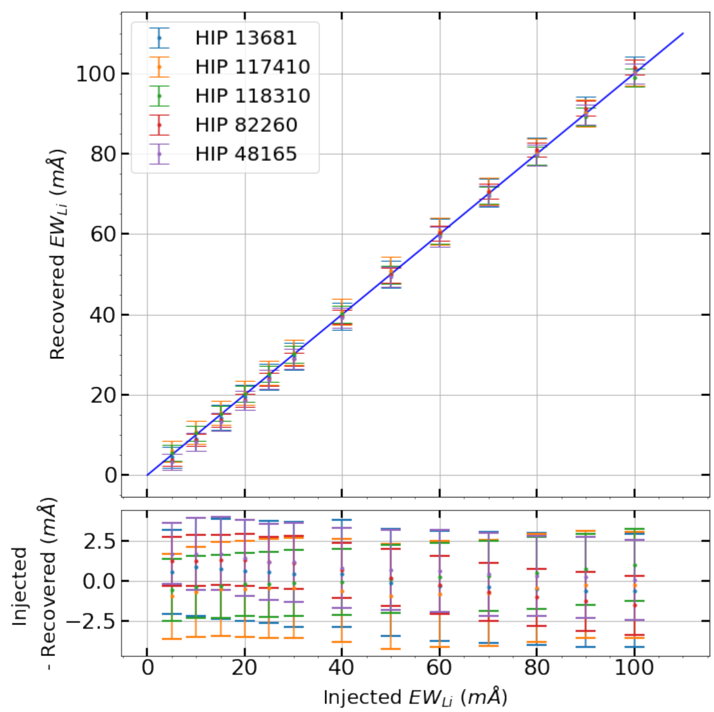

Fig. 1.6.— We confirm that our lithium fitting routine reliably recovers accurate equivalent widths with injection/recovery tests on 5 stars. We chose these stars since they had no detectable lithium, and because they cover a range of SNR and spectral types. The recovered equivalent widths are all within $\lesssim$1 mÅ of the injected value, and consistent with reported error bars.

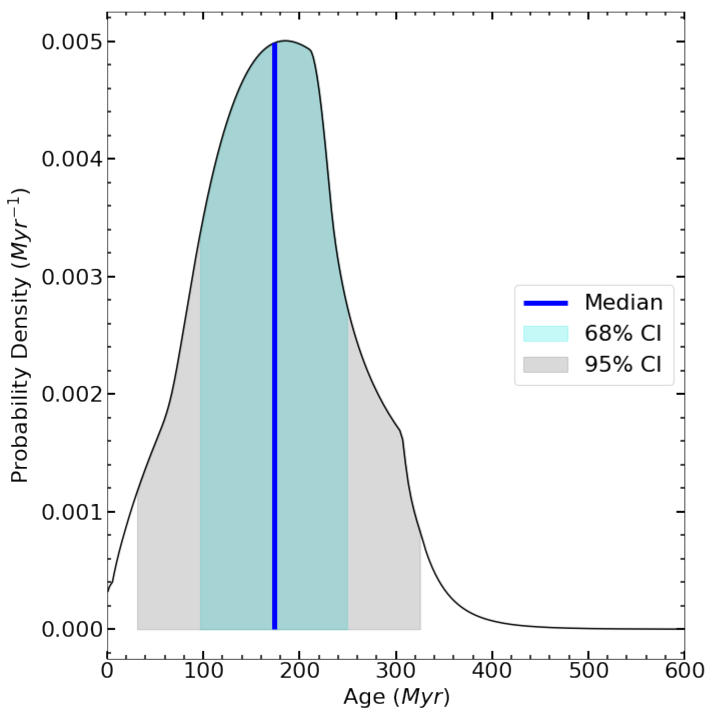

Fig. 1.7.— An example of an age posterior for one star in our sample, HIP 41277, using `BAFFLES` (Stanford-Moore et al. 2020). From an adopted B-V of 1.03 ± 0.015 and a lithium equivalent width of 125 ± 5 mÅ, we derive an age posterior with a median age of 174 Myr and 68% confidence interval between 97 and 253 Myr. Lithium measurements and median ages, and 68% and 95% confidence intervals for the full sample are reported in Table A.0.1.

1.4. Calcium Emission

We also use `BAFFLES` to derive age posteriors from calcium H (3968.47 Å) and K (3933.67 Å) emission (R'_{HK}) for stars with spectral types between F5 and K2. R'_{HK} is derived from the S-value index, which is the ratio of calcium emission flux to nearby continuum. S-values were initially measured photometrically following the methods of the Mt. Wilson Ca II H & K survey that ran from 1966 to 1983 (Duncan et al. 1991). The emission flux is measured through two triangular bandpasses, and the continuum flux is measured through two rectangular bandpasses on either side of the emission cores, as shown in Figure 1.8.

Subsequently, the photometric S-value index is most commonly measured from spectra with the S-value index calibrated for each instrument. To calibrate APO/ARCES, we observe 99 stars with published S-values from Pace (2013), which compiled published S-values from the literature including Duncan et al. (1991), Wright et al. (2004), and Gray et al. (2003). We follow a prescription similar to that in Wright et al. (2004) with some modifications to calibrate the APO S-values. We use a 1-parameter fit (compared to the 4-parameter fit of Wright et al. 2004) with the form

$$S = a\left(\frac{H+K}{R+V}\right) \tag{1.5}$$

H and K are the summed counts from convolving the spectrum with the triangu-

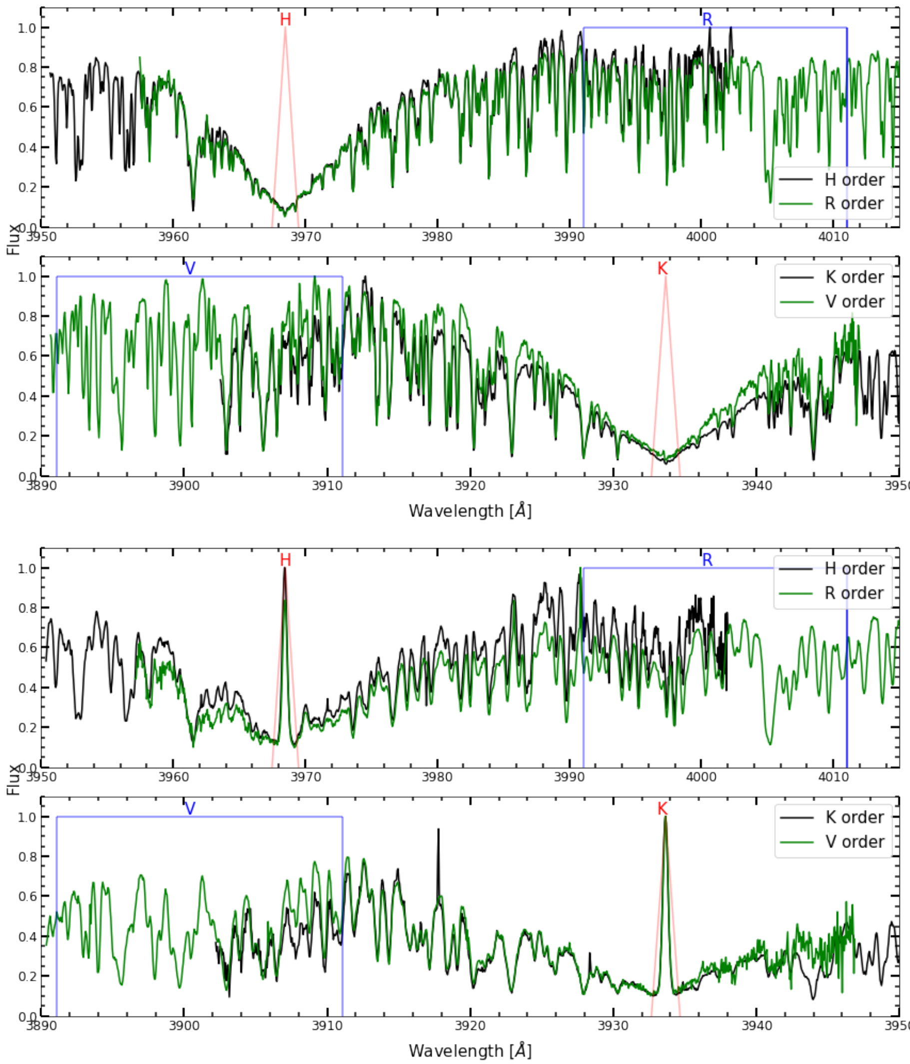

Fig. 1.8.— The S-value is measured from spectra using triangular transmission profiles (red lines) centered on the Ca II H and K emission lines, and rectangular profiles (blue lines) to measure the nearby continuum level. The top two panels show an example of a low-activity star with no significant H and K emission (HIP 102040), and the bottom two panels show HIP 115147, a star with strong H and K emission.

lar transmission profiles, while R and V are the summed counts from convolving the continuum with the rectangular transmission profiles (taken from Duncan et al. 1991).

We shift the spectra to the rest frame so the calcium emission cores or the centers of the absorption lines are centered in the narrow triangular band passes. Using the $G_{BP} - G_{RP}$ to $B - V$ conversion described in Section 1.2, and the grid from Pecaut & Mamajek (2013) we find effective temperatures, masses, and radii for our targets, assuming they are on the main sequence. We find the synthetic spectrum with solar metallicity from Husser et al. (2013) with the closest effective temperature and surface gravity. We convolve the synthetic spectrum with a Gaussian to match the resolution of ARCES. Then, we shift the ARCES spectra to the rest frame by cross correlating the observed spectrum with the theoretical spectral template.

Literature S-values are often not given with error bars, and generally variation over the star's activity cycle is expected to be larger than measurement errors. Given the lack of reliable error bars we do not attempt to derive a posterior probability distribution on the fit parameter. Instead, we assume error bars of unity for each measurement, and find the value of a in Equation 1.5 that minimizes the χ^2. We compute this minimum value over a uniform grid for a with 10^6 elements between 0 and 40, finding a best-fit value of $a = 18.3$, which results in a $\sim$15% scatter on the relationship between APO S-values and the literature S-values. The fit is shown in Figure 1.9, and we attribute this $\sim$15% scatter mainly to variable emission over a stellar cycle.

We investigated more complicated fits, such as including additional parameters to weight the H and K bands or the R and V bands differently, or attempting to further correct the continuum by comparing to theoretical spectra, or performing different fits for different spectral type bins. None of these more complicated methods significantly improved the scatter, and some resulted in systematic offsets from the fit at larger S-values. As a result, we adopt the more straightforward approach outlined above.

The transformation from S-value to R'_{HK} is

$$R'_{HK} = 1.34 \times 10^{-4} C_{Cf} S - R_{phot} \tag{1.6}$$

where C_{Cf} is the color correction factor, a third-order polynomial in $B-V$ (Middelkoop 1982), and R_{phot} is another third-order polynomial in $B-V$ that corrects for the photospheric contribution to the emission in the Ca H and K lines (Hartmann et al. 1984). S and R'_{HK} are given for each of our target stars in Table A.0.2, as well as median ages and 68% and 95% confidence intervals for the 34 stars between F5 and K2 derived using `BAFFLES` (Stanford-Moore et al. 2020). Similar to lithium, `BAFFLES` calculates age posteriors from R'_{HK} values in this spectral type range based on empirical fits to R'_{HK} as a function of age and the intrinsic scatter as observed in clusters and associations of known ages. Figure 1.10 shows an example age posterior for HIP 669, with a $log(R'_{HK})$ of -4.88 that

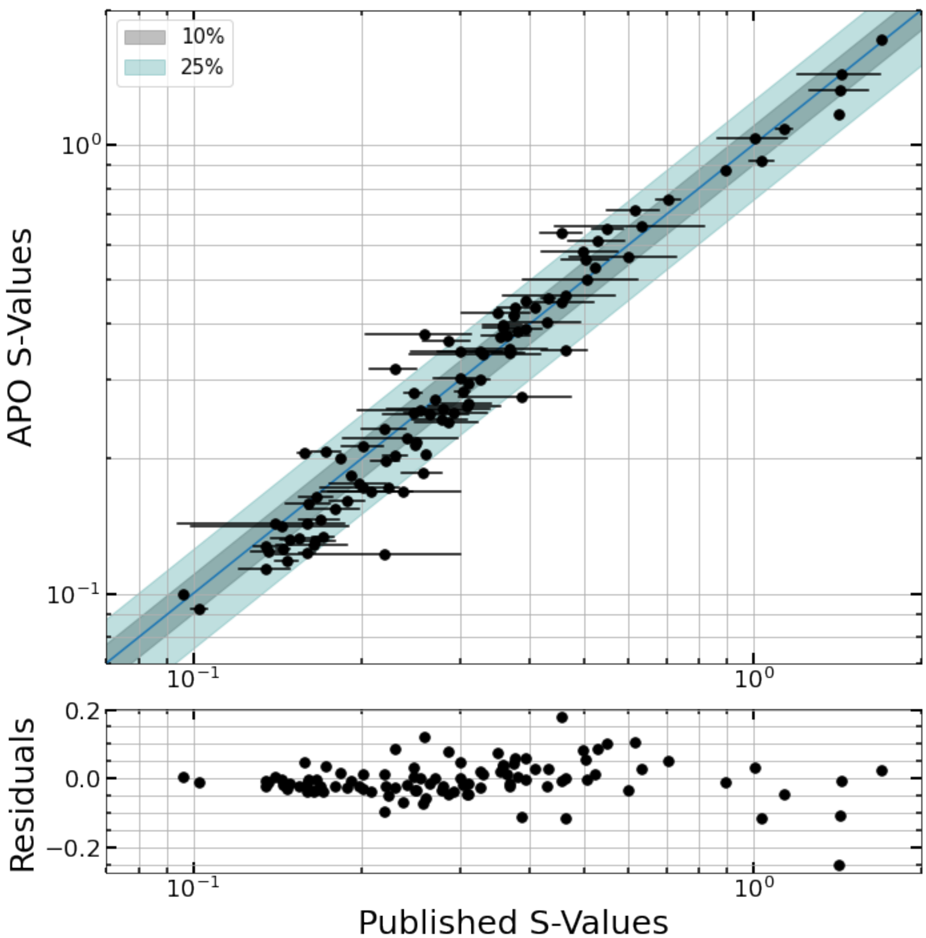

Fig. 1.9.— We use a 1 parameter fit to calibrate chromospheric S-values from APO/ARCES to values reported in the literature. We observe a $\sim 15\%$ scatter in the relationship, which we largely attribute to the intrinsic variability in chromospheric activity for each star. Error bars represent the maximum and minimum S-value reported in the literature for each star. We report S-values and $log(R'_{HK})$ for all the accelerating stars in Table A.0.2.

gives an age posterior of $6.5^{+3.2}_{-2.7}$ Gyr.

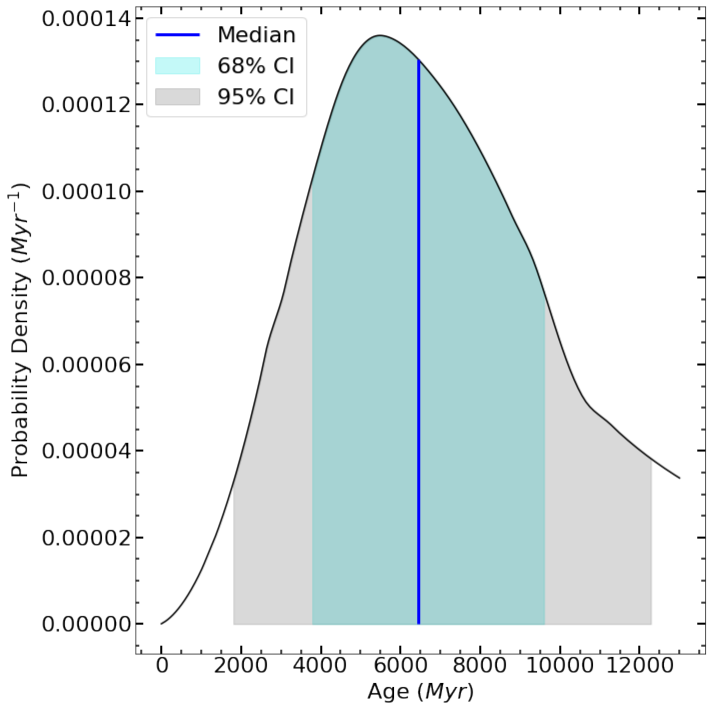

Fig. 1.10.— An example age posterior for HIP 669, using our APO/ARCES measurement and `BAFFLES` (Stanford-Moore et al. 2020). For this relatively inactive star, with $log(R'_{HK}) = -4.88$ and $B-V = 0.62 \pm 0.015$, we derive an age posterior with a median age of 6.5 Gyr and 68% confidence interval between 3.8 and 9.7 Gyr.

1.5. Gyrochronology

We calculate gyrochronological ages using the python package `gyro-interp` which computes age posteriors by interpolating between rotation sequence models for open clusters with well-defined ages (Bouma et al. 2023). `gyro-interp` is calibrated for stars with $3800K < T_{eff} < 6200K$. We measure rotation periods from $TESS$ light curves processed with the SPOC pipeline (Jenkins et al. 2016), which were available for 129 of our target stars. We visually inspect each light curve and its corresponding Lomb-Scargle periodogram (Lomb 1976; Scargle 1982) to make an initial estimate for the rotation period. Then, to further validate and refine the selected period, we use the non-linear least squares fitting python package *LMFIT* (Newville et al. 2016) to fit a sinusoid to the light curve. We run *LMFIT* to find the sinusoid's amplitude, phase, vertical offset, and period. We limit the period to a narrow range around the initial estimate to prevent the fit from getting stuck at a local minimum at a harmonic of the true period. We do not use MCMC or another sampling algorithm to find the rotation period because a sinusoid is only an approximation of spot modulation and because photometric measurements are correlated over small time spans.

To ensure we select an actual rotation signal we compare our adopted period to peaks in the corrected (PDCSAP) and the uncorrected (SAP) periodograms. The period must appear as a local maximum in both periodograms to be accepted. We do not require the period to be the global maximum of either periodogram.

We differentiate between "flat" light curves and robust variability signals by comparing χ^2 for a flat line and the best-fitting sinusoid. Based on visual inspection of light curves from our target stars, we identify light curves with $\Delta\chi^2 \gtrsim 500$ as having a variability signal, and classify all other light curves as flat. Regardless of their $\Delta\chi^2$, 25 stars have variability amplitudes $\lesssim 0.1\%$ and are taken to be flat. We also visually confirm variability in each light curve. We do not measure rotation periods for 8 stars that share a *TESS* pixel with another star that is $\gtrsim 2.5\%$ as bright as the target star in the *Gaia* G band. We also visually inspect the target-pixel files (TPFs) for bright stars on the edges of nearby pixels.

For stars observed in multiple *TESS* sectors, we adopt the period from the light curve with the fewest and/or smallest down-link gaps, which we determine by visual inspection. If more than one curve has similar amounts of data gaps, we select the light curve with the highest signal-to-noise. We use the other sectors to further validate our adopted rotation periods. Some stars are not variable in some sectors but are highly variable in others. We attribute this to spot evolution, such that spots are present in one sector but not present in another. We adopt the rotation period from the sector displaying variability.

Measuring rotation period uncertainties can be challenging, and there are several different methods commonly used. For example, Howard et al. (2021); Santos et al. (2019, 2021) estimate rotation uncertainties by the width of the periodogram peak, while Healy & McCullough (2020) use the width of the autocorrelation peak. We estimate uncertainties on the measured rotation periods with a similar method, by fitting a Gaussian to the exponentiated Lomb-Scargle periodogram peak, which is proportional to the likelihood of a single-sinusoid model at each frequency (VanderPlas 2018; Ivezić et al. 2020). VanderPlas (2018) cautions that this method does not take into account the possibility of having selected the wrong periodogram peak. Therefore, we validate our period selection in two ways: visual inspection and false alarm probability (FAP). For all our measured periods, the largest FAP we calculate is 0.5%, and all the light curves cover more than one period making visual identification of the period relatively robust. We propagate the σ of the best fitting Gaussian in frequency space to period, such that

$$\sigma_P = \sigma_f P^2 \tag{1.7}$$

We use `gyro-interp` (Bouma et al. 2023) to turn these rotation periods and uncertainties into age posteriors, with an example gyrochronological age posterior shown in Figure 1.12. `gyro-interp` generates age posteriors from rotational periods in a similar manner to how `BAFFLES` generates posteriors from lithium and calcium measurements. A reference sample of stars in clusters and associations of known ages was used to find the distribution in rotation period as a function of age and temperature, as well as the astrophysical scatter. The generated age posterior is then a combination of the rotation period and errors and the astrophysical scatter. As with lithium, the dominant source of uncertainty in the final posterior is astrophysical scatter rather than measurement uncertainty on the period.

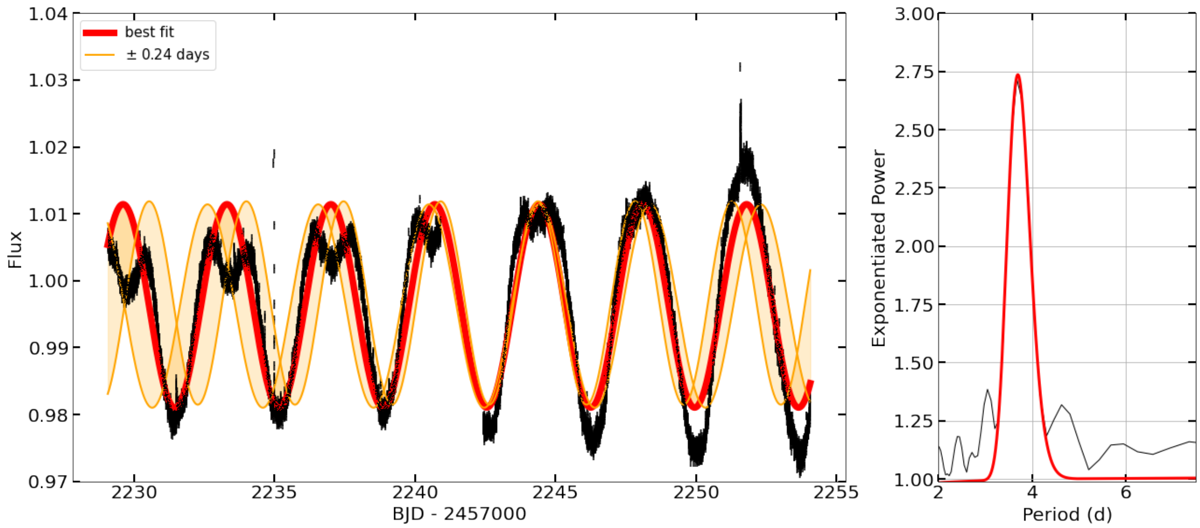

Fig. 1.11.— We measure rotation periods for the accelerating stars by analyzing their periodograms and fitting sinusoids to their lightcurves. The *TESS* light curve for HIP 41277 is shown in black with the best fitting sinusoid with a period of 3.7 days overplotted in red. The orange curves represent the best-fitting period adjusted by the uncertainty measured from the width of the exponentiated periodogram peak. The right-hand panel shows the exponentiated Lomb-Scargle periodogram with a significant peak at ~ 3.7 days and the best fitting Gaussian.

1.5.1. Complex light curves

Several *TESS* light curves of our targets do not exhibit straightforward rotation signals. In this section, we briefly describe the different types of variability we observe (e.g. McQuillan et al. 2013).

Double periods - HIP 117410 shows two distinct periods in its light curves, but is not on obviously contaminated pixels. We interpret this as a rotation signal of two stars on the same pixels. To determine each period, we visually assess each light curve and make an initial estimate for each period. Then, we fit a double

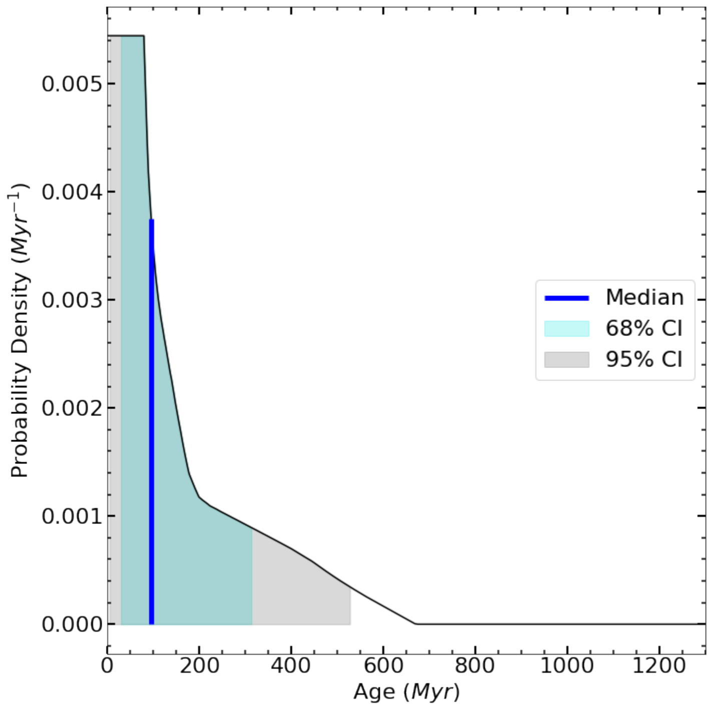

Fig. 1.12.— We calculate age posteriors from rotation periods using `gyro-interp` (Bouma et al. 2023). The age posterior for HIP 41277 with an adopted T_{eff} of 4551 ± 195K and a measured rotation period of 3.70 ± 0.24 days has a median age of 96 Myr and a 68% confidence interval between 30 and 316 Myr.

sinusoid to the flux of the form,

$$F = A_1 \sin(2\pi f_1 t + \phi_1) + A_2 \sin(2\pi f_2 t + \phi_2) \tag{1.8}$$

to each light curve using *LMFIT*. f_1 and f_2 are the frequencies, A_1 and A_2 are the amplitudes, and ϕ_1 and ϕ_2 are the phase shifts of each of the component sinusoids. We find the best fitting periods for each component to be 10.65 ± 1.07 days and 0.69 ± 0.07 days. The best-fitting double sinusoid for HIP 117410 is shown in Figure 1.13.

HIP 117410 (WDS 23484-1259) is a ~ 1" double star with a flux ratio of ~ 4 in V band (Mason et al. 2001), so this pair is most likely the source of the two signals. This pair was not flagged as a contaminated pixel due to the absence of a *Gaia* G band measurement for the secondary. Early analysis of the APO/ARCES radial velocities of HIP 117410 (to be discussed in detail in a future paper) suggests that the system is not a short-period stellar binary.

We cannot determine which period corresponds to which star, or directly measure $B - V$ for the secondary with the current data. Instead, we derive an age posterior for each period with the B-V of the primary, then marginalize over each star by summing the two posteriors and normalizing the resulting posterior so that it integrates to 1 (Table 1.1).

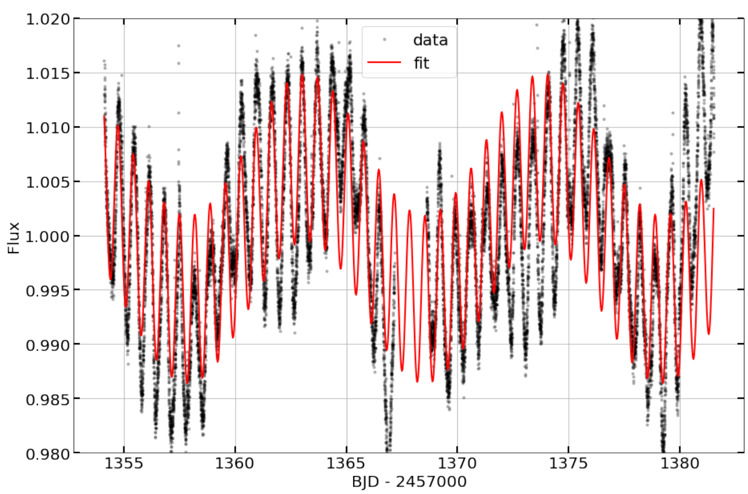

Fig. 1.13.— We fit a two-component sinusoid model (red) to a light curve (black) for HIP 117410. We find the periods of the two components to be 10.65 ± 1.07 days and 0.69 ± 0.07 days

P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
10.65 ± 1.07	415.99	214.20–967.37	146.59–11492.34
0.69 ± 0.07	105.19	32.77–337.67	5.56–542.49
Combined	254.07	64.24–625.41	10.11–1400.23

Table 1.1: The age posteriors for HIP 117410 for both periods and the combined posteriors are listed. We adopt the combined age posterior for this system.

Eclipsing Binaries - We identify HIP 16247, HIP 45731, and HIP 49577 as eclipsing binaries. HIP 45731 and HIP 16247 were previously analyzed in Prša et al. (2022) and Gallenne et al. (2023), respectively.

We calculate the orbital period and rotational periods separately for each of these three systems. To find the orbital period we phase-fold the light curve based on the highest periodogram peak, verifying that the eclipses overlap, and the primary and secondary eclipses are not repeated in the folded light curve. If the phase folded curve has too many eclipses or the eclipses do not overlap, we try the next harmonic. We estimate the uncertainty of the orbital period by altering the period until the eclipses no longer overlap in the phase-folded light curve. To measure the underlying rotational period, we mask out the eclipses and then use the method described above for single stars. Table 1.2 gives the orbital periods (P_o) and rotational periods (P_r) for each system as well as approximate depths of the primary (D_p) and secondary (D_s) eclipses. Figure 1.14 shows the light curve for the eclipsing binary HIP 16247 with the best fitting sinusoid to the underlying rotation plotted in red. The data and the sinusoid fit are out of phase with each other between $1420 < BJD - 2457000 < 1425$. We attribute this to the fact that a simple sinusoid does not account for spot evolution and changing amplitudes, which can shift the phase of the periodicity, but a sinusoid is sufficient for capturing the periodicity of the underlying rotation rate.

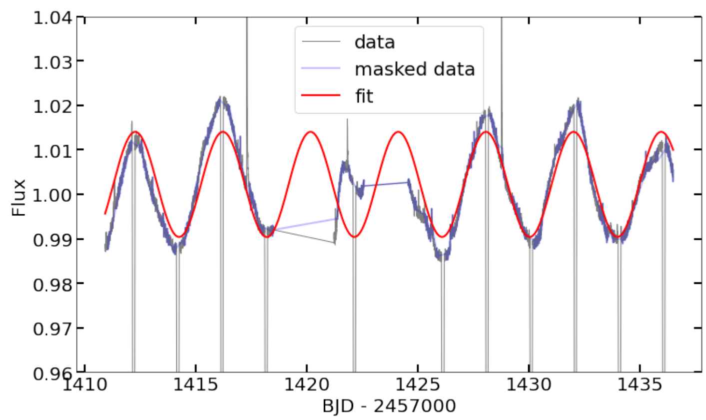

Fig. 1.14.— We fit a sinusoid to the underlying rotation signal of an eclipsing binary system. The light curve for HIP 16247 is shown in gray. The masked data with the eclipses removed is shown in purple, and the best-fitting sinusoid is shown in red. This star is tidally locked with $P_{orb} \approx P_{rot} \approx 3.98$ days.

Target	WDS	P_o (d)	D_p	D_s	P_r (d)
HIP 16247	03294-2406	3.98 ± 0.004	$\sim 28\%$	$\sim 22\%$	3.95 ± 0.4
HIP 45731	09194+6203	2.92 ± 0.003	$\sim 15\%$	$\sim 14\%$	2.92 ± 0.3
HIP 49577	–	16.85 ± 0.17	$\sim 24\%$	$\sim 10\%$	7.2 ± 0.72

Table 1.2: The estimated parameters of the three eclipsing binaries in our sample.

In close binaries, the stars can be spun up by tidal forces, so that the rotation period reflects tidal evolution rather than spin-down following the empirical gyrochronology relations. For HIP 1627 and HIP 45731, the rotation periods and orbital periods we measured are approximately the same suggesting that both systems are tidally locked. Therefore, we do not derive gyrochronological ages for these systems. For HIP 49577, the rotation rate we derived is significantly different than the orbital period we derived. From the rotation period and $B-V$ of the primary star, we measure a median age of 211 Myr and 68% confidence interval of $105-422$ Myr. However, while this system is not tidally locked, we

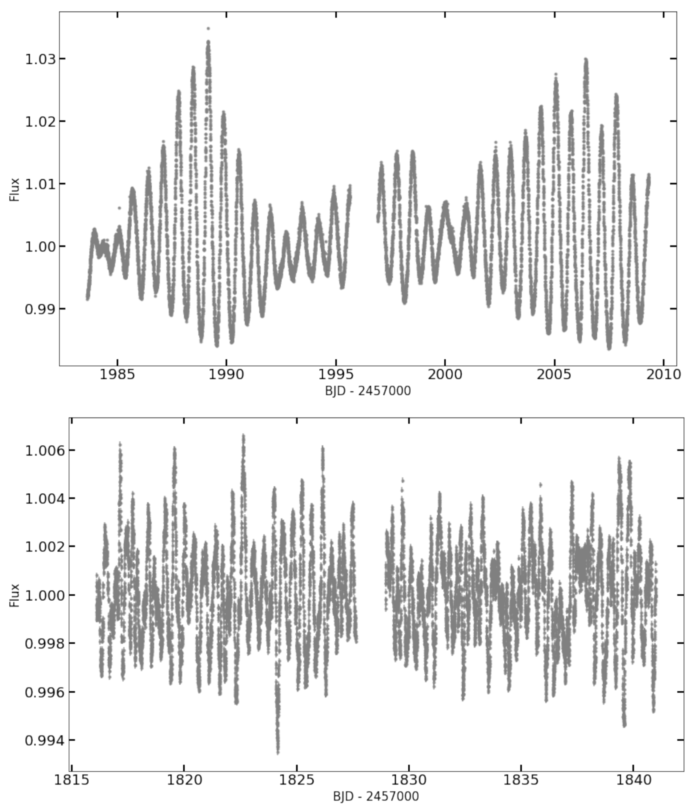

Fig. 1.15.— Several of our target stars exhibit photometric variability that is inconsistent with rotation. **Top:** A light curve of HIP 59767, a γ Doradus variable (Kahraman Aliçavuş et al. 2016), shows a beat pattern making it difficult to measure an underlying rotation period. **Bottom:** A light curve of HIP 22361 shows stochastic variability that is not attributed to rotation.

cannot rule out tidal forces affecting the rotational evolution of the system.

Other Variable Stars - We identify several stars that exhibit variability that does not resemble the rotational modulation of star spots. We use the examples in McQuillan et al. (2013) to visually classify these stars as stochastic or beat pattern variables. HIP 11090 and HIP 59767 both display a beat pattern in their light curves. HIP 21066, HIP 41274, HIP 22361, and HIP 59199 exhibit stochastic variability. An example of each is shown in Figure 1.15. Along with not displaying clear rotation signals, most of these stars are outside the `gyro-interp` temperature range and are therefore not candidates for deriving gyrochronological ages. HIP 41274 is within the `gyro-interp` temperature range, but it is a RS-CVn variable (Martínez et al. 2022), so it is also not a candidate for deriving a gyrochronological age. The close binary to HIP 41274 could be the source of its observed acceleration.

1.6. Combined Age Posteriors

For stars with $B - V \leq 0.58$ (corresponding to an F9.5 spectral type and approximate main-sequence mass of $1.1 M_{\odot}$), we apply additional constraints on their ages based on stellar lifetimes and their CMD positions (e.g. Nielsen et al. 2013, 2019). Using this technique, we also produce age posteriors for stars too massive for Li, R'_{HK}, or gyrochronology. CMD age posteriors are especially useful in cases where we can only derive a lower limit on the age from a lithium equivalent width upper limit because the stellar lifetime can be used to constrain the upper bound on the age (e.g. late F stars). To calculate CMD position-based age posteriors, we follow the method from Nielsen et al. (2019), which uses a Bayesian approach to compute ages from stellar models and optical photometry. For each star, MCMC is used to explore the 5-dimensional parameter space in stellar mass, age, metallicity, initial rotation, inclination of the rotation axis, and parallax. We use the same priors on these parameters as in Nielsen et al. (2019), and a Gaussian likelihood for the *Gaia* G_B, G, and G_R catalog photometry and errors. The model predictions at each MCMC step are generated, as in Nielsen et al. (2019), from a combination of MESA (Paxton et al. 2011) evolutionary models with ATLAS9 model atmospheres (Castelli & Kurucz 2004). As in Nielsen et al. (2019), for stars with $B - V \leq 0.27$, we include a prior based on the metallicity distribution of stars < 1 Gyr from Casagrande et al. (2011).

For 55 stars more than one age tracer is available, and we improve our constraints by taking the product of each age posterior for a given star. In Table 1.3 we present our full 166-star sample, including basic properties and final median ages and 68% and 95% confidence intervals for each star. Figure 1.16 shows the age posteriors for the sample as a function of $B - V$.

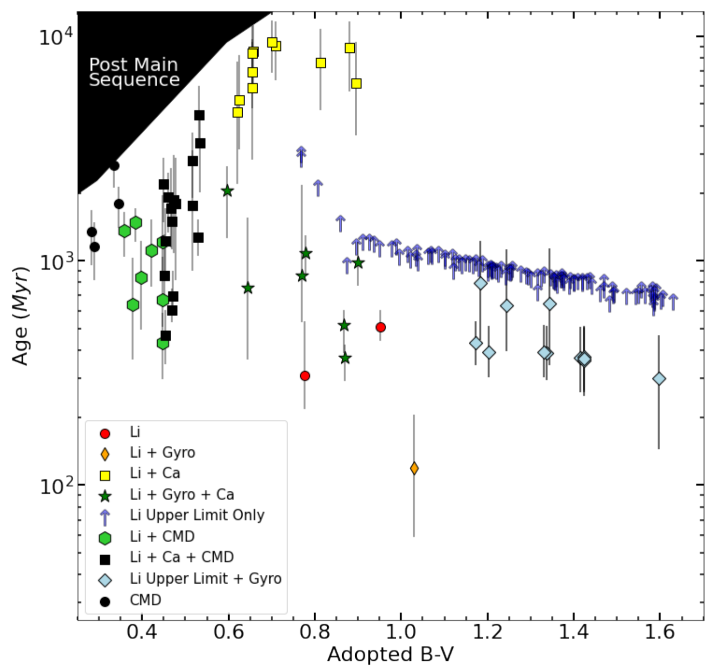

Fig. 1.16.— The medians and 68% confidence intervals for our derived age posteriors for the full sample of 166 stars, as a function of $B-V$. The symbol and color of each star represent which age tracer (or tracers) was used to derive the age posterior. The error bars represent the 68% confidence intervals.

Table 1.3:: Our complete sample of accelerating stars with age posteriors and age tracers used. Ages come from lithium (Li), lithium upper limits (Li UL), calcium emission (R'HK), rotation periods (Gyro), and CMD positions (CMD).

Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 143	00 01 49.35	+21 27 45.01	9.9811 ± 0.0002	1.033 ± 0.019	K3	43.4 ± 0.59	Li UL	≥ 4442.08	≥ 1034.76	≥ 484.61
HIP 669	00 08 16.36	-14 49 28.16	6.8956 ± 0.0002	0.624 ± 0.015	G1V	27.31 ± 0.02	Li, R'HK	5234.81	3132.71–8290.74	1783.08–11668.88
HIP 1068	00 13 15.83	+69 19 36.76	12.0476 ± 0.0015	1.315 ± 0.1	M4	20.52 ± 0.05	Li UL	≥ 4217.04	≥ 720.14	≥ 138.08
HIP 1412	00 17 40.90	-08 40 56.14	10.2711 ± 0.0006	1.414 ± 0.005	K7V	32.05 ± 0.09	Li UL, Gyro	367.99	258.82–508.19	148.22–612.04
HIP 1539	00 19 12.39	-03 03 13.01	10.2808 ± 0.0004	1.375 ± 0.005	K5V	31.19 ± 0.02	Li UL	≥ 4294.84	≥ 828.91	≥ 280.46
HIP 1771	00 22 25.19	+06 40 03.16	10.6176 ± 0.0005	1.489 ± 0.1	K7V	43.88 ± 0.09	Li UL	≥ 4234.75	≥ 744.83	≥ 164
HIP 2426	00 30 55.58	+73 22 14.89	10.0247 ± 0.0002	1.359 ± 0.02	K	29.48 ± 0.01	Li UL	≥ 4298.94	≥ 834.64	≥ 285.77
HIP 2843	00 36 01.85	-05 34 14.58	6.6114 ± 0.0002	0.466 ± 0.007	F6V	44.84 ± 0.06	Li UL, R'HK, CMD	1715.26	1150.23–2585.26	808.98–3587.63
HIP 3008	00 38 15.29	+52 19 55.52	9.8115 ± 0.0003	1.424 ± 0.02	M0.0V	23.02 ± 0.01	Li UL, Gyro	360.24	250.15–502.94	128.65–609.82
HIP 3293	00 42 01.42	+40 41 17.71	6.9164 ± 0.0003	0.452 ± 0.009	F5	49.58 ± 0.07	Li, R'HK, CMD	862.22	651.1–1311.35	521.75–2077.47
HIP 3633	00 46 34.04	+14 19 31.61	9.6005 ± 0.0001	0.814 ± 0.061	K3V	46.51 ± 0.21	Li UL, R'HK	7664.88	4696.93–10795.26	2383.87–12584.18

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 3724	00 47 48.47	+06 01 54.92	10.963 ± 0.0004	1.4 ± 0.02	M0.0V	40.62 ± 0.05	Li UL	≥ 4277.49	≥ 804.65	≥ 255.04
HIP 4024	00 51 34.56	-14 53 26.46	9.4072 ± 0.0002	1.069 ± 0.009	K3.5IV–V	43.46 ± 0.15	Li UL	≥ 4471.25	≥ 1075.56	≥ 530.93
HIP 5319	01 08 01.34	+32 00 43.66	6.1346 ± 0.0003	0.398 ± 0.01	F5IV	42.9 ± 0.06	Li UL, CMD	843.22	495.26–1230.45	219.62–1644.67
HIP 5763	01 13 58.86	+16 29 40.26	9.3939 ± 0.0002	1.22 ± 0.015	K6V	32.01 ± 0.03	Li UL	≥ 4348.13	≥ 903.42	≥ 352.3
HIP 6776	01 27 06.32	+34 22 39.18	6.1623 ± 0.0003	0.453 ± 0.007	F7IVsv	42.05 ± 0.05	Li UL, R'HK, CMD	1220.64	830.65–1719.17	617.88–2140.42
HIP 6796	01 27 27.60	+23 14 40.42	9.4186 ± 0.0004	1.157 ± 0.074	K7V	37.17 ± 0.04	Li UL	≥ 4400.69	≥ 976.89	≥ 409.71
HIP 6833	01 27 55.56	-01 59 29.35	8.4502 ± 0.0002	0.656 ± 0.002	G5V	48.1 ± 0.06	Li, R'HK	6929.95	4767.61–9685.66	2971.34–12173.29
HIP 6943	01 29 26.45	+42 16 54.85	7.078 ± 0.0003	0.453 ± 0.011	F5	48.19 ± 0.06	Li, R'HK, CMD	466.38	344.95–600.94	208.59–1059.12
HIP 7287	01 33 53.62	+27 14 08.74	10.5178 ± 0.0002	1.2 ± 0.02	K5	48.57 ± 0.22	Li UL	≥ 4334.42	≥ 884.25	≥ 332.72
HIP 8653	01 51 31.19	-07 44 23.57	8.4607 ± 0.0003	0.644 ± 0.019	G5V	47.89 ± 1.09	Li, Gyro, R'HK	759.94	364.45–1561.09	185.93–2455.98
HIP 9067	01 56 43.73	+29 48 39.04	10.8658 ± 0.0003	1.47 ± 0.1	K5V	47.3 ± 0.03	Li UL	≥ 4275.7	≥ 801.85	≥ 231.94
HIP 9989	02 08 40.08	-00 21 35.61	10.4039 ± 0.0003	1.248 ± 0.002	K5V	39.45 ± 0.03	Li UL	≥ 4340.02	≥ 892.08	≥ 341.68
HIP 10050	02 09 23.11	+17 13 27.11	6.2937 ± 0.0003	0.471 ± 0.001	F5V	36.32 ± 0.04	Li, R'HK, CMD	694.63	612.9–897.62	509.72–1344.86
HIP 10532	02 15 41.72	-09 00 47.23	8.877 ± 0.0002	0.88 ± 0.033	K0V	44.19 ± 0.03	Li UL, R'HK	8922.3	5685.95–11625.28	2863.63–12794.61
HIP 11090	02 22 50.30	+41 23 46.65	5.7433 ± 0.0009	0.289 ± 0.006	F0III–IV	48.33 ± 0.14	CMD	1155.1	820.23–1479.52	520.53–1744.73

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 11815	02 32 23.18	+03 22 57.48	11.2101 ± 0.0014	1.357 ± 0.02	K5	49.26 ± 0.05	Li UL	≥ 4276.65	≥ 803.47	≥ 253.41
HIP 12837	02 45 01.09	-22 09 59.51	7.0178 ± 0.0002	0.517 ± 0.008	F7V	38.9 ± 0.03	Li, R'HK, CMD	1771.7	899.96–3111.02	386.64–4669.22
HIP 13681	02 56 14.10	-11 50 49.84	9.8618 ± 0.0003	1.062 ± 0.015	—	47.51 ± 0.04	Li UL	≥ 4456.4	≥ 1054.8	≥ 509.53
HIP 13782	02 57 23.76	-23 51 43.79	5.399 ± 0.0005	0.238 ± 0.003	A5IV/V	48.69 ± 0.15	CMD	757.31	595.43–917.22	443.02–1075.26
HIP 15211	03 16 05.89	-05 51 15.85	9.9066 ± 0.0002	0.936 ± 0.007	K3	49.29 ± 0.04	Li UL	≥ 4551.82	≥ 1187.49	≥ 609.82
HIP 16247	03 29 22.87	-24 06 03.09	8.8107 ± 0.0005	1.02 ± 0.037	K3VkFe+0.4	31.08 ± 0.02	Li UL	≥ 4428.57	≥ 1015.84	≥ 461.81
HIP 16709	03 34 59.64	+34 36 39.51	10.3002 ± 0.0004	1.225 ± 0.02	M0	41.1 ± 0.65	Li UL	≥ 4331.28	≥ 879.86	≥ 328.75
HIP 17118	03 39 58.81	+63 52 13.47	6.7675 ± 0.0002	0.478 ± 0.023	F5	42.81 ± 0.03	Li, R'HK, CMD	1790.5	818.76–2885.78	405.89–3903.25
HIP 17213	03 41 13.71	+75 23 56.51	9.6211 ± 0.0006	1.1 ± 0.015	K0	43.63 ± 0.08	Li UL	≥ 4420.43	≥ 1004.51	≥ 453.14
HIP 18527	03 57 43.92	-20 16 04.05	8.6088 ± 0.0004	0.899 ± 0.021	K2VkFe–0.5	37.38 ± 0.03	Li UL, Gyro, R'HK	981.67	770.11–1216.88	619.34–1469.49
HIP 19739	04 13 55.87	+82 55 06.98	10.1025 ± 0.0005	1.424 ± 0.02	M0.0V	36.94 ± 0.17	Li UL, Gyro	372.38	265.37–510.32	199.63–612.75
HIP 19912	04 16 18.56	+08 43 09.96	9.4283 ± 0.0002	1.148 ± 0.002	K7V	38.96 ± 0.26	Li UL	≥ 4408.38	≥ 987.64	≥ 433.87
HIP 20419	04 22 25.69	+11 18 20.56	9.3585 ± 0.0004	1.183 ± 0.005	K2	44.05 ± 0.48	Li UL, Gyro	797.7	444.12–1230.22	313.95–1599.99
HIP 20648	04 25 29.38	+17 55 40.46	4.2957 ± 0.001	0.049 ± 0.007	A2IV–Vs	47.42 ± 0.31	CMD	602.61	530.07–670.92	456.89–742.38
HIP 21066	04 30 57.17	+10 45 06.36	6.9211 ± 0.0002	0.472 ± 0.013	F7V	46.51 ± 0.06	Li UL, R'HK, CMD	1864.19	1085.7–2959.28	650.55–3268.37
HIP 21152	04 32 04.80	+05 24 36.14	6.2678 ± 0.0002	0.42 ± 0.014	F5V	43.27 ± 0.05	Li UL, CMD	1116.14	765.3–1521.58	423.86–1861.12

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 22361	04 48 50.35	+75 56 28.39	5.8984 ± 0.0004	0.283 ± 0.004	A9IV	47.6 ± 0.06	CMD	1350.34	959.72–1684.76	602.75–1979.1
HIP 23208	04 59 38.25	-10 59 02.26	9.3009 ± 0.0004	1.18 ± 0.006	K5Vk	34.82 ± 0.05	Li UL	≥ 4365.63	≥ 927.89	≥ 376.02
HIP 24017	05 09 44.50	+64 55 10.16	6.2661 ± 0.0002	0.448 ± 0.004	F3V	39.09 ± 0.03	Li UL, CMD	1216.34	828.62–1964.68	633.89–2692.15
HIP 24177	05 11 29.73	+10 07 15.66	11.156 ± 0.0005	1.461 ± 0.005	M0V	44.19 ± 0.18	Li UL	≥ 4216.67	≥ 719.62	≥ 159.14
HIP 24292	05 12 47.95	+16 18 25.43	10.1675 ± 0.0003	1.23 ± 0.015	K7V	44.01 ± 0.03	Li UL	≥ 4357.55	≥ 916.6	≥ 365.54
HIP 24457	05 14 49.83	+00 42 31.67	9.4575 ± 0.0002	1.118 ± 0.005	K5V	39.6 ± 0.02	Li UL	≥ 4445.1	≥ 1038.96	≥ 482.73
HIP 25606	05 28 14.72	-20 45 33.99	2.6272 ± 0.0015	0.807 ± 0.027	G5II–IIIa	47.92 ± 0.47	Li UL	≥ 5405.2	≥ 2111.49	≥ 1021.51
HIP 27021	05 43 52.79	-21 16 47.04	6.9723 ± 0.0001	0.531 ± 0.006	F8/G0V	43.2 ± 0.03	Li, R'HK, CMD	4464.59	3159.78–5991.22	2077.47–7143.76
HIP 28823	06 05 03.38	+42 58 53.88	5.8206 ± 0.0006	0.358 ± 0.005	F3V	49.78 ± 0.22	Li UL, CMD	1360.86	1038.87–1642.63	703.98–1867.14
HIP 29208	06 09 35.91	+05 40 08.05	8.194 ± 0.0002	0.895 ± 0.022	G0	30.19 ± 0.02	Li UL	≥ 4526.8	≥ 1152.02	≥ 568.66
HIP 33282	06 55 23.46	+44 25 19.30	10.7232 ± 0.0004	1.344 ± 0.035	K6V	48.22 ± 0.03	Li UL, Gyro	647.7	341.31–1139.51	248.03–1546.97
HIP 33368	06 56 22.10	+46 59 26.75	10.4419 ± 0.0005	1.346 ± 0.1	K7V	47.17 ± 0.06	Li UL	≥ 4308.04	≥ 847.34	≥ 288
HIP 34804	07 12 17.20	+59 15 51.95	10.2697 ± 0.0003	1.591 ± 0.082	K4	38.93 ± 0.7	Li UL	≥ 4224.76	≥ 730.29	≥ 157.52
HIP 36081	07 26 03.61	-15 39 57.28	6.7409 ± 0.0002	0.62 ± 0.005	G1/2V	49.68 ± 0.46	Li, R'HK	4585.75	2189.37–7747.86	1010.09–11446.52
HIP 37267	07 39 10.33	+29 54 32.16	10.4391 ± 0.0006	1.277 ± 0.022	K5V	48.9 ± 0.03	Li UL	≥ 4328.24	≥ 875.61	≥ 325.41

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 40518	08 16 23.06	-17 22 03.83	10.0726 ± 0.0003	1.134 ± 0.004	—	46.43 ± 0.6	Li UL	≥ 4406.56	≥ 985.1	≥ 430.98
HIP 41274	08 25 14.08	-07 10 12.85	7.0892 ± 0.0006	0.952 ± 0.011	K0V	49.14 ± 0.07	Li	505.8	439.44–602.43	335.92–820.22
HIP 41277	08 25 16.90	+04 15 12.72	9.9158 ± 0.002	1.03 ± 0.015	K8V	41.68 ± 0.08	Li, Gyro	119.62	58.63–206.48	17.47–300.01
HIP 41319	08 25 49.87	+17 02 46.57	5.9986 ± 0.0003	0.448 ± 0.005	F5IIIm?	45.5 ± 0.07	Li, CMD	668.73	509.36–1040.76	400.75–1505.21
HIP 42231	08 36 37.96	+39 02 13.55	11.2749 ± 0.0003	1.241 ± 0.015	—	43.88 ± 0.15	Li UL	≥ 4308.4	≥ 847.87	≥ 296.87
HIP 42343	08 37 57.55	-08 52 53.72	6.9078 ± 0.0002	0.516 ± 0.001	F6V	48.73 ± 0.05	Li, R'HK, CMD	2791.29	1896.43–3724.4	1184.85–4152.26
HIP 42507	08 40 00.26	-06 28 33.06	9.2424 ± 0.0003	1.361 ± 0.001	K6V	25.46 ± 0.09	Li UL	≥ 4259.1	≥ 778.95	≥ 226.84
HIP 43745	08 54 35.72	+63 45 29.74	10.4509 ± 0.0002	1.423 ± 0.02	K5V	34.26 ± 0.02	Li UL, Gyro	366.55	260.53–505.31	168.77–610.43
HIP 44162	08 59 39.91	-19 12 28.95	6.0845 ± 0.0003	0.461 ± 0.005	F5V	40.24 ± 0.13	Li UL, R'HK, CMD	1925.01	1340.03–2478.25	901.96–3477.51
HIP 44953	09 09 30.16	-00 23 54.35	8.4254 ± 0.0002	0.767 ± 0.002	G8V	34.95 ± 0.03	Li UL	≥ 5993.75	≥ 2846.19	≥ 1323.58
HIP 45621	09 17 55.38	-03 23 14.09	7.5442 ± 0.0003	0.869 ± 0.013	K1/2V	32.28 ± 0.04	Li, Gyro, R'HK	371.47	291.48–423.05	144.29–489.92
HIP 45731	09 19 22.86	+62 03 16.89	10.5142 ± 0.0006	1.586 ± 0.02	M1.0Ve	37.94 ± 0.03	Li UL	≥ 4167.6	≥ 650.87	≥ 91.63
HIP 46385	09 27 30.56	+50 39 12.45	10.9763 ± 0.0005	1.586 ± 0.1	M2.5V	31.16 ± 0.05	Li UL	≥ 4175.04	≥ 661.4	≥ 92.44
HIP 47013	09 34 53.52	+72 12 20.44	5.6501 ± 0.0005	0.53 ± 0.003	F7V	49.6 ± 0.08	Li UL, R'HK, CMD	1275.94	1054.99–1523.22	861.32–1758.82
HIP 47261	09 37 58.33	+22 31 23.15	9.4588 ± 0.0003	1.21 ± 0.015	K4V	30.89 ± 0.02	Li UL	≥ 4351.49	≥ 908.11	≥ 356.79

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 47300	09 38 21.77	+40 14 23.25	5.237 ± 0.0007	0.223 ± 0.004	F0V	38.49 ± 0.12	CMD	718.62	500.33–934.73	298.93–1134.22
HIP 47387	09 39 14.59	+66 50 58.14	9.6883 ± 0.0002	1.07 ± 0.047	K2	41.63 ± 0.54	Li UL	≥ 4450.55	≥ 1046.57	≥ 489.6
HIP 47539	09 41 31.44	-06 23 35.83	11.2695 ± 0.0004	1.353 ± 0.011	—	44.42 ± 0.37	Li UL	≥ 4288.12	≥ 819.51	≥ 270.07
HIP 47741	09 43 55.61	+26 58 08.40	10.8486 ± 0.0004	1.542 ± 0.039	M3.5V	14.37 ± 0.01	Li UL	≥ 4202.01	≥ 698.97	≥ 130.85
HIP 48165	09 49 03.16	+07 49 08.68	9.5048 ± 0.0002	0.925 ± 0.015	G5	45.77 ± 0.48	Li UL	≥ 4574.64	≥ 1219.3	≥ 638.59
HIP 48629	09 54 59.84	+40 23 07.28	10.7594 ± 0.0002	1.581 ± 0.197	K5V	37.37 ± 0.04	Li UL	≥ 4234.47	≥ 744.26	≥ 152.31
HIP 49046	10 00 26.71	+27 16 01.61	10.5279 ± 0.0002	1.372 ± 0.015	M0.5V	36.01 ± 0.02	Li UL	≥ 4292.26	≥ 825.31	≥ 276.52
HIP 49526	10 06 43.80	+41 42 52.65	10.5229 ± 0.0003	1.492 ± 0.02	M1.0V	22.28 ± 0.02	Li UL	≥ 4210.41	≥ 710.86	≥ 147.69
HIP 49577	10 07 13.74	-14 18 16.54	9.6168 ± 0.0004	1.331 ± 0.014	K5V	39.53 ± 0.64	Li UL	≥ 4322.44	≥ 867.5	≥ 318.63
HIP 49910	10 11 23.70	-01 39 44.69	11.1457 ± 0.0004	1.391 ± 0.023	—	46.55 ± 0.04	Li UL	≥ 4286.95	≥ 817.89	≥ 268.72
HIP 51208	10 27 34.85	+12 58 48.14	10.2039 ± 0.0003	1.2 ± 0.015	K4V	46.38 ± 0.46	Li UL	≥ 4361.89	≥ 922.66	≥ 371.01
HIP 52339	10 41 48.44	+12 06 31.90	10.9365 ± 0.0003	1.42 ± 0.015	K7V	37.04 ± 0.03	Li UL	≥ 4281.04	≥ 809.62	≥ 260.19
HIP 53175	10 52 39.61	+00 29 01.95	10.1771 ± 0.0004	1.293 ± 0.005	K4V	39.35 ± 0.03	Li UL	≥ 4312.99	≥ 854.28	≥ 304.58
HIP 53869	11 01 16.35	+29 35 27.47	9.4421 ± 0.0003	0.868 ± 0.045	K3V	49.58 ± 0.06	Li UL, Gyro, R'HK	515.87	388.61–604.35	245.7–648.74

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 54094	11 04 07.15	+53 22 55.47	9.7081 ± 0.0001	1.1 ± 0.02	—	46.51 ± 0.03	Li UL	≥ 4441.41	≥ 1033.83	≥ 481.2
HIP 54199	11 05 16.57	+65 48 49.51	8.2868 ± 0.0001	0.874 ± 0.016	G5	39.08 ± 0.45	Li UL	≥ 4383.16	≥ 952.11	≥ 383.88
HIP 55192	11 18 00.48	+35 26 41.81	8.0635 ± 0.0003	0.776 ± 0.015	G5	39.7 ± 0.17	Li	308.96	218.51–539.13	66.28–1459.69
HIP 55409	11 20 51.76	-23 13 02.42	7.8415 ± 0.0002	0.658 ± 0.003	G3/5V	42.16 ± 0.04	Li UL, R'HK	8592.32	5708.12–11271.52	3349.2–12735.77
HIP 57361	11 45 34.44	-20 21 12.38	10.8203 ± 0.0004	1.482 ± 0.015	M2.5V	19.2 ± 0.06	Li UL	≥ 4164.34	≥ 646.46	≥ 85.43
HIP 57571	11 48 01.50	+71 21 28.14	10.754 ± 0.0002	1.369 ± 0.02	—	45.93 ± 0.35	Li UL	≥ 4295.72	≥ 830.14	≥ 281.31
HIP 57645	11 49 13.15	-20 20 34.67	8.9933 ± 0.0002	0.895 ± 0.015	G3V	38.12 ± 0.03	Li UL, R'HK	6204.18	3637.04–9422.01	1781.04–12308.42
HIP 57984	11 53 37.20	+49 36 01.07	10.5512 ± 0.0002	1.316 ± 0.015	K5	45.0 ± 0.15	Li UL	≥ 4310.95	≥ 851.43	≥ 302.07
HIP 59000	12 05 50.66	-18 52 30.92	9.3933 ± 0.0007	1.336 ± 0.014	K5V	23.06 ± 0.05	Li UL, Gyro	388.45	293.22–514.42	219.44–613.01
HIP 59198	12 08 22.20	-00 28 57.36	10.5473 ± 0.0004	1.409 ± 0.015	K4V	29.13 ± 0.08	Li UL	≥ 4283.83	≥ 813.52	≥ 264.26
HIP 59199	12 08 24.81	-24 43 43.95	3.9466 ± 0.0013	0.334 ± 0.015	F1V	14.98 ± 0.04	CMD	2665.78	2119.54–3334.23	1360.6–3849.28
HIP 59247	12 09 13.08	+19 28 03.61	10.4409 ± 0.0003	1.29 ± 0.015	K7V	41.84 ± 0.03	Li UL	≥ 4317.08	≥ 860.01	≥ 310.2
HIP 59767	12 15 20.27	+87 42 00.42	6.1886 ± 0.0008	0.346 ± 0.011	F1IV	48.38 ± 0.05	CMD	1793.47	1401.11–2138.93	945.46–2464.7
HIP 59905	12 17 15.20	-07 48 43.93	10.0396 ± 0.0003	1.125 ± 0.015	K4V	47.71 ± 0.05	Li UL	≥ 4406.67	≥ 985.26	≥ 430.87

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 59953	12 17 48.39	+46 37 20.73	10.7356 ± 0.0002	1.602 ± 0.1	M0.5V	35.68 ± 0.02	Li UL	≥ 4177.73	≥ 665.13	≥ 96.84
HIP 60829	12 28 04.06	-15 39 07.60	8.1604 ± 0.0002	0.71 ± 0.012	G6V	47.06 ± 0.06	Li UL, R'HK	9106.19	6391.7–11635.89	4156.82–12789.88
HIP 61947	12 41 46.81	+43 02 26.34	11.1443 ± 0.0003	1.582 ± 0.082	K5V	44.66 ± 0.04	Li UL	≥ 4183.93	≥ 673.77	≥ 104.87
HIP 62325	12 46 22.54	+09 32 22.86	5.3697 ± 0.0004	0.989 ± 0.007	G9	43.71 ± 0.12	Li UL	≥ 4529.3	≥ 1156.33	≥ 585.8
HIP 62627	12 49 56.23	+71 11 39.30	9.707 ± 0.0001	0.995 ± 0.003	K8	49.26 ± 0.07	Li UL	≥ 4447.71	≥ 1042.53	≥ 480.59
HIP 63419	12 59 45.50	-04 25 49.04	8.4878 ± 0.0004	0.779 ± 0.021	K1V	37.11 ± 0.04	Li, Gyro, R'HK	1081.58	874.63–1294.12	662.26–1603.98
HIP 63421	12 59 47.34	-06 51 37.39	11.1775 ± 0.0005	1.354 ± 0.01	K5V	43.46 ± 0.04	Li UL	≥ 4285.0	≥ 815.16	≥ 265.57
HIP 63661	13 02 51.36	-02 40 26.25	10.9351 ± 0.0003	1.468 ± 0.1	K5V	48.05 ± 0.06	Li UL	≥ 4236.36	≥ 747.11	≥ 166.58
HIP 65120	13 20 43.83	+10 52 33.31	10.8352 ± 0.0003	1.49 ± 0.02	M0V	31.35 ± 0.03	Li UL	≥ 4206.97	≥ 706.06	≥ 142.13
HIP 65602	13 27 02.90	-24 17 25.56	8.4679 ± 0.0002	0.911 ± 0.002	K2+V	29.87 ± 0.03	Li UL	≥ 4574.65	≥ 1219.41	≥ 641.92
HIP 65651	13 27 35.55	-01 50 16.98	10.3186 ± 0.0002	1.203 ± 0.02	K4	45.52 ± 0.05	Li UL, Gyro	392.63	301.17–513.07	126.99–611.37
HIP 65706	13 28 17.76	+30 02 45.98	10.5091 ± 0.0003	1.577 ± 0.082	K5V	40.49 ± 0.09	Li UL	≥ 4215.51	≥ 717.64	≥ 144.27
HIP 66262	13 34 49.79	+47 22 34.35	9.8141 ± 0.0001	1.165 ± 0.067	K4V	36.82 ± 0.36	Li UL	≥ 4386.44	≥ 956.97	≥ 393.3
HIP 66315	13 35 29.15	+46 33 30.88	10.4651 ± 0.0002	1.039 ± 0.121	K3	43.23 ± 0.03	Li UL	≥ 4412.74	≥ 985.51	≥ 395.09

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 66587	13 38 58.68	-06 14 12.46	10.0215 ± 0.0003	1.419 ± 0.009	M0.5V	24.19 ± 0.01	Li UL	≥ 4273.57	≥ 799.18	≥ 249.51
HIP 66828	13 41 47.23	+21 29 08.18	10.3186 ± 0.0002	1.33 ± 0.02	K5V	38.3 ± 0.03	Li UL, Gyro	393.32	301.47–515.92	223.58–613.17
HIP 69311	14 11 12.64	+43 06 23.96	9.5996 ± 0.0002	1.172 ± 0.07	K5V	35.34 ± 0.01	Li UL, Gyro	431.78	342.15–536.81	251.18–620.33
HIP 69333	14 11 24.19	+30 05 02.44	9.8627 ± 0.0002	1.173 ± 0.004	K5V	43.8 ± 0.03	Li UL	≥ 4402.4	≥ 979.3	≥ 426.36
HIP 69860	14 17 47.84	+21 26 01.58	10.8113 ± 0.0005	1.629 ± 0.1	M1.0V	41.84 ± 0.21	Li UL	≥ 4171.9	≥ 656.97	≥ 92.18
HIP 69963	14 18 59.04	+38 38 26.33	10.7754 ± 0.0004	1.39 ± 0.02	M1.0V	31.69 ± 0.05	Li UL	≥ 4252.64	≥ 769.92	≥ 217.68
HIP 70472	14 24 49.86	-17 27 08.09	10.1555 ± 0.0004	1.257 ± 0.01	K7V	34.2 ± 0.05	Li UL	≥ 4358.91	≥ 918.5	≥ 368.06
HIP 71243	14 34 11.70	+32 32 04.12	6.2142 ± 0.0001	0.45 ± 0.003	F5V	41.36 ± 0.03	Li UL, R'HK, CMD	2202.78	1488.73–2873.15	911.32–3350.92
HIP 71515	14 37 33.07	-16 32 27.62	8.4606 ± 0.0005	0.95 ± 0.002	K2/3V	44.11 ± 0.3	Li UL	≥ 4516.62	≥ 1138.55	≥ 566.76
HIP 71899	14 42 23.10	+21 17 35.10	7.3501 ± 0.0002	0.533 ± 0.01	F8	45.58 ± 0.04	Li, R'HK, CMD	3344.07	2011.77–4550.69	935.24–5494.66
HIP 71957	14 43 03.62	-05 39 29.53	3.7671 ± 0.002	0.385 ± 0.006	F2V	19.16 ± 0.14	Li, CMD	1483.42	1211.67–1716.6	893.61–1957.38
HIP 73121	14 56 40.64	+21 04 16.72	9.1024 ± 0.0002	1.035 ± 0.009	K	40.88 ± 0.16	Li UL	≥ 4483.97	≥ 1093.34	≥ 544.57
HIP 73183	14 57 25.90	+55 54 39.01	9.9311 ± 0.0006	1.555 ± 0.197	K5V	37.04 ± 0.4	Li UL	≥ 4236.44	≥ 747.08	≥ 152.29
HIP 73252	14 58 15.67	+59 35 00.43	9.5255 ± 0.0002	1.565 ± 0.082	K5V	25.75 ± 0.01	Li UL	≥ 4205.13	≥ 703.31	≥ 129.34
HIP 73787	15 04 53.98	-18 35 27.36	9.006 ± 0.0004	1.21 ± 0.008	K5+Vk	33.0 ± 0.64	Li UL	≥ 4358.02	≥ 917.25	≥ 366.04
HIP 74926	15 18 39.55	-18 37 35.71	9.8387 ± 0.0003	1.214 ± 0.009	K4V	27.35 ± 0.1	Li UL	≥ 4350.77	≥ 907.11	≥ 356.02

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 79203	16 09 55.27	-18 20 26.25	6.3266 ± 0.0003	0.447 ± 0.007	F3V	49.7 ± 0.17	Li, CMD	430.94	297.69–584.21	170.71–1095.7
HIP 79593	16 14 20.73	-03 41 39.56	2.0164 ± 0.0055	1.584 ± 0.01	M0.5III	49.0 ± 1.33	Li UL	≥ 4144.8	≥ 619.13	≥ 62.8
HIP 81655	16 40 48.88	+36 19 00.08	10.5661 ± 0.0003	1.605 ± 0.1	M2V	21.47 ± 0.02	Li UL	≥ 4191.94	≥ 684.91	≥ 114.09
HIP 82260	16 48 28.31	-16 20 04.33	7.4515 ± 0.0002	0.767 ± 0.002	G8IV–V	37.01 ± 0.18	Li UL	≥ 6115.02	≥ 2980.3	≥ 1383.07
HIP 83676	17 06 08.20	-06 10 02.07	8.489 ± 0.0003	1.015 ± 0.015	K3V	30.22 ± 0.02	Li UL	≥ 4468.19	≥ 1071.18	≥ 513.21
HIP 83929	17 09 19.61	+41 26 09.85	10.4487 ± 0.0004	1.597 ± 0.197	K4V	39.68 ± 0.06	Li UL, Gyro	298.37	145.14–467.64	84.98–595.32
HIP 87370	17 51 07.73	-22 55 15.61	6.991 ± 0.0001	0.656 ± 0.015	G3V	44.27 ± 0.19	Li, R'HK	8392.46	5631.85–11082.4	3331.8–12696.46
HIP 88962	18 09 33.26	-12 02 19.98	9.8511 ± 0.0005	1.372 ± 0.004	K7V	27.27 ± 0.01	Li UL	≥ 4310.61	≥ 850.97	≥ 302.95
HIP 89320	18 13 30.98	+81 04 32.78	10.5168 ± 0.0003	1.586 ± 0.197	K8	43.84 ± 0.02	Li UL	≥ 4209.43	≥ 709.43	≥ 119.63
HIP 89449	18 15 18.29	+18 29 59.98	9.5074 ± 0.0003	1.323 ± 0.019	K7V	29.19 ± 0.01	Li UL	≥ 4298.3	≥ 833.75	≥ 284.08
HIP 90306	18 25 31.92	+38 21 12.67	10.5152 ± 0.0003	1.442 ± 0.02	K6V	23.6 ± 0.01	Li UL	≥ 4254.11	≥ 771.96	≥ 220.21
HIP 93072	18 57 32.65	-19 02 46.86	10.3501 ± 0.0005	1.437 ± 0.004	K9Vk	25.79 ± 0.01	Li UL	≥ 4285.25	≥ 815.5	≥ 266.15
HIP 98007	19 55 01.54	+04 03 43.66	9.0905 ± 0.0001	0.978 ± 0.003	K3	34.94 ± 0.21	Li UL	≥ 4510.78	≥ 1130.48	≥ 561.44
HIP 99969	20 16 55.42	+06 55 18.29	9.2486 ± 0.0001	1.14 ± 0.015	K4V	42.09 ± 0.11	Li UL	≥ 4388.19	≥ 959.42	≥ 405.74
HIP 100133	20 18 45.79	-00 39 26.33	10.2561 ± 0.0003	1.43 ± 0.012	M0V	27.31 ± 0.27	Li UL	≥ 4281.86	≥ 810.76	≥ 261.29

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 101852	20 38 21.10	-04 30 42.73	7.6339 ± 0.0003	0.597 ± 0.012	G0V	43.59 ± 0.09	Li UL, Gyro, R'HK	2050.45	1262.36–2646.59	580.98–3021.11
HIP 102119	20 41 42.24	-22 19 20.51	9.3352 ± 0.0013	1.121 ± 0.009	K6Ve	24.2 ± 0.02	Li UL	≥ 4347.61	≥ 902.7	≥ 349.81
HIP 104687	21 12 22.60	-15 00 00.36	8.0059 ± 0.0007	0.654 ± 0.041	G3V	46.49 ± 0.05	Li, R'HK	5895.89	2829.05–9452.36	855.62–12158.07
HIP 105504	21 22 07.77	-10 30 47.89	9.7045 ± 0.0002	1.255 ± 0.015	K7	41.95 ± 0.06	Li UL	≥ 4341.38	≥ 893.98	≥ 343.5
HIP 107317	21 44 12.99	+06 38 29.28	10.9909 ± 0.0006	1.522 ± 0.021	M3.0Ve	20.57 ± 0.02	Li UL	≥ 4199.9	≥ 696.16	≥ 131.93
HIP 108036	21 53 17.77	-13 33 06.36	4.9792 ± 0.0009	0.378 ± 0.011	F2V	26.78 ± 0.08	Li, CMD	641.55	361.52–1031.51	166.74–1407.73
HIP 109807	22 14 26.74	+02 42 24.12	9.8969 ± 0.0003	1.252 ± 0.005	K5V	33.51 ± 0.02	Li UL	≥ 4335.15	≥ 885.27	≥ 334.91
HIP 110106	22 18 13.19	-03 10 19.84	10.5093 ± 0.0003	1.399 ± 0.02	K7V	36.11 ± 0.09	Li UL	≥ 4272.52	≥ 797.7	≥ 247.79
HIP 110401	22 21 43.34	-17 09 40.90	10.2378 ± 0.0005	1.208 ± 0.013	K4V	47.48 ± 0.13	Li UL	≥ 4364.59	≥ 926.43	≥ 374.95
HIP 110663	22 25 04.26	-13 59 53.92	8.4368 ± 0.0007	0.77 ± 0.005	G8/K0V	43.29 ± 0.05	Li UL, Gyro, R'HK	860.94	529.79–2168.83	392.82–2847.0
HIP 111976	22 40 53.96	+64 54 26.04	11.2184 ± 0.0003	1.548 ± 0.1	K5	45.83 ± 0.03	Li UL	≥ 4206.41	≥ 705.19	≥ 126.14
HIP 113159	22 54 54.77	-22 22 08.23	6.6536 ± 0.0004	0.469 ± 0.005	F7V	43.5 ± 0.05	Li, R'HK, CMD	600.81	533.5–751.18	358.69–1086.46
HIP 115280	23 20 53.26	+38 10 56.36	5.6536 ± 0.0004	0.468 ± 0.004	F5V	42.05 ± 0.06	Li UL, R'HK, CMD	1499.09	1124.39–1920.51	825.97–2277.7
HIP 115411	23 22 43.20	-19 41 25.93	8.5252 ± 0.0003	0.7 ± 0.005	G6V	42.88 ± 0.04	Li UL, R'HK	9476.12	6862.42–11765.94	4396.19–12807.11

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Target	R.A.	Dec.	G	$B-V$	Spec. Type	Dist (pc)	Tracers	Median Age	68% CI	95% CI
HIP 116973	23 42 43.81	-19 52 57.32	9.719 ± 0.0004	1.085 ± 0.008	K3V	41.93 ± 0.06	Li UL	≥ 4468.03	≥ 1071.06	≥ 525.19
HIP 117159	23 45 09.93	+29 33 42.72	8.147 ± 0.0002	0.859 ± 0.006	K0	29.75 ± 0.13	Li UL	≥ 4766.76	≥ 1475.94	≥ 801.54
HIP 117235	23 46 17.95	+10 16 41.79	11.1245 ± 0.0002	1.376 ± 0.133	K7V	47.8 ± 0.08	Li UL	≥ 4278.54	≥ 806.1	≥ 233.62
HIP 117410	23 48 25.69	-12 59 14.85	9.3062 ± 0.0019	1.244 ± 0.014	K5Vke	28.57 ± 0.22	Li UL, Gyro	635.14	394.14–1127.47	309.22–12215.08
HIP 117795	23 53 19.84	+59 56 42.41	10.0403 ± 0.0002	1.21 ± 0.015	K8	26.72 ± 0.01	Li UL	≥ 4346.62	≥ 901.3	≥ 350.04
HIP 118086	23 57 14.38	-16 30 27.36	10.348 ± 0.0003	1.3 ± 0.015	K7V	36.7 ± 0.03	Li UL	≥ 4291.23	≥ 823.87	≥ 273.54
HIP 118310	23 59 47.78	+06 39 50.95	8.4993 ± 0.0002	1.187 ± 0.052	K4V	23.03 ± 0.01	Li UL	≥ 4399.71	≥ 975.53	≥ 416.39

1.7. Stars within the Lithium Dip

There are several stars in our sample that are near the lithium dip, which is a range of $B - V$ ($\sim 0.4 - 0.55$) where lithium depletes rapidly around the age of the Hyades. This rapid evolution means the feature is not well modeled (e.g. Stanford-Moore et al. 2020), we discuss the consistency of the ages for stars near the lithium dip below.

Many stars with $0.4 < B - V < 0.55$ have age posteriors from different tracers that are generally consistent, including the lithium posterior. For example, HIP 42343 has ages from lithium (4964^{+4688}_{-2766} Myr), R'_{HK} (4793^{+3345}_{-2103} Myr), and CMD position (2105^{+1309}_{-1071} Myr) that give a combined posterior of 2791^{+933}_{-895} Myr. Similarly, the posteriors mainly agree for HIP 17118, HIP 12837, HIP 71899, and HIP 27021.

HIP 47013 shows significant offset between posteriors. The lithium non-detection gives a 2σ lower limit of 1528 Myr, which is consistent with the R'_{HK} posterior of 4997^{+3340}_{-2175} Myr. However, the CMD posterior gives a significantly younger age of 1181^{+247}_{-203} Myr. This may be due to the small number of stars within the lithium dip in the $\sim 500 - 1000$ Myr clusters used to calibrate `BAFFLES`. Because of this limitation, we recommend not relying on ages from lithium alone for stars within the lithium dip. We see similar discrepancies between the posteriors for HIP 6943, HIP 113159, HIP 10050, HIP 71243, HIP 3293, HIP 6776, HIP

44162, HIP 2843, HIP 115280, and HIP 21066.

1.8. Discussion

From our full sample of 166 stars, 102 age posteriors are based on lithium upper limits alone, while 55 stars have an age posterior based on more than one tracer. 31 of our targets ($\sim 19\%$) have a median age younger than 1 Gyr, which is more than expected for a uniform star formation rate. We discuss the full age distribution of our sample below.

1.8.1. Star Formation History of the Solar Neighborhood

Rana (1990) and Cukanovaite et al. (2023) found that the local star formation rate has been near constant for the last 10 to 12 Gyr based on the luminosity distribution of white dwarfs. Other studies found slightly different histories: a maximum star formation rate ($\sim$ 2 times the typical star formation rate at other times) $\sim$ 3 Gyr ago (Cignoni et al. 2006) or two star forming events separated by $\sim$ 5 Gyr (Noguchi 2018). We assume a uniform star formation history, which also matches the priors used in `BAFFLES` (Stanford-Moore et al. 2020) and `gyro-interp` (Bouma et al. 2023).

1.8.2. Population Level Ages

We investigate how significant it is that our sample contains more stars than expected with median ages younger than 1 Gyr. Given the small size of our sample (< 200 stars), and the biased nature in which it was selected, we are not attempting here to constrain the local star formation history. Rather, we are examining whether the total age distribution we observe is broadly consistent with a uniform star formation rate in the solar neighborhood. We construct a population-level age distribution by summing (and then normalizing) the age posteriors for the entire sample. This is equivalent to marginalizing over individual stars; in the limit where each age posterior is a delta-function, this method produces a histogram of stellar ages. With a large enough volume-limited sample, we would expect this population-level distribution to match the star formation of the solar neighborhood. If the star formation rate has been uniform from 0 to 13 Gyr, we expect 1/13 (7.7%) of the summed age distribution to be younger than 1 Gyr.

The left panel of Figure 1.17 shows the population-level age distribution for our 166-star sample, which peaks at young ages ($\lesssim$2 Gyr). Part of this peak is due to early-type stars, which have short main sequence lifetimes and thus do not trace the full 13 Gyr star formation history. These stars make up $\sim 19\%$ of the sample which is an over-representation relative to the initial mass function (Salpeter 1955; Chabrier 2003; Kroupa & Weidner 2003). This is likely due to the multiplicity fractions ($\sim 50-70\%$) of these stars (Currie et al. 2023a) making them more likely to be accelerating. Removing stars with $B-V < 0.595$ (corresponding to a main sequence mass of 1.06 $M_{\odot}$) leaves us with 135 stars and produces the posterior in the right panel of Figure 1.17. The peak at young ages is still present, but less pronounced.

This general behavior is consistent with the types of age posteriors we produce here. For any given late-type star the posterior is either a well-constrained Gaussian-like peak or a lower age limit (corresponding to a lithium upper limit) with only the youngest ages ruled out. While R'_{HK} produces Gaussian-like posteriors at both young and old ages for stars earlier than K2, lithium and gyrochronology generally produce peaked posteriors for later spectral types if they are younger than ~1 Gyr. Stars older than ~1 Gyr mostly do not have detectable lithium (depending on spectral type), and either have rotation periods longer than a typical $TESS$ sector or are less spotted. For the reduced sample of 135 stars, we expect $\sim$ 10 (7.7%) stars to be younger than 1 Gyr. Following the Poisson distribution, we calculate a 65% chance of between 8 and 13 stars being younger than 1 Gyr and a 96% chance of between 5 and 17 stars being younger than 1 Gyr. Testing this prediction is complicated by the fact that we have somewhat broad age posteriors, rather than delta-function age measurements. Nevertheless, we can estimate the number of young stars in our sample by summing the probability in the posteriors for all stars. We calculate that 18% of the age posterior for the reduced sample falls below 1 Gyr. This corresponds to ~ 24 stars, which is significantly higher than expected. In fact, for a Poisson distribution with an expectation value of 10.4 stars, we only expect a value of 24 or higher 0.01% of the time.

1.8.3. Testing Individual Age Tracers

Figure 1.18 further divides the late spectral type stars. The left panel only includes posteriors from lithium detections and/or rotation period detections. The right panel only includes posteriors derived from lithium upper limits. The posterior derived exclusively from lithium upper limits, as expected, is much less peaked and is essentially following the prior. The posterior derived from lithium detections and rotation rates peaks at ages younger than $\sim$1 Gyr. This peaked posterior is expected for age tracers that only return peaked posteriors for the youngest stars.

Nevertheless, this peak seems to include more probability at ages less than 1 Gyr than we would expect from a uniform star formation rate operating between 0 and 13 Gyr. We investigate the relative contributions of lithium and gyrochronology to this result in Figure 1.19. We consider ages from lithium, calcium, and gyrochronology separately.

Lithium: We find that the combined age posterior for lithium-only derived posteriors (114 posteriors) is consistent with a uniform star formation rate. From the Poisson distribution and an assumed uniform star formation rate history, there is an 82% chance of between $5 \leq N_{stars} \leq 11$ being younger than 1 Gyr. We calculate 8.0% of the posterior (~ 9 stars) falls below 1 Gyr, consistent with this prediction.

Calcium: We derive age posteriors from R'_{HK} for 18 late-type stars. There are 7 additional late-type stars that have appropriate $B - V$ values to derive posteriors from R'_{HK} but have $R'_{HK} \leq -5$, which is outside the calibration range of `BAFFLES`, but consistent with old ages. From the Poisson distribution, there is a $\sim 60\%$ chance of measuring between 0 and 1 stars younger than 1 Gyr from a sample of 18 stars. 1% of our 18 star R'_{HK} posterior falls below 1 Gyr, which corresponds to finding 0.18 stars being less than 1 Gyr, which is again consistent with the prediction.

Gyrochronology: We have *TESS* light curves for 129 stars, 106 of which have $B - V > 0.595$. We measure rotation periods for 20 of the late-type stars. From the gyrochronology population-level age distribution, 92% of the probability is less than 1 Gyr, which corresponds to $\sim 17\%$ of the total late-type gyrochronology sample of 106 stars, or about 18 young stars. Our ages from gyrochronology do not include upper limits because flat *TESS* light curves could indicate a long rotation period (longer than a typical $TESS$ sector) or just a star without significant star spots. Overall, we would expect ~ 8 stars out of 106 to have an age of 1 Gyr or younger. From the Poisson distribution with an expectation value of 8, there is a $\sim 4\%$ chance of finding 13 or more stars below 1 Gyr, and a $\sim 0.08\%$ chance of finding 18 or more. Thus, while the number of young stars identified from lithium and calcium alone are consistent with the predictions of a uniform star formation rate, there appears to be $\sim 5 - 10$ excess young stars identified by rotation.

This result, that our combined age posterior suggests 18.0 stars are younger than 1 Gyr, does not change significantly even if we increase the size of the rotational period errors when computing age posteriors. Our period errors are between 0.11 days and 2.02 days, with a median of 0.55 days. Setting a minimum error bar to 1 day, 2 days, or 3 days still results in an estimate that 18.4, 17.8, and 16.8 stars are younger than 1 Gyr, respectively. As a result, we conclude that this excess of probability at young ages cannot be attributed solely to a systematic underestimate of the rotation period uncertainties.

There are several possible explanations for the higher-than-expected fraction of young stars in the gyrochronological sample. Since we select stars that are accelerating, there is a greater chance of selecting a short-period stellar binary (e.g. De Rosa et al. 2019). The rotational evolution of stars in close binaries could be influenced by tides resulting in shorter rotation periods than expected from age alone. Additionally, $TESS$ pixels may be contaminated by multiple stars (either physical binaries or crowded fields). While we screen for contaminated pixels, additional components may be present in some light curves. Alternatively, this effect could be physical: if binaries become unbound over time, a sample of accelerating stars (which is more likely to contain binaries than the field) would be younger than a volume-limited sample.

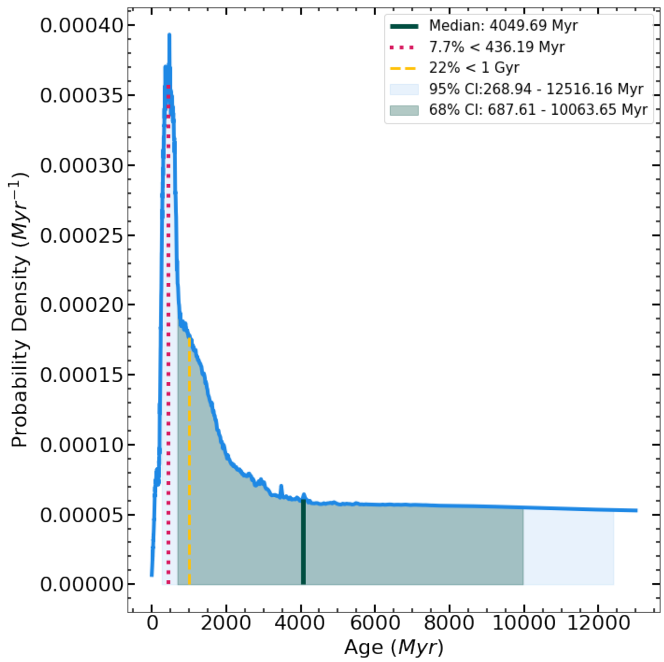
While we screened our initial accelerating star sample for binaries from WDS, SB9, and archival imaging, such a search is not complete to all binary configurations. Indeed, by constructing our sample from accelerating stars not in these catalogs we expect undiscovered binaries to be overrepresented in our sample compared to an unbiased volume-limited sample. For example, three eclipsing binaries (HIP 16247, HIP 45731, and HIP 49577) discussed in Section 1.5 were discovered to be physical binaries only after our survey began. We are currently carrying out radial velocity monitoring of the full sample with the aim of identifying additional short-period stellar binaries, and will report the results in a future paper.

1.8.4. Companions

Since the beginning of this survey in 2020, two brown dwarf companions have been discovered around two of the accelerating stars discussed in this work.

Kuzuhara et al. (2022) found a $\sim 28\ M_{\rm Jup}$ companion to HIP 21152, a likely member of the Hyades Cluster ($\sim$ 750 Myr; Gagné et al. 2018). We measure a slightly older age 1116^{+406}_{-351} Myr based on a lithium upper limit and the star's CMD position. This star, however, is in the lithium dip as discussed in section 1.6. The CMD age 855^{+1348}_{-412}, is consistent with the age of the Hyades. The lithium lower limit suggest an age a factor of two older than the Hyades (>1588 Myr), though this may be an overestimate due to how `BAFFLES` models the lithium dip.

Swimmer et al. (2022) found a $\sim 31\ M_{\rm Jup}$ companion to HIP 5319. With $B - V < 0.45$, HIP 5319 is not a candidate for deriving an age from R'_{HK} using `BAFFLES`. Casagrande et al. (2011) report an age between $1.23^{+0.54}_{-0.6}$ Gyr by fitting isochrones from Bertelli et al. (2008, 2009). We measure an age of 843^{+387}_{-348} Myr from a lithium upper limit and the star's CMD position, which is consistent with the previously reported age. These detections underscore how effective astrometric selection can be at identifying stars hosting substellar companions.



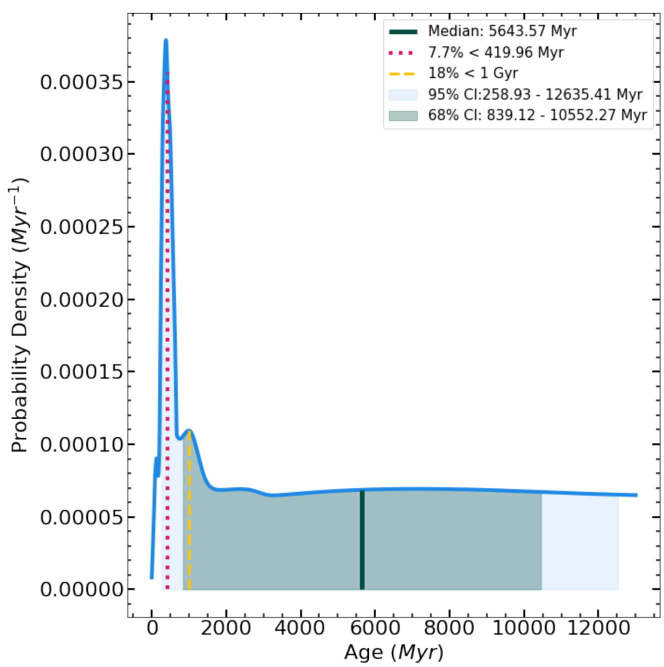


Fig. 1.17.— **Left**: Sum of all the age posteriors for the full sample. The distribution of ages is peaked at young ages, but this is partially due to the inclusion of early-type stars with shorter main sequence lifetimes. **Right**: Sum of age posteriors for only stars with $B-V>0.595$, where there is still a peak at young ages, if less pronounced than for the full sample.

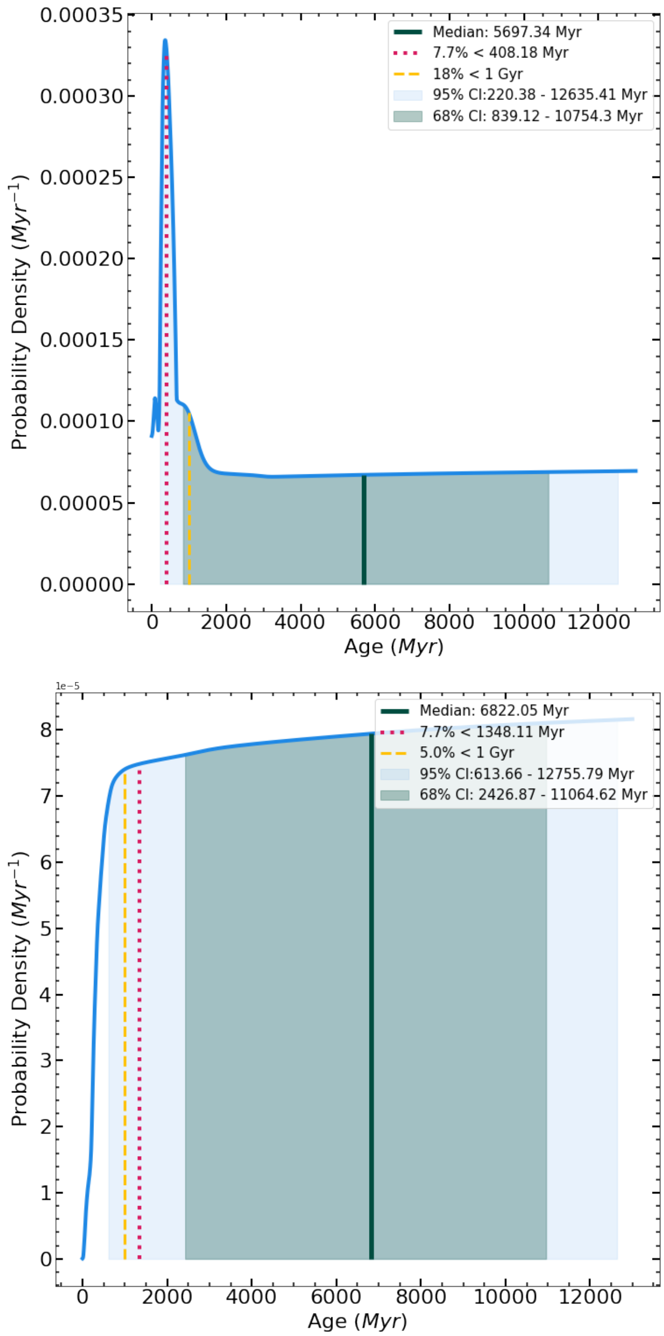


Fig. 1.18.— **Left**: The combined age posterior from lithium detections and/or gyrochronology for the late-type stars in the sample. The peak at young ages is consistent with stars that are < 1 Gyr being most likely to have detectable lithium or clear rotation signals. **Right**: For stars that only have lithium upper limits, the age posterior is mostly flat, following the prior, with only the youngest ages excluded.

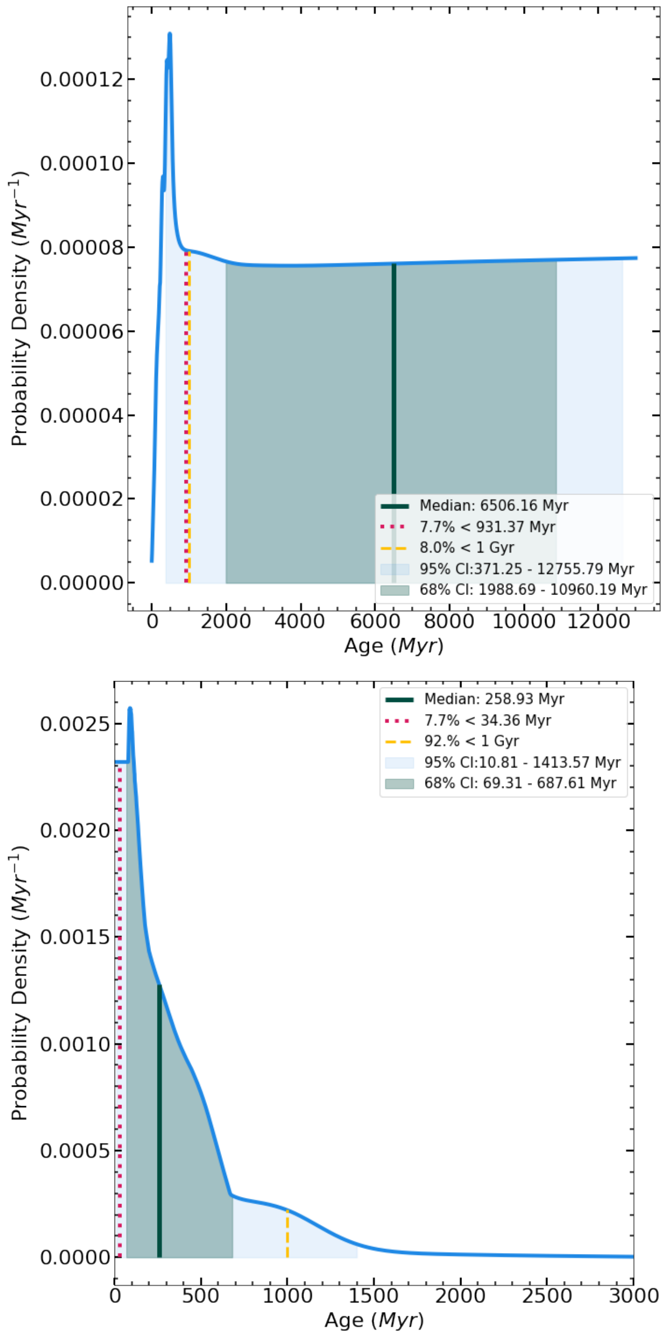


Fig. 1.19.— **Left:** Ages derived from lithium alone yield an age distribution consistent with a constant star formation rate. 8% of the combined distribution falls below 1 Gyr, which is consistent with the expected 7.7%. **Right:** Ages derived from gyrochronology alone yield a significantly younger age distribution than expected. For the 20 late-type stars with rotation periods $\lesssim$ 15 days, 92% of the combined distribution of ages falls below 1 Gyr, which corresponds to $\sim$ 17% of the full gyrochronology sample.

1.9. Conclusions

We identified a sample of 166 accelerating stars that may host substellar companions and do not have archival spectra. From APO/ARCES spectra of the full 166 star sample we measured lithium equivalent widths (or upper limits) and R'_{HK}. We acquired TESS light curves for 129 stars and measured rotation rates for 23 out of the 129 stars. For the earlier spectral types, we also constrain the ages from color-magnitude diagram positions. Combining these methods we calculated median ages and 68% and 95% confidence intervals for the full sample. We are continuing to collect APO/ARCES spectra of the full sample over at least 3 epochs per star to identify stellar binaries (which could be the source of the astrometric acceleration).

We find approximately twice as many young stars from gyrochronology than expected from a uniform star formation rate. We identify two possible explanations for this result. First, there could be unidentified short-period binaries in the sample: either short period binaries that spin each other up, or visual binaries that overlap in *TESS* images. Second, if binaries tend to dissolve over time, then young stars may be more likely to be part of binaries (and possess astrometric accelerations) than older stars. We expect our multi-epoch radial velocity monitoring to identify additional binaries in the sample, and we will report on the results of this monitoring in a future paper.

Looking ahead, *Gaia* DR4 is expected to be released in mid-2026. Unlike DR3, which only contains a single position and velocity measurement, DR4 will contain individual scans over several years. These individual scans will better differentiate between short-period stellar binaries and long-period substellar companions. Future *TESS* data releases will provide light curves for more accelerating stars, allowing for additional gyrochronological age measurements. Additional light curves for accelerating stars that already have light curves will allow for monitoring of spot evolution over the course of a stellar cycle.

The youngest of the accelerating stars in our sample are high-priority targets for direct imaging since these substellar companions will be bright enough to be imaged. In addition to continued radial velocity monitoring, we have begun a direct imaging campaign targeting the youngest stars. Results from both studies will be presented in future work. Substellar companions found in this sample will become key benchmarks for substellar evolution models.

2. HD 143811 AB

HD 143811 AB is the host star to the directly imaged planet HD 143811 AB b, which was recently discovered using data from the Gemini Planet Imager and Keck NIRC2. A member of the Sco-Cen star-forming region with an age of 13 ± 4 Myr, HD 143811 AB is somewhat rare among hosts of directly imaged planets as it is a close stellar binary, with an $\sim$18 day period. Accurate values for the orbital and stellar parameters of this binary are needed to understand the formation and evolutionary history of the planet in orbit. We utilize archival high-resolution spectroscopy from FEROS on the MPG/ESO 2.2-meter telescope to fit the orbit of the binary, and combine with unresolved photometry to derive the basic stellar properties of the system. From the orbit, we derive precise values of orbital period of 18.59090 ± 0.00007 days, and mass ratio of 0.886 ± 0.003. When combined with stellar evolutionary models, we find masses of both components of $M_A = 1.30^{+0.03}_{-0.05}$ $M_\odot$ and $M_B = 1.15^{+0.03}_{-0.04}$ $M_\odot$. While the current data are consistent with the planet and stellar orbits being coplanar, the 3D orientations of both systems are currently poorly constrained, with additional observations required to more rigorously test for coplanarity.

The data and results described in this chapter were published in the *Astrophysical Journal Letters* (Peck et al. 2025b). To cite this chapter, please use Peck et al. (2025b).

2.1. Introduction

A vital part of studying exoplanets is understanding their stellar hosts. The key parameter of age, especially important for self-luminous planets, is more readily measured from the host star than the planet, especially if the star is part of a moving group or association (e.g. Lagrange et al. 2010, Rameau et al. 2013, Gagné et al. 2018). The mass of the star is also of great interest, both to constrain orbital motion (e.g. De Rosa et al. 2015) and to determine how exoplanet occurrence rate depends on stellar mass (e.g. Johnson et al. 2010; Nielsen et al. 2019).

Exoplanets in binary (and other multiple) systems represent a particularly interesting dataset. Typically, single stars are targeted by exoplanet surveys given observational constraints; however, some binary configurations still allow for planet detection. In the case of direct imaging, exoplanets have been identified orbiting a single star in a multiple system (e.g 51 Eri b, Macintosh et al. 2015) as well as circumbinary planets orbiting far from an inner binary (e.g. HD 106906 b, Bailey et al. 2014). In addition to providing constraints on planet formation (e.g. Kraus et al. 2016), these planetary systems are also more likely to experience complex 3-body dynamics, such as Kozai-Lidov oscillations (e.g. Cheetham et al. 2018).

Using Gemini-South/GPI and Keck/NIRC2 data, a giant exoplanet was recently confirmed to be orbiting the binary star HD 143811 AB (HIP 78663). The planet is at a separation of $\sim$59 au and has a mass of $\sim$5.6 M_{Jup} (Jones et al. 2025, submitted ApJL). In this companion paper we characterize the stellar binary host using a new analysis of archival and catalog data. We combine a new orbit fit with SED fitting of the system to derive the individual masses and orbital properties of the binary, in order to understand HD 143811 AB b in the broader context of planet formation within both single stars and binaries.

2.2. HD 143811 AB

HD 143811 AB, before being determined to be a binary, was identified as a member of the Sco-Cen star forming region (de Zeeuw et al. 1999). In particular, the star was assigned to Upper Scorpius by de Zeeuw et al. (1999), Pecaut et al. (2012), and Galli et al. (2018). Luhman & Esplin (2020), however, rejected HD 143811 AB as an Upper Sco member when conducting an Upper Sco census based on *Gaia* DR2 data (Gaia Collaboration et al. 2023), instead classifying it as part of "the remainder of Sco-Cen." This detailed membership analysis was based on UVW velocities utilizing *Gaia* DR2 proper motions and parallax and, in the case of HD 143811 AB, a radial velocity of $RV = -11.3 \pm 0.3$ km/s from Chen et al. (2011). Given our analysis below, this radial velocity measurement likely captured the binary orbital motion, shifting the measurement of U from its true value by $\sim$10 km/s.

BANYAN Σ (Gagné et al. 2018) gives a 69.5% chance that HD 143811 AB is a member of Upper Sco, a 30.3% chance of being part of UCL (Upper Centaurus-Lupus), and a 0.2% chance of being in the field. This analysis is based on *Gaia* DR3 (Gaia Collaboration et al. 2023) parallax and proper motion and the DR2 (Gaia Collaboration et al. 2023) radial velocity (1.68 ± 0.96 km/s). Substituting the system radial velocity we derive below (0.48 ± 0.02 km/s) we find similar probabilities, but with a slightly higher chance of Upper Sco membership: 72.1%, compared to 27.8% and 0.2% chances of UCL and the field, respectively.

Given ages of Upper Sco of 10$\pm$3 Myr and UCL of 16 ± 2 Myr (Pecaut et al. 2012), and the uncertainty in the subgroup membership, we adopt an age for the system of 13 ± 4 Myr. This is consistent with the 2D age map of Pecaut & Mamajek (2016), where ages between $\sim$10–17 Myr are close to the 2D location of HD 143811 AB.

More recently, HD 143811 AB was identified as a spectroscopic binary from analysis of high resolution spectroscopy. Zakhozhay et al. (2022) obtained 15 epochs of FEROS spectra, and note the target is a double lined spectroscopic binary (SB2) with a period between 20-80 days with a companion mass in the stellar regime. We utilize these archival FEROS data, as well as archival spectra from additional epochs, below in our analysis of the system. Similarly, Grandjean et al.

(2023) obtained three HARPS (High Accuracy Radial velocity Planet Searcher) spectra of the system within one day, noting the cross-correlation function of HD 143811 AB is double-peaked, suggesting a stellar binary.

While HD 143811 AB has an infrared excess, and so likely contains a cold debris belt, the parameters of the debris disk are not currently well constrained. Analysis by Chen et al. (2011), McDonald et al. (2012), Ballering et al. (2013), Chen et al. (2014), and Cotten & Song (2016) all find evidence for a disk using either Spitzer or ALLWISE data (or both). However, the estimated disk radius varies from 8.5 AU (Cotten & Song 2016), to 23.8 AU (Chen et al. 2014), to 90 AU (Chen et al. 2011). Going forward, a reanalysis of the disk properties of this system will be necessary, especially to consider the binary nature of the host star, which was not known at the time of these past studies. Of particular interest will be whether the disk is between the orbits of the stellar binary (∼0.2 AU) and the planet (∼60 AU), or if instead the disk lies beyond the planet's orbit.

2.3. Analysis

2.3.1. Data

HD 143811 AB was observed by the MPG/ESO 2.2-meter telescope/FEROS (Fiber-fed Extended Range Optical Spectrograph, Kaufer et al. 1999) on 22 epochs between 17 Apr 2018 and 18 Apr 2022. These R∼48,000 spectra clearly show two sets of stellar lines, identifying the system as an SB2, and in many epochs the two sets of lines are clearly separated. At other epochs, near conjunction, the velocities of the two stars are more comparable, and the lines overlap. The combination of multiple years of coverage and the ~1 day cadence of many observations allows excellent time sampling of the orbital motion. Since FEROS obtains spectra of both the science target and a wavelength calibration lamp simultaneously, the spectra have exceptional wavelength stability with radial velocity precision as low as $\sim$ 6 m/s (Zakhozhay et al. 2022). We retrieve the reduced spectra from the ESO Phase 3 archive.

The two stellar components of HD 143811 AB are near equal brightness, and are close enough (~1 mas) that the data contain combined spectra of both stars in all epochs. Nevertheless, the large RV semi-amplitude of each star in the system (~25 km/s) and the high spectral resolution of FEROS result in significant separation of spectral lines at some epochs.

2.3.2. Spectral Fitting

We extract radial velocities and $v\sin(i)$ for each star by generating a model of the combined spectrum and comparing to the observations, using a grid of PHOENIX stellar atmosphere models (Hauschildt & Baron 1999) from Husser et al. (2013). We explore the parameter space of the model with Markov Chain

Monte Carlo (MCMC) as implemented in the *emcee* package (Foreman-Mackey et al. 2013). Our model consists of 11 parameters: RV offsets for both the primary and the secondary, effective temperature for both components, log(g) for both components, $v\sin(i)$ for both components, system metallicity, a single flux ratio between the two stars, and an additional broadening parameter σ. This last term accounts for resolution differences between FEROS and PHOENIX, but also broadening due to atmospheric seeing. The PHOENIX grid is linearly interpolated in effective temperature, metallicity, and log(g), generating two spectra, one for each component of the binary. The resolution of the PHOENIX template is degraded to match that of the FEROS spectra using a Gaussian kernel with a width set to σ, and the model spectrum is further broadened using a $v\sin(i)$ kernel. Next, we scale the model spectrum for the B component using the flux ratio and then combine the two spectra before normalizing to the continuum level of the data. We perform the fit on 9 regions (~3-8 Å in width) of the spectrum with well-modeled narrow lines (mostly Fe I and Fe II lines) between 6,000 and 8,000 Å. Error bars are assigned to each spectral channel based on the channel-to-channel variation seen in continuum regions, and then inflated by a factor of two to account for spectral covariance and model mismatch. Our final error bars are about 3% for each spectral channel.

We identify 11 epochs with well separated lines by visual inspection. In particular, we classify spectra based on whether the flux returns to the continuum level when moving from a line belonging to HD 143811 A to the same line in HD 143811 B. We find our reference lines to be well-separated when the two stars have a relative radial velocity ΔRV $\gtrsim$ 20 km/s, and are more heavily blended otherwise. This is consistent with many of these lines having a width of $\sim$0.4 Å, or about $\sim$20 km/s. For these 11 epochs we perform a fit with uniform priors in $v\sin(i)$, RV, $T_{\rm eff}$, metallicity, and σ. These broad priors have -50 km/s $\leq$ RV $\leq$ 50 km/s, 4000 K $\leq T_{\rm eff} \leq$ 8000 K, 1 km/s $\leq v\sin(i) \leq$ 10 km/s, $-0.5 \leq$ Z ≤ 0.5, $0.1 \leq$ FR ≤ 1.0, and $0.01 \leq \sigma \leq 100$. Since our method does not reliably fit $\log(g)$ (see Appendix), we select priors for $\log(g)$ of each star based on the grids from Baraffe et al. (2015) for 10-20 Myr stars at 1.3$M_\odot$ and 1.15$M_\odot$. We use Gaussian priors on $\log(g)$ of 4.1 ± 0.1 and 4.2 ± 0.1 for the primary and secondary, respectively. As expected, the posteriors on $\log(g)$ match the priors.

For epochs where the relative RVs are smaller than 20 km/s (11 epochs) and the lines mostly overlap, using wide priors results in poor fits with degeneracies between the multiple parameters. Therefore, for these blended epochs we use narrower priors on the stellar parameters, based on the fit results to epochs with more separated lines. We sum the posteriors on the stellar parameters from the 11 widely-separated epochs (to better account for systematic errors). This produces Gaussian priors of $T_{\rm eff,A} = 6439 \pm 132$K, $T_{\rm eff,A} = 5900 \pm 157$K, $v\sin(i)_A = 6.8 \pm 0.6$ km/s, $v\sin(i)_B = 2.8 \pm 1.4$ km/s, Z$= -0.18 \pm 0.1$, FR$= 0.76 \pm 0.05$, and $\sigma = 33 \pm 3.6$, which we use when fitting the blended epochs. The radial velocity prior remains the same: -50 km/s $\leq$ RV ≤ 50 km/s.

The temperatures and metallicities of HD 143811 A and B are not well constrained by our spectral fitting method (see Appendix). To ensure that our RVs are accurate, we perform a second round of spectral fitting with more precise priors on temperature and metallicity from the SED fit (described below in subsection 2.3.4), which in turn depends on the mass ratio from the orbit fit (subsection 2.3.3). The RVs from this second round of spectral fitting are used for a second round of orbit fits. There is a negligible change in the RVs and the mass ratio between the initial and final round of fitting. For example, the RVs for the first epoch changed from 12.47 ± 0.12 km/s and -13.37 ± 0.13 km/s to 12.46 ± 0.12 km/s and -13.35 ± 0.13, while the posterior on mass ratio changed from 0.8852 ± 0.0025 to 0.8856 ± 0.0026. As a result, we conclude our results are self-consistent after this second round of fitting, and adopt the second set of RVs and orbital parameters going forward. The final RVs for the 22 epochs are given in Table 2.3.1. Figure 2.3.1 shows all 9 segments of the spectrum from one epoch, with FEROS data in black, and fits in red. Overall there is good agreement between model and data, though as expected there are mismatches between the model and data for some lines.

While the parameters derived from each epoch generally agree, final values for T_{eff}, $v\sin(i)$, and metallicity are calculated from the summed 1D posteriors across all epochs to better incorporate systematic errors. Error bars are taken to be the standard deviations of the combined posterior, with an additional error of 107.2 K added in quadrature to the uncertainty on temperature. The derivation of this additional error term, from a comparison of our method to literature results on 8 stars, is presented in the Appendix. Stellar parameters from this fit, and additional fits described below, are given in Table 2.3.2.

2.3.3. Orbit Fitting

We perform an orbit fit (using the same Metropolis-Hastings MCMC method as Nielsen et al. 2020) to the SB2 based on our measured velocities, fitting for the RV semi-amplitude of both stars, period, eccentricity, argument of periastron, epoch of periastron passage, and system RV. Uniform priors are assumed on all parameters, with eccentricity bounded between $0 \leq e \leq 0.95$. Ten MCMC chains are run in parallel for 10^7 steps each, with every 100th step saved. We assess convergence both by visual inspection of the posteriors and by considering the Gelman-Rubin statistics, which is below 1.001 for all parameters. Since the data contain both high-cadence observations (15 epochs over two weeks) and long baselines (four years between the first and last epoch) the constraints on the orbital parameters are excellent.

Figure 2.3.2 shows the RVs of both components and draws from the posterior.

The best-fitting orbit has a reduced χ^2 close to unity, indicating our RV error bars are appropriate. The posteriors on this fit are shown in Figure 2.3.3. Period is constrained to approximately six seconds: $P = 18.59090 \pm 0.00007$ days, and mass ratio constrained to less than a percent: $q = 0.886 \pm 0.003$. The significant eccentricity of the orbit is well-determined, $e = 0.4941 \pm 0.0013$, and the $\sim$90% mass ratio is consistent with the similarity of the two spectra, as well as the similar depths of the lines.

2.3.4. Stellar Parameters

The SB2 orbit does not provide direct measurements of the individual masses, semi-major axis, or inclination angle. Nevertheless, with the unresolved photometry of the system and an estimate for the binary mass ratio, we can derive model-dependent values of the stellar parameters. We follow the same procedure outlined in Nielsen et al. (2019) for stars with $B - V > 0.35$, using a joint evolutionary-atmospheric model to fit the blended photometry of the system to estimate the masses of the two components. As in Nielsen et al. (2019), we use a modified version of the MESA Isochrones and Stellar Tracks (MIST) model grid (Dotter 2016; Choi et al. 2016) combined with ATLAS9 model atmospheres (Castelli & Kurucz 2003). With this approach, we can estimate the spectral energy distribution (SED) of the two stars given the masses of the two components (M_1, M_2)

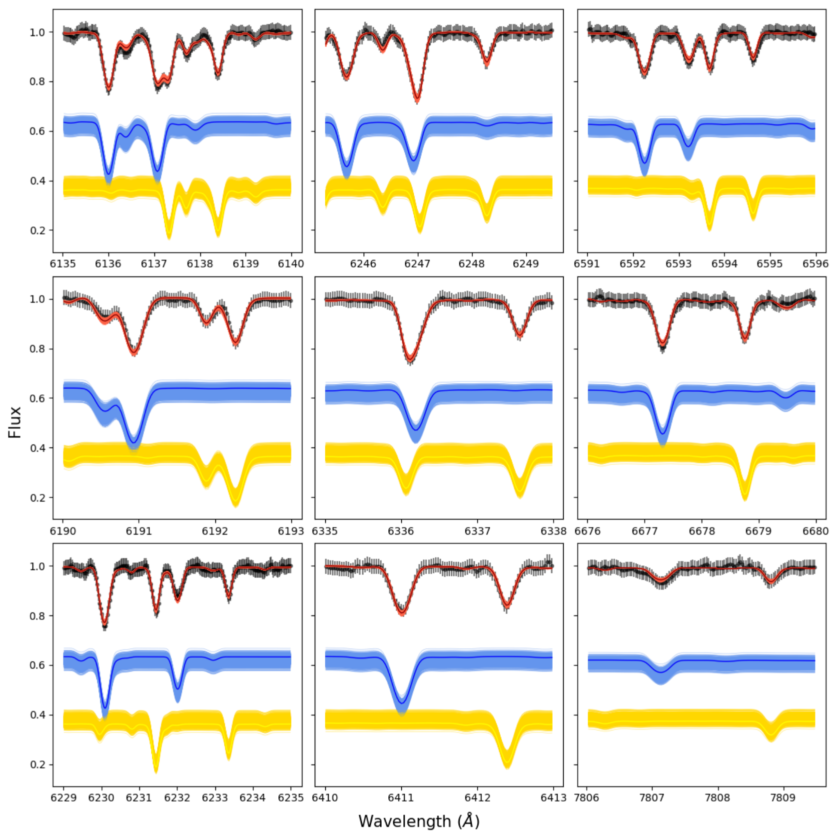


Fig. 2.3.1.— FEROS spectra of HD 143811 AB taken on 2018-04-22 (black points with error bars), showing two sets of well-resolved lines. Red tracks are draws from the posterior (lowest chi-square solution in dark red) of a two-star model that best fits the data. The blue and yellow lines are draws from single star models for A and B, respectively. Given the high resolution and SNR of these data we are able to recover both the radial velocities of both components as well as $v\sin(i)$.

and an age (t), metallicity ($[M/H]$), parallax (ϖ), extinction (A_V) for the system. The joint evolutionary-atmospheric model allows us to translate these parameters into the observable SED of the system, the two components being unresolved and blended together due to their very small angular separation. This can then be used to fit the observed optical photometry from *Gaia* (Gaia Collaboration et al. 2023) and near-infrared photometry from 2MASS (Skrutskie et al. 2006). While this system has a debris disk, the observed excess is at ∼24 microns, and so the disk should not impact the bluer photometry. The observed and modeled SED is shown in Figure 2.3.4, while the triangle plot for this fit is given in the Appendix in Figure B.0.5.

With our combined model in hand, we use an MCMC-based approach for parameter estimation. We use Gaussian priors for the age ($t = 13 \pm 4\,$Myr; synthesizing the age for the US and UCL sub-groups from Pecaut & Mamajek 2016), parallax ($\varpi = 7.3065 \pm 0.0204\,$mas; Gaia Collaboration et al. 2023), metallicity ($[M/H] = -0.05 \pm 0.11\,$dex; Nielsen et al. 2013), and mass ratio ($q = 0.885 \pm 0.003$). From this, we find individual masses of the two stars of $M_A = 1.30^{+0.03}_{-0.05}$ M$_\odot$ and $M_B = 1.15^{+0.03}_{-0.04}$ M$_\odot$, although the two masses are highly covariant given the tight constraint on the mass ratio.

Based on the sum of these individual masses and the orbital period, we find a semi-major axis (for the orbit of B around A) of $0.1854^{+0.0014}_{-0.0024}$ au. Combined with

Table 2.3.1: Radial Velocities of HD 143811 A and B. Widely separated epochs are where the two sets of lines are distinct from each other, which corresponds to $\Delta RV \gtrsim 20 km/s$

Epoch (BJD)	UTC	RV_A (km/s)	RV_B (km/s)	Widely Separated?
2458225.8481768593	2018-04-17 08:21:22.480645	12.457±0.125	-13.349±0.129	Y
2458226.840259357	2018-04-18 08:09:58.408435	11.382±0.125	-11.445±0.136	Y
2458227.8335984894	2018-04-19 08:00:22.909485	8.041±0.135	-8.285±0.152	N
2458228.831037526	2018-04-20 07:56:41.642248	1.962±0.508	-0.994±0.578	N
2458228.9179943185	2018-04-20 10:01:54.709121	1.462±0.592	-0.456±0.65	N
2458229.8972283546	2018-04-21 09:32:00.529836	-12.01±0.13	14.416±0.133	Y
2458230.8849802855	2018-04-22 09:14:22.296671	-29.906±0.123	34.725±0.126	Y
2458231.870725756	2018-04-23 08:53:50.705322	-29.459±0.125	34.198±0.129	Y
2458233.8412399422	2018-04-25 08:11:23.131009	-9.609±0.119	11.851±0.132	Y
2458235.851854297	2018-04-27 08:26:40.211249	1.438±0.558	-0.442±0.644	N
2458236.710698977	2018-04-28 05:03:24.391623	4.191±0.381	-3.368±0.264	N
2458236.7799807256	2018-04-28 06:43:10.334693	4.561±0.561	-3.431±0.306	N
2458236.8734692587	2018-04-28 08:57:47.743948	4.693±0.349	-3.851±0.262	N
2458237.8811232424	2018-04-29 09:08:49.048146	7.296±0.196	-6.654±0.21	N
2458238.8585430426	2018-04-30 08:36:18.118877	8.852±0.128	-9.01±0.136	N
2459432.578191188	2021-08-06 01:52:35.718657	12.993±0.122	-13.352±0.132	Y
2459435.6336258487	2021-08-09 03:12:25.273326	10.156±0.132	-10.816±0.131	Y
2459438.5826384313	2021-08-12 01:58:59.960468	-17.199±0.113	20.502±0.147	Y
2459682.817435831	2022-04-13 07:37:06.455784	-19.745±0.117	23.526±0.137	Y
2459684.8084221077	2022-04-15 07:24:07.670107	-4.349±0.382	5.28±0.344	N
2459687.784224547	2022-04-18 06:49:17.000859	6.681±0.249	-6.123±0.274	N
2459690.8318373566	2022-04-21 07:57:50.747606	11.342±0.126	-12.101±0.145	Y

the eccentricity of the orbit and the distance to the system, apastron should be ~2 mas. While this is too close for traditional imaging, it should be reachable with an interferometer like VLTI/GRAVITY (GRAVITY Collaboration et al. 2017). This semi-major axis is a factor of ~300 smaller than the planet separation; such a large difference would be expected for a stable orbit of a planet around a binary star.

An RV orbit does not constrain the orbital inclination angle, but we can use the SED-derived masses to estimate the inclination angle of the stellar binary orbit. In particular, semi-amplitude is a function of inclination angle, as well as masses of both components, period, and eccentricity. By combining posteriors from the orbit fit for semi-amplitude for HD 143811 A, period, and eccentricity with posteriors on the two masses from the SED fit we then obtain a posterior on orbital inclination angle. We repeat this process with the semi-amplitude of HD 143811 B to obtain a second posterior on the orbital inclination angle. As expected for self-consistent results, both semi-amplitudes return similar values of inclination angle: $i_A = 22.9^{+0.3\circ}_{-0.2}$ and $i_B = 22.8^{+0.3\circ}_{-0.2}$. We adopt the first value as the model-dependent inclination angle of the system. Since radial velocity measurements cannot distinguish between clockwise and counterclockwise orbits, a value of $i = 157.1^{+0.2\circ}_{-0.3}$ is also possible.

As expected for young F-stars we detect significant lithium absorption in both stars near 6707.8 Å. We follow the fitting procedure described in Peck et al. (2025a), but we adapt it to measure lithium equivalent widths for both components of the stellar binary rather than a single star by scaling the continuum based on the flux ratio. We measure equivalent widths of 74.3±2.7mÅ and 105.0±3.5mÅ for the primary and secondary respectively. Since we do not have accurate B-V values for either component, we do not attempt to calculate age posteriors using `BAFFLES` (Stanford-Moore et al. 2020). However, the measured equivalent widths are consistent with those observed for F stars in β Pic, IC 2602, α Per, and the Pleiades. We do not observe signification emission cores in Calcium H and K or in Hα.

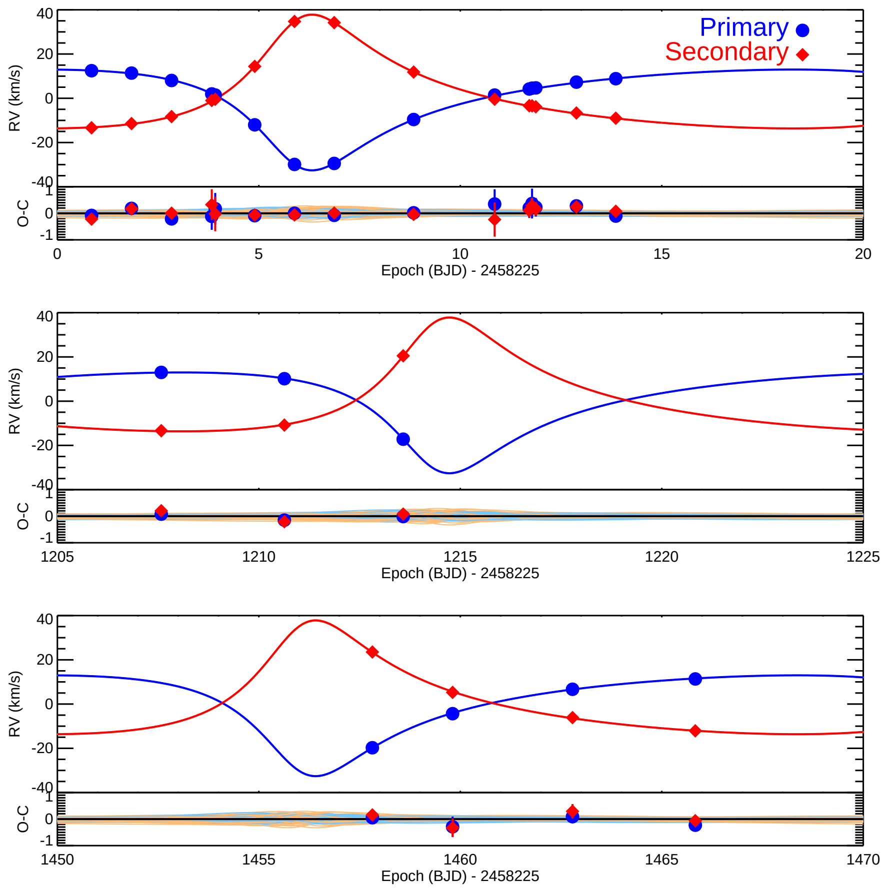


Fig. 2.3.2.— Radial velocity measurements of both HD 143811 A and HD 143811 B with orbits drawn from the posteriors, with the darkest blue and red lines representing the lowest χ^2 orbit. The Observed-Calculated (O-C) panels show the residuals with respect to the lowest χ^2 orbit (black line at 0), as well as the offset of other draws from the posterior with respect to the best-fitting orbit. The $\sim$18 day orbit is an excellent fit to the data, and the combination of multiple observation over 2 weeks and a multi-year baseline strongly constrain the orbital parameters.

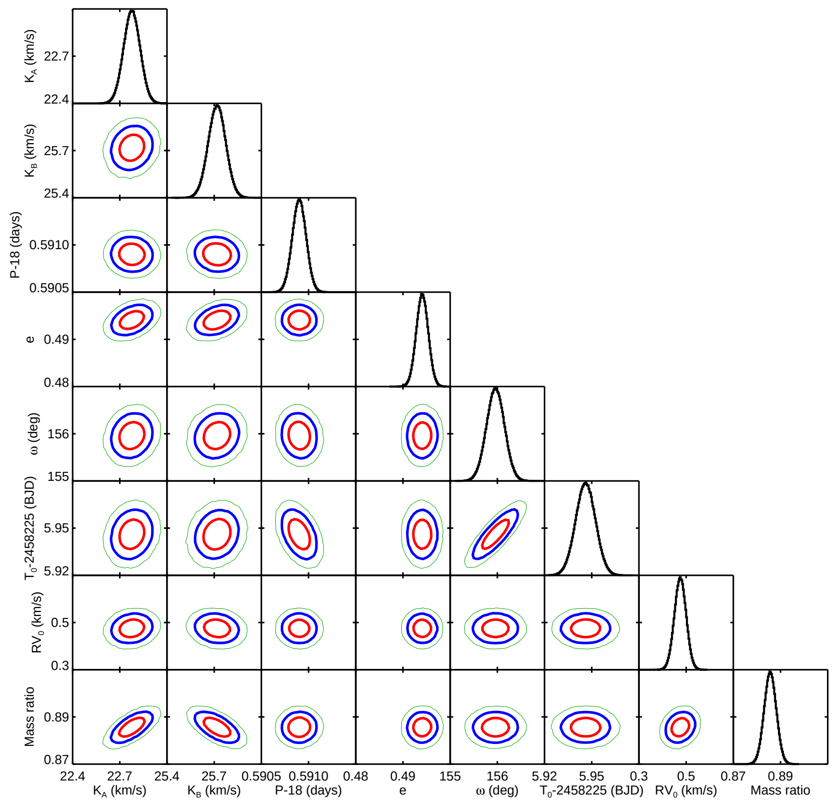


Fig. 2.3.3.— Posterior on the SB2 radial velocity orbit of HD 143811 AB. The orbit is very well constrained by the FEROS data, with orbital period known to 6 second precision. Direct measurement of both semi-amplitudes allows a measurement of mass ratio at the sub-1% level: 0.886 ± 0.003.

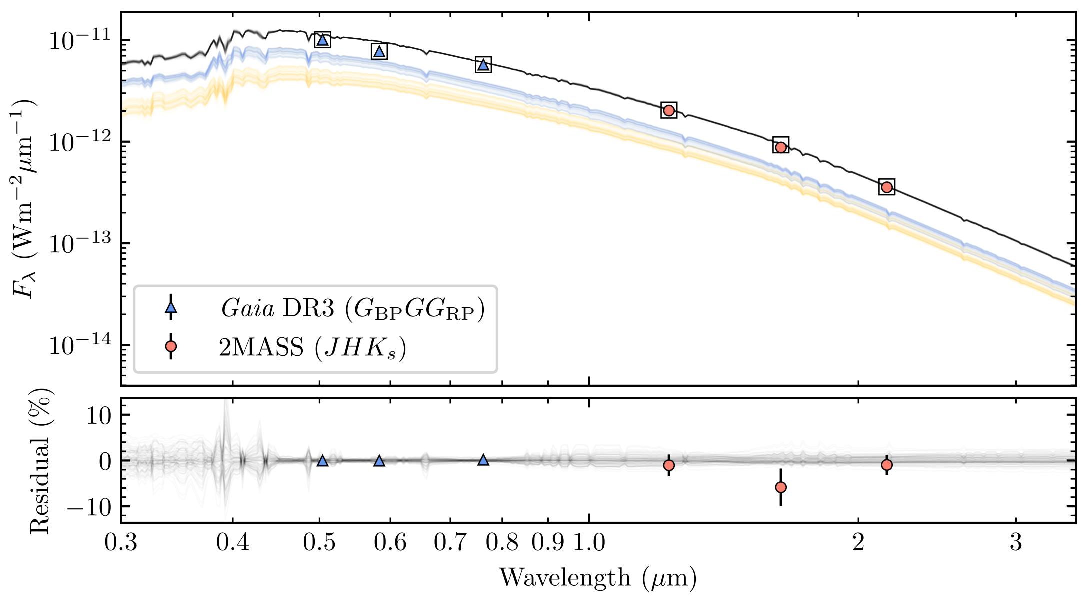


Fig. 2.3.4.— The results of a two-star fit to the unresolved optical and near-infrared photometry of the system from *Gaia* and 2MASS. Plotted are the spectral energy distribution of the blended system (black) and the two components (blue, yellow) drawn from the posterior distributions estimated using MCMC. Also shown are the synthetic photometry of the blended system derived from the median SED (black squares), and the observed photometry (circles).

2.4. Discussion

2.4.1. Physical Parameters

The relatively large eccentricity of HD 143811 AB (~0.5) is not unexpected given the orbital period of ~18 days. Torres et al. (2021), examining Pleiades binaries, find that tidal circularization is most prominent for orbital periods below ~7 days. The binary period itself is typical of FGK binaries, though the peak of the distribution is closer to 10^5 days (~300 years), the log-normal distribution is quite broad (Duquennoy & Mayor 1991, Raghavan et al. 2010). The discovery of a planet around such a binary is partially the result of an important selection effect: wider-separation binaries (~0.1-1") would be excluded from the GPIES target list, or aborted once the binary became visible. This is due both to the expected lack of dynamical space for a planet with similar orbital separation to a stellar binary, and the practical effect that a star not occulted by the coronagraph would saturate the detector. As a result, binary stars identified as planet hosts by direct imaging surveys tend to either have very small orbits (so that both stars fit comfortably under the coronagraph), or very large orbits (so that the other star is well outside the field of view of the detector).

There is a significant offset between the temperatures we derive for the two components from fitting the SB2 spectrum (6572 ± 139 and 6134 ± 127 K) compared to fitting the SED and the mass ratio (6751^{+207}_{-148} and 6349^{+122}_{-133}K). The SED results are systematically $\sim$200 K higher than the spectral fitting results; the $\sim$1 σ confidence intervals touch for HD 143811 A and B. In the SED fit, there is a strong covariance between the extinction and both temperatures, with lower values of extinction corresponding to lower temperatures. There is a range of extinction values across Sco-Cen, both across subgroups (the mean A_V is 0.73 and 0.17 for Upper Sco and UCL, respectively, de Geus et al. 1989; Wright & Mamajek 2018) and within subgroups. Pecaut & Mamajek (2016), for example, find multiple members of both Upper Sco and UCL with large extinctions ($A_V \sim 1$), as well as stars with extinctions as low as $A_V = 0$. Setting the extinction in the SED fit to $A_V = [0.00, 0.05, 0.10]$ moves the median temperatures of A and B to $[6530, 6564, 6636]$ K and $[6104, 6216, 6297]$ K, respectively; the lowest of these values are more in line with the spectral fitting results.

Alternatively, there may be a systematic offset in the spectral fitting when fitting an SB2 compared to fitting single stars. A significant issue is the incompleteness of models, especially lines that appear in the model but not the data or vice versa. These are visible when comparing FEROS data to the PHOENIX models, and we chose the spectral regions for the fit to isolate well-modeled lines. However, this can become more important when the RV offset of an SB2 results in a well-modeled line in one star overlaps with a poorly-modeled line in the other.

When applying a single star version of the spectral fit to well characterized stars, we find that for temperatures above $\sim$ 6000 K, our method systematically underpredicts temperature and does not reliably measure metallicity, though it does recover accurate radial velocities (see Appendix). Therefore, we adopt the temperatures and metallicities from the SED fit rather than the spectral fit. While the masses of the two components have some covariance with temperature in the SED fit, this effect is relatively small. As a result, our derived masses of the two components ($1.30^{+0.03}_{-0.05}$ and $1.15^{+0.03}_{-0.04}$ $M_{\odot}$) are generally better constrained than the temperatures.

2.4.2. Sco-Cen Membership

The rotation and metallicity of these stars are generally consistent with Sco-Cen, as expected. We find relatively low $v\sin(i)$ for both A and B: 6.8 ± 0.6 km/s and 2.9 ± 1.2 km/s, respectively. Compared to other Sco-Cen stars of similar $B - V$ color from Grandjean et al. (2023) (their Figure 1b), these values are at the low end of their HARPS Sco-Cen sample, which span $v\sin(i)$ from $\sim$3 to $\sim$100 km/s. It is possible the two components of HD 143811 AB are being viewed more pole-on compared to other Sco-Cen members, or that tidal forces affected their rotational evolution. Slower overall rotation would be consistent with the lack of strong calcium emission we observe in the spectra. Our derived metallicity for the system from the SED fit, $-0.15^{+0.07}_{-0.08}$, is within the distribution of 0.0 ± 0.18

measured by Grandjean et al. (2023).

The system radial velocity of $RV_0 = 0.48 \pm 0.02$ km/s, obtained from our orbit fit, is $\sim 1\sigma$ lower than the *Gaia* DR2 velocity of 1.68 ± 0.96 km/s, and significantly higher than the Chen et al. (2011) value of -11.3 ± 0.3 km/s. This large difference is to be expected given the semi-amplitude of each component of $\sim$25 km/s. By combining our system radial velocity with *Gaia* DR3 astrometry we find space motions of the HD 143811 AB system of $(U, V, W) = (-2.68 \pm 0.04, -18.10 \pm 0.05, -4.68 \pm 0.02)$. This places HD 143811 AB $(2.1, 1.1, 0.9)\sigma$ from the central UVW of Upper Sco, as given by Wright & Mamajek (2018). The same calculation places the system slightly further from the central UVW of UCL, also from Wright & Mamajek (2018): $(1.6, 2.6, 0.9)\sigma$. This is in line with the BANYAN Σ results, when we combine *Gaia* DR3 astrometry with our radial velocity we find 72.1% probability of Upper Sco membership, 27.8% for UCL, and 0.2% for the field. The SED fit seems generally more consistent with UCL than Upper Sco, with the $20.9^{+1.8}_{-1.7}$ Myr age from that fit closer to the $\sim$16 Myr UCL than the $\sim$10 Myr Upper Sco. Similarly, a lower value of extinction (which would bring the SED and spectral fit temperatures more in line with each other) would also suggest UCL membership. Going forward, better age constraints on HD 143811 AB will allow for a more definitive determination of subgroup membership.

It is also interesting to compare HD 143811 AB b to another Sco-Cen directly imaged planet: HD 95086 b (Rameau et al. 2013). Both planets formed in the same association, are about the same age, orbit at roughly the same distance ($\sim$60 au), and are about the same mass ($\sim$5 M_{Jup}). The major difference is the host star: HD 95086 is a $\sim$1.6 $M_\odot$ A8 single star, while HD 143811 AB is a close binary containing a 1.30 $M_\odot$ mid-F star and a 1.15 $M_\odot$ late-F star. A key question is whether both planets formed at their present location, or if they moved outward after formation, either from early interactions with the protoplanetary disk or due to gravitational scattering. Gravitational scattering is expected to play an important role in another Sco-Cen planet orbiting a tight binary, HD 106906 b (Bailey et al. 2014), which is at a projected separation of $\sim$740 au. This very large separation is thought to result from gravitational interactions between the planet, the inner binary, and an additional Sco-Cen star that passed close to this system (De Rosa & Kalas 2019; Nguyen et al. 2021). If both HD 143811 AB b and HD 95086 b are close to their formation location it might suggest that wide-separation giant planet formation around close binary stars proceeds in a similar way as around single stars. The role of gravitational scattering can be tested with future orbit monitoring of the planet. In particular, if HD 143811 AB b has a more eccentric orbit than most directly imaged planets (Bowler et al. 2020) a scattering event might be more likely compared to in-situ formation. Further analysis of the binary orbit of HD 143811 AB, in particular the extent to which it is aligned (or not aligned) with the planet's orbit, will also provide dynamical constraints on the origins of HD 143811 AB b.

With the current data it is not yet possible to robustly determine if the three-dimensional orbits of both the inner stellar binary and the outer planet are coplanar. In particular, the radial velocity data on the stellar binary provides no information on the position angle of nodes of the inner orbit. Similarly, with only a few degrees of orbital motion, the planet's position angle of nodes is not well-determined, where despite four peaks in the posterior there is non-zero probability for all values between 0 and 360° (Jones et al. 2025, submitted ApJL). Without inclination angle and position angle of nodes measurements for both orbits, mutual inclination cannot be directly computed. Nevertheless, coplanarity cannot be ruled out with the current observations, with a model-dependent inclination angle of the binary orbit of $i_{AB} = 22.9^{+0.3}_{-0.2}{}^{\circ}$, and an inclination angle for the planet's orbit of $i_b = 38 \pm 17^{\circ}$ (Jones et al. 2025, submitted ApJL). Thus, both inclination angle posteriors overlap at the $\sim 1\sigma$ level. Retrograde orbits are still possible, however, if $i_{AB} \approx 157$ is the actual inclination of the stellar binary. Additional orbital monitoring of the planet will allow a more precise inclination angle measurement and a quantitative comparison of the two inclination angles (e.g. Kennedy et al. 2013). Measurements of both orbits' position angles of nodes, which for the inner binary would likely require VLTI/GRAVITY or *Gaia* DR4, would allow for a more direct test of coplanarity.

2.5. Conclusions

We report the characterization of the HD 143811 AB binary, the host of the directly imaged planet HD 143811 AB b. We extract RVs of both components from archival spectra at 22 epochs and derive orbital parameters, consistent with an eccentric ~18 day orbit, and a mass ratio of ~0.9. Combining with the age of the system, unresolved photometry, and evolutionary models we derive masses of $M_A = 1.30^{+0.03}_{-0.05}$ $M_\odot$ and $M_B = 1.15^{+0.03}_{-0.04}$ $M_\odot$ for the two components.

Future observations of the HD 143811 AB system will refine the orbital parameters of both orbits. While the RV orbit of HD 143811 AB is very well constrained, interferometry of the system will make it possible to resolve the ~1 mas visual orbit of the binary, and so directly measure the full 3D orbit. In addition, *Gaia* DR 4 will contain individual astrometric scans, which may have the precision to detect the ~0.1 mas photocenter orbit of the binary. Similarly, additional orbital monitoring of the planet will better constrain its 3D orbit, allowing a direct measurement of the mutual inclination of the two orbits. In addition to aiding 3-body modeling of the system, these improved orbits will place valuable constraints on the formation mechanism and dynamical evolution of this system.

An improved exoplanet orbit will be especially interesting when coupled with future long wavelength observations of the star, such as from JWST MRS. More accurately determining the debris disk's radius and extent, especially given the physical parameters of the stellar binary, will better inform the dynamics of the disk, and its interactions with the planet.

HD 143811 AB b joins a growing list of directly imaged planets in binary systems including 51 Eri b (Macintosh et al. 2015) and HD 106906 b (Bailey et al. 2014; De Rosa & Kalas 2019). With additional detections, it will become feasible to determine how the occurrence rate of wide-separation giant planets depends on binary properties, and how it compares to the occurrence rate for single stars. This comparison has traditionally been difficult, given the relatively low occurrence rate of wide-separation giant planets and the tendency of imaging surveys to screen for binaries (e.g. Bonavita et al. 2016). Improvements to high contrast imaging, such as the upcoming Gemini Planet Imager 2.0 (Chilcote et al. 2024), will enable better sensitivity to giant planets orbiting single stars, orbiting one star in wide binaries, and orbiting both stars in close binaries. *Gaia* DR4 will also allow a comparison of closer-in giant planet populations between binaries and single stars, especially in cases where astrometry from both components of the binary can be extracted.

Table 2.3.2: Properties of HD 143811 AB. Adopted parameters are indicated with a Y. We use an age for the system of 13 ± 4 Myr based on the uncertain Upper Sco or UCL membership, though the posterior from the SED fit favors somewhat older ages. Temperature and metallicity are adopted from the SED fit rather than the spectral fit.

Parameter	HD 143811 A	HD 143811 B	Reference			Adopted?
Age (Myr)	13 ± 4		1			Y
Distance (pc)	136.9 ± 0.4		2			Y
SED Fitting				*Prior*	*Prior Reference*	
$[M/H]$ (dex)	$-0.15^{+0.07}_{-0.08}$		1	0.05 ± 0.11	3	Y
Age (Myr)	$20.9^{+1.8}_{-1.7}$		1	13 ± 4	1	N
A_V (mag)	0.16 ± 0.01		1	Uniform	1	Y
Mass ($M_\odot$)	$1.30^{+0.03}_{-0.05}$	$1.15^{+0.03}_{-0.04}$	1	$q = 0.885 \pm 0.003$	1 (orbit fit)	Y
$T_{\rm eff}$ (K)	6751^{+207}_{-148}	6349^{+122}_{-133}	1			Y
Spectral Fitting				*Prior*	*Prior Reference*	
$[M/H]$ (dex)	-0.078 ± 0.037		1	$-0.15^{+0.07}_{-0.08}$	1 (SED Fit)	N
$T_{\rm eff}$ (K)	6572 ± 139	6134 ± 127	1	6751^{+207}_{-148}, 6349^{+122}_{-133}	1 (SED Fit)	N
$v\sin(i)$ (km/s)	$6.73^{+0.70}_{-0.82}$	$2.89^{+1.41}_{-1.29}$	1	Uniform, 1–10	1	Y
Flux Ratio	$0.75^{+0.14}_{-0.05}$		1	Uniform, 0.01–1	1	Y
Orbit Fitting				*Prior*	*Prior Reference*	
K (km/s)	22.78 ± 0.05	25.72 ± 0.06	1	Uniform	1	Y
Period (days)	18.59090 ± 0.00007		1	Uniform	1	Y
Eccentricity	0.4941 ± 0.0013		1	Uniform, 0–0.95	1	Y
ω (deg)	155.96 ± 0.19	335.96 ± 0.19	1	Uniform	1	Y
T_0 (BJD)	2458230.946 ± 0.006		1	Uniform	1	Y
RV_0 (km/s)	0.48 ± 0.02		1	Uniform	1	Y
$q = \frac{M_B}{M_A}$	0.886 ± 0.003		1			Y
Combined Results						
Semi-major axis (AU)	$0.1854^{+0.0014}_{-0.0024}$		1			Y
Orbit Inclination Angle (deg)	$22.9^{+0.3}_{-0.2}$ or $157.1^{+0.2}_{-0.3}$		1			Y

References. — 1: this work, 2: Gaia Collaboration et al. (2023), 3: Nielsen et al. (2013).

APPENDIX A.

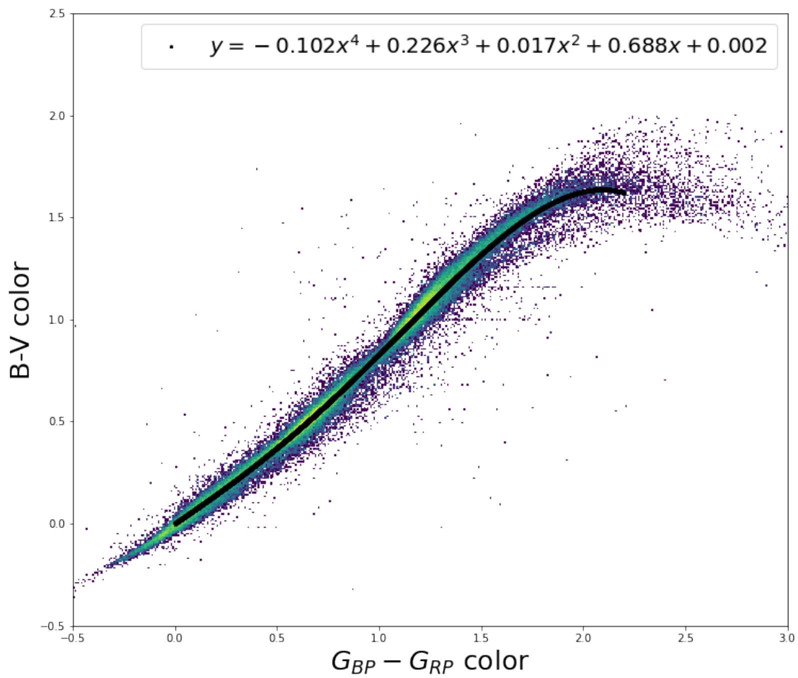


Fig. A.0.1.— We transform *Gaia* $G_{BP} - G_{RP}$ color to Johnson $B - V$ using a polynomial fit to a large sample of $Gaia - Hipparcos$ stars (van Leeuwen et al. 1997; Gaia Collaboration et al. 2023). We use our derived transformation when calculating median ages 68% and 95% confidence intervals from `BAFFLES` (Stanford-Moore et al. 2020) for stars without precise $B - V$ measurements.

Table A.0.1:: We present our lithium equivalent width measurements or upper limits for each star and its corresponding BAFFLES median age and confidence intervals where appropriate. For stars with $B-V$ outside the BAFFLES range, we do not derive age posteriors.

Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)
HIP 143	1.033 ± 0.019	≤ 7.64	–	–	–	≥ 484.61	≥ 1034.76	≥ 4442.08
HIP 669	0.624 ± 0.015	31.45 ± 1.89	4612.67	2164.81-9235.42	859.53-12375.61	–	–	–
HIP 1068	1.315 ± 0.1	≤ 68.37	–	–	–	≥ 138.08	≥ 720.14	≥ 4217.04
HIP 1412	1.414 ± 0.005	≤ 13.65	–	–	–	≥ 254.02	≥ 803.55	≥ 4276.7
HIP 1539	1.375 ± 0.005	≤ 12.17	–	–	–	≥ 280.46	≥ 828.91	≥ 4294.84
HIP 1771	1.489 ± 0.1	≤ 15.03	–	–	–	≥ 164.0	≥ 744.83	≥ 4234.75
HIP 2426	1.359 ± 0.02	≤ 10.93	–	–	–	≥ 285.77	≥ 834.64	≥ 4298.94
HIP 2843	0.466 ± 0.007	≤ 7.97	–	–	–	≥ 716.16	≥ 1300.72	≥ 4632.99
HIP 3008	1.424 ± 0.02	≤ 14.98	–	–	–	≥ 246.95	≥ 797.04	≥ 4272.04
HIP 3293	0.452 ± 0.009	13.22 ± 4.98	726.52	573.26-1064.21	224.61-1842.72	–	–	–
HIP 3633	0.814 ± 0.061	≤ 11.11	–	–	–	≥ 571.54	≥ 1376.96	≥ 4795.91
HIP 3724	1.4 ± 0.02	≤ 16.09	–	–	–	≥ 255.04	≥ 804.65	≥ 4277.49
HIP 4024	1.069 ± 0.009	≤ 4.33	–	–	–	≥ 530.93	≥ 1075.56	≥ 4471.25
HIP 5319	0.398 ± 0.01	≤ 35.97	–	–	–	≥ 247.18	≥ 1301.88	≥ 4825.13
HIP 5763	1.22 ± 0.015	≤ 8.14	–	–	–	≥ 352.3	≥ 903.42	≥ 4348.13
HIP 6776	0.453 ± 0.007	≤ 16.62	–	–	–	≥ 573.96	≥ 1158.02	≥ 4530.53
HIP 6796	1.157 ± 0.074	≤ 5.35	–	–	–	≥ 409.71	≥ 976.89	≥ 4400.69
HIP 6833	0.656 ± 0.002	13.12 ± 2.19	7124.14	4221.16-10604.78	1831.07-12609.77	–	–	–
HIP 6943	0.453 ± 0.011	45.34 ± 4.14	361.64	185.75-490.47	50.47-662.73	–	–	–

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Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)
HIP 7287	1.2 ± 0.02	≤ 12.3	–	–	–	≥ 332.72	≥ 884.25	≥ 4334.42
HIP 8653	0.644 ± 0.019	47.98 ± 2.44	2101.63	818.22-7067.98	347.5-11896.89	–	–	–
HIP 9067	1.47 ± 0.1	≤ 6.96	–	–	–	≥ 231.94	≥ 801.85	≥ 4275.7
HIP 9989	1.248 ± 0.002	≤ 7.92	–	–	–	≥ 341.68	≥ 892.08	≥ 4340.02
HIP 10050	0.471 ± 0.001	14.76 ± 2.12	641.25	536.76-745.32	166.15-976.21	–	–	–
HIP 10532	0.88 ± 0.033	≤ 4.38	–	–	–	≥ 647.91	≥ 1305.19	≥ 4648.61
HIP 11090	0.289 ± 0.006	≤ 37.8	–	–	–	–	–	–
HIP 11815	1.357 ± 0.02	≤ 21.0	–	–	–	≥ 253.41	≥ 803.47	≥ 4276.65
HIP 12837	0.517 ± 0.008	82.5 ± 4.08	2947.74	458.78-8919.31	118.81-12361.88	–	–	–
HIP 13681	1.062 ± 0.015	≤ 5.91	–	–	–	≥ 509.53	≥ 1054.8	≥ 4456.4
HIP 13782	0.238 ± 0.003	≤ 25.62	–	–	–	–	–	–
HIP 15211	0.936 ± 0.007	≤ 3.61	–	–	–	≥ 609.82	≥ 1187.49	≥ 4551.82
HIP 16247	1.02 ± 0.037	≤ 9.35	–	–	–	≥ 461.81	≥ 1015.84	≥ 4428.57
HIP 16709	1.225 ± 0.02	≤ 11.19	–	–	–	≥ 328.75	≥ 879.86	≥ 4331.28
HIP 17118	0.478 ± 0.023	40.49 ± 10.51	3616.84	512.59-9572.67	172.65-12482.35	–	–	–
HIP 17213	1.1 ± 0.015	≤ 6.23	–	–	–	≥ 453.14	≥ 1004.51	≥ 4420.43
HIP 18527	0.899 ± 0.021	≤ 4.12	–	–	–	≥ 633.73	≥ 1222.56	≥ 4577.49
HIP 19739	1.424 ± 0.02	≤ 11.89	–	–	–	≥ 257.67	≥ 807.4	≥ 4279.45
HIP 19912	1.148 ± 0.002	≤ 4.65	–	–	–	≥ 433.87	≥ 987.64	≥ 4408.38
HIP 20419	1.183 ± 0.005	≤ 13.39	–	–	–	≥ 334.12	≥ 885.61	≥ 4335.39
HIP 20648	0.049 ± 0.007	≤ 3.04	–	–	–	–	–	–
HIP 21066	0.472 ± 0.013	≤ 31.26	–	–	–	≥ 391.93	≥ 1076.4	≥ 4478.33
HIP 21152	0.42 ± 0.014	≤ 13.2	–	–	–	≥ 566.24	≥ 1588.14	≥ 5230.07

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Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)
HIP 22361	0.283 ± 0.004	≤ 26.75	–	–	–	–	–	–
HIP 23208	1.18 ± 0.006	≤ 7.62	–	–	–	≥ 376.02	≥ 927.89	≥ 4365.63
HIP 24017	0.448 ± 0.004	≤ 9.48	–	–	–	≥ 669.48	≥ 1254.06	≥ 4599.58
HIP 24177	1.461 ± 0.005	≤ 33.19	–	–	–	≥ 159.14	≥ 719.62	≥ 4216.67
HIP 24292	1.23 ± 0.015	≤ 6.29	–	–	–	≥ 365.54	≥ 916.6	≥ 4357.55
HIP 24457	1.118 ± 0.005	≤ 3.42	–	–	–	≥ 482.73	≥ 1038.96	≥ 4445.1
HIP 25606	0.807 ± 0.027	≤ 3.03	–	–	–	≥ 1021.51	≥ 2111.49	≥ 5405.2
HIP 27021	0.531 ± 0.006	44.96 ± 6.8	6549.83	2786.96-10753.71	830.61-12664.69	–	–	–
HIP 28823	0.358 ± 0.005	≤ 38.81	–	–	–	≥ 199.04	≥ 1161.78	≥ 4649.16
HIP 29208	0.895 ± 0.022	≤ 6.74	–	–	–	≥ 568.66	≥ 1152.02	≥ 4526.8
HIP 33282	1.344 ± 0.035	≤ 13.59	–	–	–	≥ 278.21	≥ 827.95	≥ 4294.15
HIP 33368	1.346 ± 0.1	≤ 8.69	–	–	–	≥ 288.0	≥ 847.34	≥ 4308.04
HIP 34804	1.591 ± 0.082	≤ 8.55	–	–	–	≥ 157.52	≥ 730.29	≥ 4224.76
HIP 36081	0.62 ± 0.005	49.7 ± 1.89	2660.91	1006.18-8005.41	403.71-12139.12	–	–	–
HIP 37267	1.277 ± 0.022	≤ 8.48	–	–	–	≥ 325.41	≥ 875.61	≥ 4328.24
HIP 40518	1.134 ± 0.004	≤ 5.26	–	–	–	≥ 430.98	≥ 985.1	≥ 4406.56
HIP 41274	0.952 ± 0.011	7.47 ± 1.22	505.8	439.44-602.43	335.92-820.22	–	–	–
HIP 41277	1.03 ± 0.015	125.14 ± 5.01	173.84	96.87-252.49	31.29-327.98	–	–	–
HIP 41319	0.448 ± 0.005	29.11 ± 2.49	489.33	249.94-579.52	68.41-782.82	–	–	–
HIP 42231	1.241 ± 0.015	≤ 16.83	–	–	–	≥ 296.87	≥ 847.87	≥ 4308.4
HIP 42343	0.516 ± 0.001	35.12 ± 2.7	4964.37	2198.09-9633.86	776.39-12466.6	–	–	–
HIP 42507	1.361 ± 0.001	≤ 30.52	–	–	–	≥ 226.84	≥ 778.95	≥ 4259.1
HIP 43745	1.423 ± 0.02	≤ 12.37	–	–	–	≥ 256.35	≥ 806.11	≥ 4278.53

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Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)
HIP 44162	0.461 ± 0.005	≤ 6.86	–	–	–	≥ 735.01	≥ 1316.0	≥ 4643.77
HIP 44953	0.767 ± 0.002	≤ 3.52	–	–	–	≥ 1323.58	≥ 2846.19	≥ 5993.75
HIP 45621	0.869 ± 0.013	28.92 ± 4.34	409.4	336.01-508.61	157.11-770.0	–	–	–
HIP 45731	1.586 ± 0.02	≤ 23.11	–	–	–	≥ 91.63	≥ 650.87	≥ 4167.6
HIP 46385	1.586 ± 0.1	≤ 30.6	–	–	–	≥ 92.44	≥ 661.4	≥ 4175.04
HIP 47013	0.53 ± 0.003	≤ 9.22	–	–	–	≥ 215.33	≥ 1527.67	≥ 5866.68
HIP 47261	1.21 ± 0.015	≤ 8.15	–	–	–	≥ 356.79	≥ 908.11	≥ 4351.49
HIP 47300	0.223 ± 0.004	≤ 59.79	–	–	–	–	–	–
HIP 47387	1.07 ± 0.047	≤ 5.05	–	–	–	≥ 489.6	≥ 1046.57	≥ 4450.55
HIP 47539	1.353 ± 0.011	≤ 15.01	–	–	–	≥ 270.07	≥ 819.51	≥ 4288.12
HIP 47741	1.542 ± 0.039	≤ 15.13	–	–	–	≥ 130.85	≥ 698.97	≥ 4202.01
HIP 48165	0.925 ± 0.015	≤ 3.19	–	–	–	≥ 638.59	≥ 1219.3	≥ 4574.64
HIP 48629	1.581 ± 0.197	≤ 9.94	–	–	–	≥ 152.31	≥ 744.26	≥ 4234.47
HIP 49046	1.372 ± 0.015	≤ 12.76	–	–	–	≥ 276.52	≥ 825.31	≥ 4292.26
HIP 49526	1.492 ± 0.02	≤ 24.48	–	–	–	≥ 147.69	≥ 710.86	≥ 4210.41
HIP 49577	1.331 ± 0.014	≤ 6.65	–	–	–	≥ 318.63	≥ 867.5	≥ 4322.44
HIP 49910	1.391 ± 0.023	≤ 12.9	–	–	–	≥ 268.72	≥ 817.89	≥ 4286.95
HIP 51208	1.2 ± 0.015	≤ 7.14	–	–	–	≥ 371.01	≥ 922.66	≥ 4361.89
HIP 52339	1.42 ± 0.015	≤ 11.84	–	–	–	≥ 260.19	≥ 809.62	≥ 4281.04
HIP 53175	1.293 ± 0.005	≤ 11.01	–	–	–	≥ 304.58	≥ 854.28	≥ 4312.99
HIP 53869	0.868 ± 0.045	≤ 5.27	–	–	–	≥ 628.48	≥ 1339.17	≥ 4698.29
HIP 54094	1.1 ± 0.02	≤ 4.44	–	–	–	≥ 481.2	≥ 1033.83	≥ 4441.41
HIP 54199	0.874 ± 0.016	≤ 34.44	–	–	–	≥ 383.88	≥ 952.11	≥ 4383.16

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Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)
HIP 55192	0.776 ± 0.015	75.4 ± 1.89	308.96	218.51-539.13	66.28-1459.69	–	–	–
HIP 55409	0.658 ± 0.003	≤ 5.57	–	–	–	≥ 892.89	≥ 3214.05	≥ 7741.72
HIP 57361	1.482 ± 0.015	≤ 70.46	–	–	–	≥ 85.43	≥ 646.46	≥ 4164.34
HIP 57571	1.369 ± 0.02	≤ 11.45	–	–	–	≥ 281.31	≥ 830.14	≥ 4295.72
HIP 57645	0.895 ± 0.015	≤ 3.69	–	–	–	≥ 659.12	≥ 1242.26	≥ 4591.27
HIP 57984	1.316 ± 0.015	≤ 10.03	–	–	–	≥ 302.07	≥ 851.43	≥ 4310.95
HIP 59000	1.336 ± 0.014	≤ 10.5	–	–	–	≥ 292.85	≥ 842.01	≥ 4304.21
HIP 59198	1.409 ± 0.015	≤ 11.98	–	–	–	≥ 264.26	≥ 813.52	≥ 4283.83
HIP 59199	0.334 ± 0.015	47.3 ± 3.74	–	–	–	–	–	–
HIP 59247	1.29 ± 0.015	≤ 10.15	–	–	–	≥ 310.2	≥ 860.01	≥ 4317.08
HIP 59767	0.346 ± 0.011	32.5 ± 3.95	–	–	–	–	–	–
HIP 59905	1.125 ± 0.015	≤ 5.68	–	–	–	≥ 430.87	≥ 985.26	≥ 4406.67
HIP 59953	1.602 ± 0.1	≤ 25.28	–	–	–	≥ 96.84	≥ 665.13	≥ 4177.73
HIP 60829	0.71 ± 0.012	≤ 3.99	–	–	–	≥ 1515.85	≥ 3828.19	≥ 7411.7
HIP 61947	1.582 ± 0.082	≤ 21.91	–	–	–	≥ 104.87	≥ 673.77	≥ 4183.93
HIP 62325	0.989 ± 0.007	≤ 1.63	–	–	–	≥ 585.8	≥ 1156.33	≥ 4529.3
HIP 62627	0.995 ± 0.003	≤ 7.1	–	–	–	≥ 480.59	≥ 1042.53	≥ 4447.71
HIP 63419	0.779 ± 0.021	7.65 ± 2.67	2340.25	1276.27-5485.81	757.54-10722.15	–	–	–
HIP 63421	1.354 ± 0.01	≤ 16.29	–	–	–	≥ 265.57	≥ 815.16	≥ 4285.0
HIP 63661	1.468 ± 0.1	≤ 17.8	–	–	–	≥ 166.58	≥ 747.11	≥ 4236.36
HIP 65120	1.49 ± 0.02	≤ 26.81	–	–	–	≥ 142.13	≥ 706.06	≥ 4206.97
HIP 65602	0.911 ± 0.002	≤ 3.47	–	–	–	≥ 641.92	≥ 1219.41	≥ 4574.65
HIP 65651	1.203 ± 0.02	≤ 17.22	–	–	–	≥ 309.3	≥ 860.85	≥ 4317.68

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Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)
HIP 65706	1.577 ± 0.082	≤ 10.88	–	–	–	≥ 144.27	≥ 717.64	≥ 4215.51
HIP 66262	1.165 ± 0.067	≤ 6.35	–	–	–	≥ 393.3	≥ 956.97	≥ 4386.44
HIP 66315	1.039 ± 0.121	≤ 12.91	–	–	–	≥ 395.09	≥ 985.51	≥ 4412.74
HIP 66587	1.419 ± 0.009	≤ 15.03	–	–	–	≥ 249.51	≥ 799.18	≥ 4273.57
HIP 66828	1.33 ± 0.02	≤ 8.89	–	–	–	≥ 303.69	≥ 852.86	≥ 4311.97
HIP 69311	1.172 ± 0.07	≤ 9.39	–	–	–	≥ 360.75	≥ 923.47	≥ 4362.47
HIP 69333	1.173 ± 0.004	≤ 4.28	–	–	–	≥ 426.36	≥ 979.3	≥ 4402.4
HIP 69860	1.629 ± 0.1	≤ 25.8	–	–	–	≥ 92.18	≥ 656.97	≥ 4171.9
HIP 69963	1.39 ± 0.02	≤ 34.7	–	–	–	≥ 217.68	≥ 769.92	≥ 4252.64
HIP 70472	1.257 ± 0.01	≤ 5.02	–	–	–	≥ 368.06	≥ 918.5	≥ 4358.91
HIP 71243	0.45 ± 0.003	≤ 12.35	–	–	–	≥ 624.45	≥ 1202.77	≥ 4562.54
HIP 71515	0.95 ± 0.002	≤ 4.52	–	–	–	≥ 566.76	≥ 1138.55	≥ 4516.62
HIP 71899	0.533 ± 0.01	90.22 ± 3.36	3446.91	535.05-9289.75	124.97-12428.99	–	–	–
HIP 71957	0.385 ± 0.006	62.96 ± 6.43	742.99	198.67-2858.42	57.32-9369.38	–	–	–
HIP 73121	1.035 ± 0.009	≤ 3.93	–	–	–	≥ 544.57	≥ 1093.34	≥ 4483.97
HIP 73183	1.555 ± 0.197	≤ 10.92	–	–	–	≥ 152.29	≥ 747.08	≥ 4236.44
HIP 73252	1.565 ± 0.082	≤ 14.68	–	–	–	≥ 129.34	≥ 703.31	≥ 4205.13
HIP 73787	1.21 ± 0.008	≤ 7.17	–	–	–	≥ 366.04	≥ 917.25	≥ 4358.02
HIP 74926	1.214 ± 0.009	≤ 8.04	–	–	–	≥ 356.02	≥ 907.11	≥ 4350.77
HIP 79203	0.447 ± 0.007	52.15 ± 2.24	316.16	170.59-447.18	48.25-603.54	–	–	–
HIP 79593	1.584 ± 0.01	≤ 64.58	–	–	–	≥ 62.8	≥ 619.13	≥ 4144.8
HIP 81655	1.605 ± 0.1	≤ 16.7	–	–	–	≥ 114.09	≥ 684.91	≥ 4191.94
HIP 82260	0.767 ± 0.002	≤ 2.66	–	–	–	≥ 1383.07	≥ 2980.3	≥ 6115.02

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Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)
HIP 83676	1.015 ± 0.015	≤ 5.23	–	–	–	≥ 513.21	≥ 1071.18	≥ 4468.19
HIP 83929	1.597 ± 0.197	≤ 7.89	–	–	–	≥ 164.45	≥ 754.63	≥ 4242.01
HIP 87370	0.656 ± 0.015	5.45 ± 1.73	9440.7	5779.69-11965.31	2459.94-12855.22	–	–	–
HIP 88962	1.372 ± 0.004	≤ 7.43	–	–	–	≥ 302.95	≥ 850.97	≥ 4310.61
HIP 89320	1.586 ± 0.197	≤ 17.74	–	–	–	≥ 119.63	≥ 709.43	≥ 4209.43
HIP 89449	1.323 ± 0.019	≤ 13.44	–	–	–	≥ 284.08	≥ 833.75	≥ 4298.3
HIP 90306	1.442 ± 0.02	≤ 20.2	–	–	–	≥ 220.21	≥ 771.96	≥ 4254.11
HIP 93072	1.437 ± 0.004	≤ 9.38	–	–	–	≥ 266.15	≥ 815.5	≥ 4285.25
HIP 98007	0.978 ± 0.003	≤ 4.06	–	–	–	≥ 561.44	≥ 1130.48	≥ 4510.78
HIP 99969	1.14 ± 0.015	≤ 6.74	–	–	–	≥ 405.74	≥ 959.42	≥ 4388.19
HIP 100133	1.43 ± 0.012	≤ 10.69	–	–	–	≥ 261.29	≥ 810.76	≥ 4281.86
HIP 101852	0.597 ± 0.012	≤ 5.55	–	–	–	≥ 413.67	≥ 2147.3	≥ 6660.85
HIP 102119	1.121 ± 0.009	≤ 15.56	–	–	–	≥ 349.81	≥ 902.7	≥ 4347.61
HIP 104687	0.654 ± 0.041	63.32 ± 3.17	1661.55	516.08-6885.97	216.02-11896.04	–	–	–
HIP 105504	1.255 ± 0.015	≤ 7.34	–	–	–	≥ 343.5	≥ 893.98	≥ 4341.38
HIP 107317	1.522 ± 0.021	≤ 19.01	–	–	–	≥ 131.93	≥ 696.16	≥ 4199.9
HIP 108036	0.378 ± 0.011	36.67 ± 8.24	1531.65	629.16-4486.11	186.82-10568.9	–	–	–
HIP 109807	1.252 ± 0.005	≤ 8.56	–	–	–	≥ 334.91	≥ 885.27	≥ 4335.15
HIP 110106	1.399 ± 0.02	≤ 18.94	–	–	–	≥ 247.79	≥ 797.7	≥ 4272.52
HIP 110401	1.208 ± 0.013	≤ 6.42	–	–	–	≥ 374.95	≥ 926.43	≥ 4364.59
HIP 110663	0.77 ± 0.005	≤ 30.38	–	–	–	≥ 449.42	≥ 1135.54	≥ 4534.24
HIP 111976	1.548 ± 0.1	≤ 17.49	–	–	–	≥ 126.14	≥ 705.19	≥ 4206.41
HIP 113159	0.469 ± 0.005	22.57 ± 2.34	563.71	364.7-656.67	95.73-869.36	–	–	–

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Target	B-V	Li EW (mÅ)	Age (Myr)	68% CI (Myr)	95% CI (Myr)	99.7% LL (Myr)	95% LL (Myr)	68% LL (Myr)
HIP 115280	0.468 ± 0.004	≤ 13.37	–	–	–	≥ 630.31	≥ 1208.72	≥ 4566.8
HIP 115411	0.7 ± 0.005	≤ 3.79	–	–	–	≥ 1551.06	≥ 4259.39	≥ 8276.42
HIP 116973	1.085 ± 0.008	≤ 3.36	–	–	–	≥ 525.19	≥ 1071.06	≥ 4468.03
HIP 117159	0.859 ± 0.006	≤ 2.4	–	–	–	≥ 801.54	≥ 1475.94	≥ 4766.76
HIP 117235	1.376 ± 0.133	≤ 13.91	–	–	–	≥ 233.62	≥ 806.1	≥ 4278.54
HIP 117410	1.244 ± 0.014	≤ 6.27	–	–	–	≥ 359.04	≥ 909.77	≥ 4352.67
HIP 117795	1.21 ± 0.015	≤ 8.98	–	–	–	≥ 350.04	≥ 901.3	≥ 4346.62
HIP 118086	1.3 ± 0.015	≤ 18.5	–	–	–	≥ 273.54	≥ 823.87	≥ 4291.23
HIP 118310	1.187 ± 0.052	≤ 4.14	–	–	–	≥ 416.39	≥ 975.53	≥ 4399.71

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Table A.0.2:: Each target star and its adopted $B-V$, measured S index and measured $log(R'_{HK})$ are listed. R'_{HK} is only calibrated as an age metric for F5-K2 stars for which report a median age, 68% and 98% confidence intervals.

Target	Adopted B-V	S	$log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 143	1.033 ± 0.019	0.41	-4.58	–	–	–
HIP 669	0.624 ± 0.015	0.11	-4.88	6468.35	3805.7-9673.65	1811.36-12362.31
HIP 1068	1.315 ± 0.1	0.41	-4.3	–	–	–
HIP 1412	1.414 ± 0.005	1.38	-3.7	–	–	–
HIP 1539	1.375 ± 0.005	1.25	-3.77	–	–	–
HIP 1771	1.489 ± 0.1	1.4	-3.66	–	–	–
HIP 2426	1.359 ± 0.02	0.85	-3.95	–	–	–
HIP 2843	0.466 ± 0.007	0.1	-4.82	5038.2	2848.33-8376.32	1319.5-12018.59
HIP 3008	1.424 ± 0.02	1.57	-3.64	–	–	–
HIP 3293	0.452 ± 0.009	0.06	-4.8	4744.41	2658.9-8090.39	1223.88-11927.21
HIP 3633	0.814 ± 0.061	0.28	-4.93	7627.38	4629.95-10777.13	2245.46-12580.04
HIP 3724	1.4 ± 0.02	1.95	-3.56	–	–	–
HIP 4024	1.069 ± 0.009	0.39	-4.55	–	–	–
HIP 5319	0.398 ± 0.01	0.13	-4.83	–	–	–
HIP 5763	1.22 ± 0.015	0.67	-4.16	–	–	–
HIP 6776	0.453 ± 0.007	0.25	-4.89	6703.86	3969.45-9900.36	1896.83-12406.95
HIP 6796	1.157 ± 0.074	0.58	-4.28	–	–	–
HIP 6833	0.656 ± 0.002	0.12	-4.91	7056.53	4218.1-10241.09	2027.37-12469.75
HIP 6943	0.453 ± 0.011	0.06	-4.8	4719.62	2643.05-8065.82	1215.9-11919.12
HIP 7287	1.2 ± 0.02	0.44	-4.36	–	–	–

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Target	Adopted B-V	S	$log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 8653	0.644 ± 0.019	0.22	-4.91	7289.8	4385.03-10462.51	2115.46-12502.65
HIP 9067	1.47 ± 0.1	0.74	-3.94	–	–	–
HIP 9989	1.248 ± 0.002	0.78	-4.07	–	–	–
HIP 10050	0.471 ± 0.001	0.05	-4.8	4649.09	2597.95-7995.11	1193.23-11895.76
HIP 10532	0.88 ± 0.033	0.19	-5.0	8917.93	5673.64-11623.81	2817.71-12794.38
HIP 11090	0.289 ± 0.006	0.08	-4.82	–	–	–
HIP 11815	1.357 ± 0.02	2.34	-3.51	–	–	–
HIP 12837	0.517 ± 0.008	0.13	-4.84	5572.64	3199.05-8862.9	1498.1-12163.83
HIP 13681	1.062 ± 0.015	0.55	-4.42	–	–	–
HIP 13782	0.238 ± 0.003	0.11	-4.83	–	–	–
HIP 15211	0.936 ± 0.007	0.21	-4.95	–	–	–
HIP 16247	1.02 ± 0.037	1.24	-4.13	–	–	–
HIP 16709	1.225 ± 0.02	0.46	-4.32	–	–	–
HIP 17118	0.478 ± 0.023	0.14	-4.84	5506.82	3155.38-8805.1	1475.75-12147.31
HIP 17213	1.1 ± 0.015	0.8	-4.21	–	–	–
HIP 18527	0.899 ± 0.021	0.45	-4.72	3335.89	1782.91-6529.46	789.98-11322.22
HIP 19739	1.424 ± 0.02	1.01	-3.83	–	–	–
HIP 19912	1.148 ± 0.002	0.39	-4.47	–	–	–
HIP 20419	1.183 ± 0.005	0.47	-4.35	–	–	–
HIP 20648	0.049 ± 0.007	0.1	-4.76	–	–	–
HIP 21066	0.472 ± 0.013	0.13	-4.83	5415.69	3095.17-8724.59	1445.0-12123.82
HIP 21152	0.42 ± 0.014	0.12	-4.82	–	–	–
HIP 22361	0.283 ± 0.004	0.13	-4.83	–	–	–

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Target	Adopted B-V	S	$log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 23208	1.18 ± 0.006	1.03	-4.01	–	–	–
HIP 24017	0.448 ± 0.004	0.11	-4.82	–	–	–
HIP 24177	1.461 ± 0.005	0.65	-4.0	–	–	–
HIP 24292	1.23 ± 0.015	0.7	-4.13	–	–	–
HIP 24457	1.118 ± 0.005	0.45	-4.44	–	–	–
HIP 25606	0.807 ± 0.027	0.07	-5.1	–	–	–
HIP 27021	0.531 ± 0.006	0.05	-4.81	4955.97	2795.06-8297.47	1292.55-11993.99
HIP 28823	0.358 ± 0.005	0.11	-4.83	–	–	–
HIP 29208	0.895 ± 0.022	0.11	-5.13	–	–	–
HIP 33282	1.344 ± 0.035	1.12	-3.83	–	–	–
HIP 33368	1.346 ± 0.1	1.19	-3.81	–	–	–
HIP 34804	1.591 ± 0.082	0.38	-4.18	–	–	–
HIP 36081	0.62 ± 0.005	0.06	-4.86	6087.61	3544.88-9317.97	1676.01-12283.7
HIP 37267	1.277 ± 0.022	0.72	-4.08	–	–	–
HIP 40518	1.134 ± 0.004	0.48	-4.39	–	–	–
HIP 41274	0.952 ± 0.011	0.61	-4.53	–	–	–
HIP 41277	1.03 ± 0.015	1.68	-3.98	–	–	–
HIP 41319	0.448 ± 0.005	0.22	-4.88	–	–	–
HIP 42231	1.241 ± 0.015	0.53	-4.24	–	–	–
HIP 42343	0.516 ± 0.001	0.05	-4.8	4792.54	2689.78-8138.18	1239.43-11942.73
HIP 42507	1.361 ± 0.001	1.48	-3.7	–	–	–
HIP 43745	1.423 ± 0.02	1.28	-3.73	–	–	–
HIP 44162	0.461 ± 0.005	0.1	-4.82	5091.45	2882.9-8426.86	1337.04-12034.09

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Target	Adopted B-V	S	$log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 44953	0.767 ± 0.002	0.12	-5.01	–	–	–
HIP 45621	0.869 ± 0.013	0.36	-4.83	5312.27	3027.16-8631.79	1410.33-12096.23
HIP 45731	1.586 ± 0.02	4.53	-3.11	–	–	–
HIP 46385	1.586 ± 0.1	1.07	-3.73	–	–	–
HIP 47013	0.53 ± 0.003	0.06	-4.81	4997.36	2821.85-8337.18	1306.1-12006.47
HIP 47261	1.21 ± 0.015	1.16	-3.93	–	–	–
HIP 47300	0.223 ± 0.004	0.08	-4.83	–	–	–
HIP 47387	1.07 ± 0.047	0.49	-4.46	–	–	–
HIP 47539	1.353 ± 0.011	0.95	-3.9	–	–	–
HIP 47741	1.542 ± 0.039	0.61	-3.99	–	–	–
HIP 48165	0.925 ± 0.015	0.16	-5.03	–	–	–
HIP 48629	1.581 ± 0.197	1.11	-3.72	–	–	–
HIP 49046	1.372 ± 0.015	1.26	-3.77	–	–	–
HIP 49526	1.492 ± 0.02	0.97	-3.81	–	–	–
HIP 49577	1.331 ± 0.014	0.71	-4.05	–	–	–
HIP 49910	1.391 ± 0.023	1.09	-3.82	–	–	–
HIP 51208	1.2 ± 0.015	0.59	-4.23	–	–	–
HIP 52339	1.42 ± 0.015	1.32	-3.72	–	–	–
HIP 53175	1.293 ± 0.005	1.41	-3.77	–	–	–
HIP 53869	0.868 ± 0.045	0.36	-4.83	5401.81	3086.04-8712.37	1440.37-12120.17
HIP 54094	1.1 ± 0.02	0.16	-4.9	–	–	–
HIP 54199	0.874 ± 0.016	0.13	-5.09	–	–	–
HIP 55192	0.776 ± 0.015	0.1	-5.03	–	–	–

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Target	Adopted B-V	S	$log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 55409	0.658 ± 0.003	0.12	-4.91	7119.41	4262.9-10301.28	2050.96-12479.67
HIP 57361	1.482 ± 0.015	0.64	-4.0	–	–	–
HIP 57571	1.369 ± 0.02	0.73	-4.0	–	–	–
HIP 57645	0.895 ± 0.015	0.29	-4.87	6194.96	3617.97-9416.26	1713.82-12306.8
HIP 57984	1.316 ± 0.015	1.08	-3.87	–	–	–
HIP 59000	1.336 ± 0.014	1.96	-3.6	–	–	–
HIP 59198	1.409 ± 0.015	0.93	-3.87	–	–	–
HIP 59199	0.334 ± 0.015	0.1	-4.82	–	–	–
HIP 59247	1.29 ± 0.015	0.81	-4.01	–	–	–
HIP 59767	0.346 ± 0.011	0.12	-4.83	–	–	–
HIP 59905	1.125 ± 0.015	0.58	-4.32	–	–	–
HIP 59953	1.602 ± 0.1	0.97	-3.77	–	–	–
HIP 60829	0.71 ± 0.012	0.07	-4.96	8269.16	5128.5-11264.78	2514.96-12724.17
HIP 61947	1.582 ± 0.082	1.12	-3.71	–	–	–
HIP 62325	0.989 ± 0.007	0.08	-5.23	–	–	–
HIP 62627	0.995 ± 0.003	0.2	-4.91	–	–	–
HIP 63419	0.779 ± 0.021	0.36	-4.9	6919.61	4121.07-10109.29	1976.29-12446.08
HIP 63421	1.354 ± 0.01	0.93	-3.91	–	–	–
HIP 63661	1.468 ± 0.1	0.95	-3.84	–	–	–
HIP 65120	1.49 ± 0.02	0.99	-3.81	–	–	–
HIP 65602	0.911 ± 0.002	0.24	-4.92	–	–	–
HIP 65651	1.203 ± 0.02	1.16	-3.93	–	–	–
HIP 65706	1.577 ± 0.082	0.96	-3.78	–	–	–

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Target	Adopted B-V	S	$log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 66262	1.165 ± 0.067	0.48	-4.35	–	–	–
HIP 66315	1.039 ± 0.121	0.33	-4.66	–	–	–
HIP 66587	1.419 ± 0.009	0.81	-3.93	–	–	–
HIP 66828	1.33 ± 0.02	1.17	-3.83	–	–	–
HIP 69311	1.172 ± 0.07	1.15	-3.97	–	–	–
HIP 69333	1.173 ± 0.004	0.49	-4.34	–	–	–
HIP 69860	1.629 ± 0.1	1.22	-3.66	–	–	–
HIP 69963	1.39 ± 0.02	0.57	-4.1	–	–	–
HIP 70472	1.257 ± 0.01	0.65	-4.14	–	–	–
HIP 71243	0.45 ± 0.003	0.1	-4.81	5027.51	2841.38-8366.19	1315.99-12015.47
HIP 71515	0.95 ± 0.002	0.33	-4.77	–	–	–
HIP 71899	0.533 ± 0.01	0.16	-4.86	5942.96	3446.92-9187.67	1625.41-12251.95
HIP 71957	0.385 ± 0.006	0.11	-4.82	–	–	–
HIP 73121	1.035 ± 0.009	0.19	-4.89	–	–	–
HIP 73183	1.555 ± 0.197	1.13	-3.72	–	–	–
HIP 73252	1.565 ± 0.082	1.26	-3.67	–	–	–
HIP 73787	1.21 ± 0.008	0.76	-4.11	–	–	–
HIP 74926	1.214 ± 0.009	0.57	-4.23	–	–	–
HIP 79203	0.447 ± 0.007	0.07	-4.8	–	–	–
HIP 79593	1.584 ± 0.01	0.19	-4.48	–	–	–
HIP 81655	1.605 ± 0.1	0.6	-3.98	–	–	–
HIP 82260	0.767 ± 0.002	0.08	-5.03	–	–	–
HIP 83676	1.015 ± 0.015	0.37	-4.65	–	–	–

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Target	Adopted B-V	S	$log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 83929	1.597 ± 0.197	0.73	-3.9	–	–	–
HIP 87370	0.656 ± 0.015	0.07	-4.9	6908.91	4113.47-10099.0	1972.33-12444.19
HIP 88962	1.372 ± 0.004	0.91	-3.91	–	–	–
HIP 89320	1.586 ± 0.197	1.31	-3.65	–	–	–
HIP 89449	1.323 ± 0.019	0.82	-3.99	–	–	–
HIP 90306	1.442 ± 0.02	0.78	-3.93	–	–	–
HIP 93072	1.437 ± 0.004	0.98	-3.84	–	–	–
HIP 98007	0.978 ± 0.003	0.17	-4.99	–	–	–
HIP 99969	1.14 ± 0.015	0.62	-4.27	–	–	–
HIP 100133	1.43 ± 0.012	0.82	-3.92	–	–	–
HIP 101852	0.597 ± 0.012	0.16	-4.88	6452.83	3795.0-9658.97	1805.77-12359.28
HIP 102119	1.121 ± 0.009	1.9	-3.81	–	–	–
HIP 104687	0.654 ± 0.041	0.25	-4.92	7504.1	4539.92-10663.71	2197.54-12545.56
HIP 105504	1.255 ± 0.015	1.39	-3.81	–	–	–
HIP 107317	1.522 ± 0.021	1.73	-3.55	–	–	–
HIP 108036	0.378 ± 0.011	0.14	-4.84	–	–	–
HIP 109807	1.252 ± 0.005	1.21	-3.87	–	–	–
HIP 110106	1.399 ± 0.02	1.26	-3.75	–	–	–
HIP 110401	1.208 ± 0.013	0.59	-4.23	–	–	–
HIP 110663	0.77 ± 0.005	0.35	-4.91	7152.13	4286.34-10332.43	2063.3-12484.35
HIP 111976	1.548 ± 0.1	1.07	-3.74	–	–	–
HIP 113159	0.469 ± 0.005	0.08	-4.81	4922.09	2773.17-8264.65	1281.48-11983.6
HIP 115280	0.468 ± 0.004	0.13	-4.83	5405.28	3088.32-8715.42	1441.52-12121.08

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Target	Adopted B-V	S	$log(R'_{HK})$	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 115411	0.7 ± 0.005	0.09	-4.95	8010.07	4920.15-11092.03	2401.41-12683.1
HIP 116973	1.085 ± 0.008	0.55	-4.39	–	–	–
HIP 117159	0.859 ± 0.006	0.1	-5.13	–	–	–
HIP 117235	1.376 ± 0.133	0.12	-4.78	–	–	–
HIP 117410	1.244 ± 0.014	2.99	-3.49	–	–	–
HIP 117795	1.21 ± 0.015	0.55	-4.26	–	–	–
HIP 118086	1.3 ± 0.015	0.82	-4.0	–	–	–
HIP 118310	1.187 ± 0.052	0.76	-4.14	–	–	–

Table A.0.3:: Rotation period measurements for 23 of our 166 accelerating stars. An additional 106 stars have a TESS light curve but display either no variability signal or a signal that is not due to rotation, and 37 stars do not have a TESS light curve. Median ages and 68% and 95% confidence intervals are given for stars with a rotation period measurement that are also within the `gyro-interp` T_{eff} range.

Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 143	4776 ± 200	no LC	–	–	–
HIP 669	5955 ± 343	no signal	–	–	–
HIP 1068	3401 ± 51	no signal	–	–	–
HIP 1412	3807 ± 82	5.1 ± 0.44	134.6	41.14 - 385.6	6.73 - 568.83
HIP 1539	3956 ± 153	no LC	–	–	–
HIP 1771	4055 ± 154	no signal	–	–	–
HIP 2426	4104 ± 151	no signal	–	–	–
HIP 2843	6465 ± 335	no signal	–	–	–
HIP 3008	3891 ± 121	5.11 ± 0.36	128.37	39.61 - 378.36	6.54 - 563.46
HIP 3293	6442 ± 344	no signal	–	–	–
HIP 3633	4831 ± 211	no signal	–	–	–
HIP 3724	3873 ± 104	no LC	–	–	–
HIP 4024	4692 ± 184	no signal	–	–	–
HIP 5319	6693 ± 324	no signal	–	–	–
HIP 5763	4370 ± 144	no signal	–	–	–
HIP 6776	6496 ± 324	no signal	–	–	–
HIP 6796	4590 ± 174	no signal	–	–	–
HIP 6833	5765 ± 305	no signal	–	–	–

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Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 6943	6464 ± 336	no signal	–	–	–
HIP 7287	4242 ± 123	no signal	–	–	–
HIP 8653	5755 ± 306	4.11 ± 0.34	220.38	76.93 - 423.96	11.88 - 1158.34
HIP 9067	4117 ± 150	no signal	–	–	–
HIP 9989	4279 ± 126	no signal	–	–	–
HIP 10050	6461 ± 337	no signal	–	–	–
HIP 10532	5149 ± 198	no signal	–	–	–
HIP 11090	7127 ± 426	not rotation	–	–	–
HIP 11815	3899 ± 129	2 periods	nan	nan	nan
HIP 12837	6293 ± 311	no signal	–	–	–
HIP 13681	4709 ± 188	no signal	–	–	–
HIP 13782	7615 ± 441	no signal	–	–	–
HIP 15211	4867 ± 211	no signal	–	–	–
HIP 16247	4453 ± 160	EB	–	–	–
HIP 16709	4104 ± 151	no signal	–	–	–
HIP 17118	6406 ± 343	no LC	–	–	–
HIP 17213	4650 ± 177	no LC	–	–	–
HIP 18527	5139 ± 201	9.59 ± 1.78	734.8	412.07 - 1053.55	153.71 - 1335.39
HIP 19739	3818 ± 88	7.84 ± 1.06	254.07	129.59 - 453.05	71.31 - 590.82
HIP 19912	4550 ± 170	no signal	–	–	–
HIP 20419	4438 ± 158	10.15 ± 2.02	674.7	284.69 - 1180.52	132.07 - 1568.97
HIP 20648	8718 ± 670	no signal	–	–	–
HIP 21066	6411 ± 344	not rotation	–	–	–

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Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 21152	6629 ± 311	no signal	–	–	–
HIP 22361	7155 ± 428	not rotation	–	–	–
HIP 23208	4268 ± 125	no LC	–	–	–
HIP 24017	6579 ± 307	no signal	–	–	–
HIP 24177	3687 ± 87	no signal	–	–	–
HIP 24292	4342 ± 137	no signal	–	–	–
HIP 24457	4627 ± 177	no signal	–	–	–
HIP 25606	5109 ± 184	no signal	–	–	–
HIP 27021	6123 ± 315	no signal	–	–	–
HIP 28823	6887 ± 322	no signal	–	–	–
HIP 29208	5112 ± 186	no LC	–	–	–
HIP 33282	4200 ± 137	9.87 ± 1.77	507.65	220.38 - 1073.73	127.15 - 1510.58
HIP 33368	4299 ± 128	contaminated pixel	–	–	–
HIP 34804	4517 ± 166	no LC	–	–	–
HIP 36081	5957 ± 343	no signal	–	–	–
HIP 37267	4329 ± 135	no signal	–	–	–
HIP 40518	4435 ± 157	no LC	–	–	–
HIP 41274	5058 ± 173	not rotation	–	–	–
HIP 41277	4550 ± 170	3.7 ± 0.24	95.67	30.09 - 315.98	5.16 - 532.3
HIP 41319	6511 ± 319	no signal	–	–	–
HIP 42231	4200 ± 137	no LC	–	–	–
HIP 42343	6306 ± 316	no signal	–	–	–
HIP 42507	3955 ± 153	no signal	–	–	–

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Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 43745	3936 ± 148	5.82 ± 0.56	158.14	55.2 - 396.73	8.77 - 574.25
HIP 44162	6499 ± 323	no signal	–	–	–
HIP 44953	5322 ± 239	no signal	–	–	–
HIP 45621	5150 ± 197	5.48 ± 0.51	130.82	39.99 - 287.4	6.6 - 419.96
HIP 45731	3709 ± 87	EB	–	–	–
HIP 46385	3529 ± 57	10.33 ± 1.69	–	–	–
HIP 47013	6286 ± 309	no signal	–	–	–
HIP 47261	4334 ± 136	contaminated pixel	–	–	–
HIP 47300	7661 ± 466	no signal	–	–	–
HIP 47387	4657 ± 178	no LC	–	–	–
HIP 47539	3954 ± 153	no LC	–	–	–
HIP 47741	3363 ± 57	no signal	–	–	–
HIP 48165	4854 ± 214	no signal	–	–	–
HIP 48629	3961 ± 155	no signal	–	–	–
HIP 49046	3835 ± 91	no signal	–	–	–
HIP 49526	3713 ± 87	no signal	–	–	–
HIP 49577	4133 ± 151	EB	–	–	–
HIP 49910	3902 ± 132	no signal	–	–	–
HIP 51208	4384 ± 146	no LC	–	–	–
HIP 52339	3859 ± 96	no signal	–	–	–
HIP 53175	4211 ± 126	not rotation	–	–	–
HIP 53869	4937 ± 197	5.9 ± 0.51	102.24	32.77 - 263.89	5.56 - 436.19
HIP 54094	4839 ± 212	no signal	–	–	–

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Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 54199	5114 ± 187	no LC	–	–	–
HIP 55192	5301 ± 229	no signal	–	–	–
HIP 55409	5735 ± 307	no signal	–	–	–
HIP 57361	3516 ± 54	no signal	–	–	–
HIP 57571	3933 ± 147	no signal	–	–	–
HIP 57645	5093 ± 174	no signal	–	–	–
HIP 57984	4214 ± 123	no signal	–	–	–
HIP 59000	4101 ± 151	6.56 ± 0.87	185.8	82.99 - 412.07	12.82 - 574.25
HIP 59198	3831 ± 89	no LC	–	–	–
HIP 59199	6799 ± 318	not rotation	–	–	–
HIP 59247	4166 ± 153	no signal	–	–	–
HIP 59767	6943 ± 327	not rotation	–	–	–
HIP 59905	4551 ± 170	no signal	–	–	–
HIP 59953	3801 ± 79	no signal	–	–	–
HIP 60829	5636 ± 323	no signal	–	–	–
HIP 61947	3853 ± 96	no signal	–	–	–
HIP 62325	4937 ± 197	no LC	–	–	–
HIP 62627	4799 ± 205	no signal	–	–	–
HIP 63419	5278 ± 216	9.84 ± 1.5	847.11	563.46 - 1115.23	251.67 - 1373.92
HIP 63421	3894 ± 124	no signal	–	–	–
HIP 63661	4120 ± 150	no signal	–	–	–
HIP 65120	3760 ± 78	no LC	–	–	–
HIP 65602	5060 ± 172	no signal	–	–	–

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Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 65651	4355 ± 140	4.86 ± 0.4	106.19	33.08 - 337.67	5.62 - 542.49
HIP 65706	3964 ± 156	no signal	–	–	–
HIP 66262	4470 ± 161	no signal	–	–	–
HIP 66315	4580 ± 173	no signal	–	–	–
HIP 66587	3824 ± 89	no LC	–	–	–
HIP 66828	4186 ± 150	4.39 ± 0.28	107.2	33.08 - 344.13	5.62 - 547.66
HIP 69311	4399 ± 150	6.45 ± 0.62	129.59	43.96 - 357.44	7.19 - 547.66
HIP 69333	4416 ± 153	no signal	–	–	–
HIP 69860	3729 ± 84	no signal	–	–	–
HIP 69963	3762 ± 77	no signal	–	–	–
HIP 70472	4126 ± 150	no LC	–	–	–
HIP 71243	6550 ± 310	no signal	–	–	–
HIP 71515	4999 ± 185	no LC	–	–	–
HIP 71899	6268 ± 311	2.53 ± 0.11	–	–	–
HIP 71957	6497 ± 324	no signal	–	–	–
HIP 73121	4924 ± 200	no LC	–	–	–
HIP 73183	4094 ± 152	no signal	–	–	–
HIP 73252	4045 ± 154	no signal	–	–	–
HIP 73787	4294 ± 128	no LC	–	–	–
HIP 74926	4266 ± 125	no LC	–	–	–
HIP 79203	6599 ± 305	no LC	–	–	–
HIP 79593	4260 ± 125	no LC	–	–	–
HIP 81655	3551 ± 57	no signal	–	–	–

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Target	T_{eff} (K)	P_{rot} (d)	Age (Myr)	68% CI (Myr)	95% CI (Myr)
HIP 82260	5473 ± 331	no LC	–	–	–
HIP 83676	4871 ± 211	no LC	–	–	–
HIP 83929	4177 ± 154	6.19 ± 0.63	162.7	60.69 - 392.99	9.55 - 568.83
HIP 87370	5831 ± 319	no LC	–	–	–
HIP 88962	3949 ± 151	no LC	–	–	–
HIP 89320	4206 ± 131	contaminated pixel	–	–	–
HIP 89449	4190 ± 147	no signal	–	–	–
HIP 90306	3767 ± 77	no signal	–	–	–
HIP 93072	3824 ± 89	no LC	–	–	–
HIP 98007	4765 ± 198	no signal	–	–	–
HIP 99969	4467 ± 161	no LC	–	–	–
HIP 100133	3769 ± 76	no LC	–	–	–
HIP 101852	6004 ± 335	6.87 ± 0.54	913.87	493.41 - 1914.67	276.7 - 2693.67
HIP 102119	4217 ± 121	no LC	–	–	–
HIP 104687	5832 ± 319	no LC	–	–	–
HIP 105504	4228 ± 122	no LC	–	–	–
HIP 107317	3432 ± 45	6.18 ± 0.51	–	–	–
HIP 108036	6760 ± 321	no LC	–	–	–
HIP 109807	4279 ± 126	no LC	–	–	–
HIP 110106	3841 ± 93	no signal	–	–	–
HIP 110401	4240 ± 123	no signal	–	–	–
HIP 110663	5420 ± 289	7.3 ± 0.55	457.37	263.89 - 619.51	123.59 - 1554.17
HIP 111976	3904 ± 134	no signal	–	–	–

Continued on next page

Target	T_{eff}	P_{rot}	Age	68% CI	95% CI
	(K)	(d)	(Myr)	(Myr)	(Myr)
HIP 113159	6433 ± 347	no signal	–	–	–
HIP 115280	6427 ± 348	no signal	–	–	–
HIP 115411	5515 ± 331	no signal	–	–	–
HIP 116973	4614 ± 177	no signal	–	–	–
HIP 117159	5131 ± 197	no signal	–	–	–
HIP 117235	3947 ± 151	no signal	–	–	–
HIP 117410	4174 ± 154	2 periods	254.07	64.24 - 625.41	10.11 - 1400.23
HIP 117795	4342 ± 137	no signal	–	–	–
HIP 118086	4035 ± 155	no signal	–	–	–
HIP 118310	4501 ± 165	no signal	–	–	–

APPENDIX B.

We investigate the accuracy of our stellar parameter fitting using archival FEROS spectra of 8 well-studied F and G stars. We compare our stellar parameters to those measured Worley et al. (2012) using these same FEROS spectra. Using the spectral fitting method described in this work, modified to model a single star instead of an SB2, we measure the effective temperatures and metallicities of these stars. We do not fit for $\log(g)$, because we find that $\log(g)$ values extracted from our method are significantly different from the literature values. This could be the result of the PHOENIX model grids being too coarsely sampled in $\log(g)$, our choice of lines not constraining $\log(g)$, or limitations in our science spectra such as the spectral resolution or continuum correction, all of which could prevent our method from constraining $\log(g)$. Since gravity is covariant with metallicity and temperature, we place a prior on $\log(g)$, rather than attempt to measure it from our spectral fits. We use a $\log(g)$ prior for each star of a Gaussian, from the value and error reported by Worley et al. (2012).

Our measured effective temperatures and metallicities for these 8 stars are shown in Figure B.0.1. The effective temperature residuals show a systematic offset at temperatures greater than ~ 6000 K, with our method recovering higher effective temperatures compared to Worley et al. (2012). The metallicity residuals do not show a systematic offset, however our metallicity measurements for the F stars are significantly more discrepant with the Worley et al. (2012) values compared to the better-matching G stars. The residuals in effective temperature have a median of -129.5 K and a standard deviation of 123.3 K, while the metallicity residuals have a median of -0.007 dex and a standard deviation of 0.100 dex. The offset in effective temperature is likely due to the wavelength ranges we used in our analysis, but further analysis is required to confirm this. Therefore, we adopt the effective temperatures and metallicities for HD 143811 A and B (both with effective temperatures above 6000 K) from the SED analysis rather than our spectral modeling.

For effective temperature the spread in the residuals is not consistent with the expected errors, as calculated by adding our measurement errors and those from Worley et al. (2012) in quadrature. While we do not adopt these values, in order to report them with more accurate errors we use the residuals to measure an error correction. We assume that our errors are underestimated, but do not account for the systematic offset in this step, and model the additional noise as Gaussian, using maximum likelihood estimation to determine this additional error. We find that we need to add 107.2 K quadrature to our effective temperature uncertainties, which we utilize when reporting temperatures for HD 143811 A and B from the spectral fits. Regardless of temperature and metallicity measurements, the measured RVs of both these 8 comparison stars and HD 143811 A and B are consistent across fits.

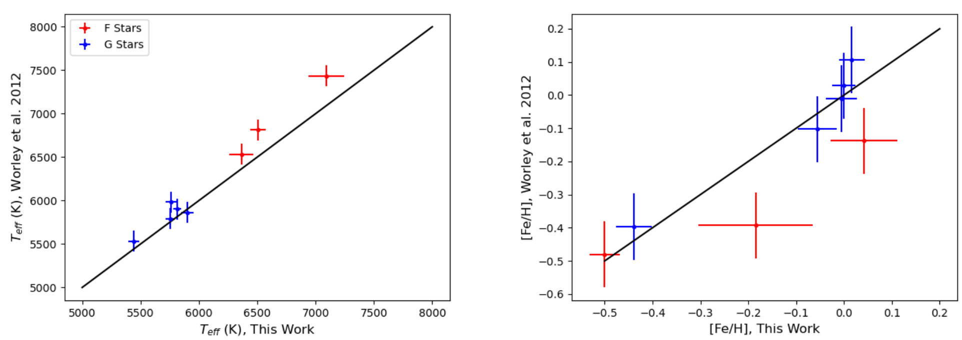

Fig. B.0.1.— We are generally able to accurately recover effective temperature and metallicity when applying our method to well-characterized F and G stars from Worley et al. (2012) However, the results for stars hotter than $\gtrsim$6000 K show a systematic offset in temperature and a larger scatter in metallicity compared to G stars. The solid black line is the 1-to-1 line, representing identical results.

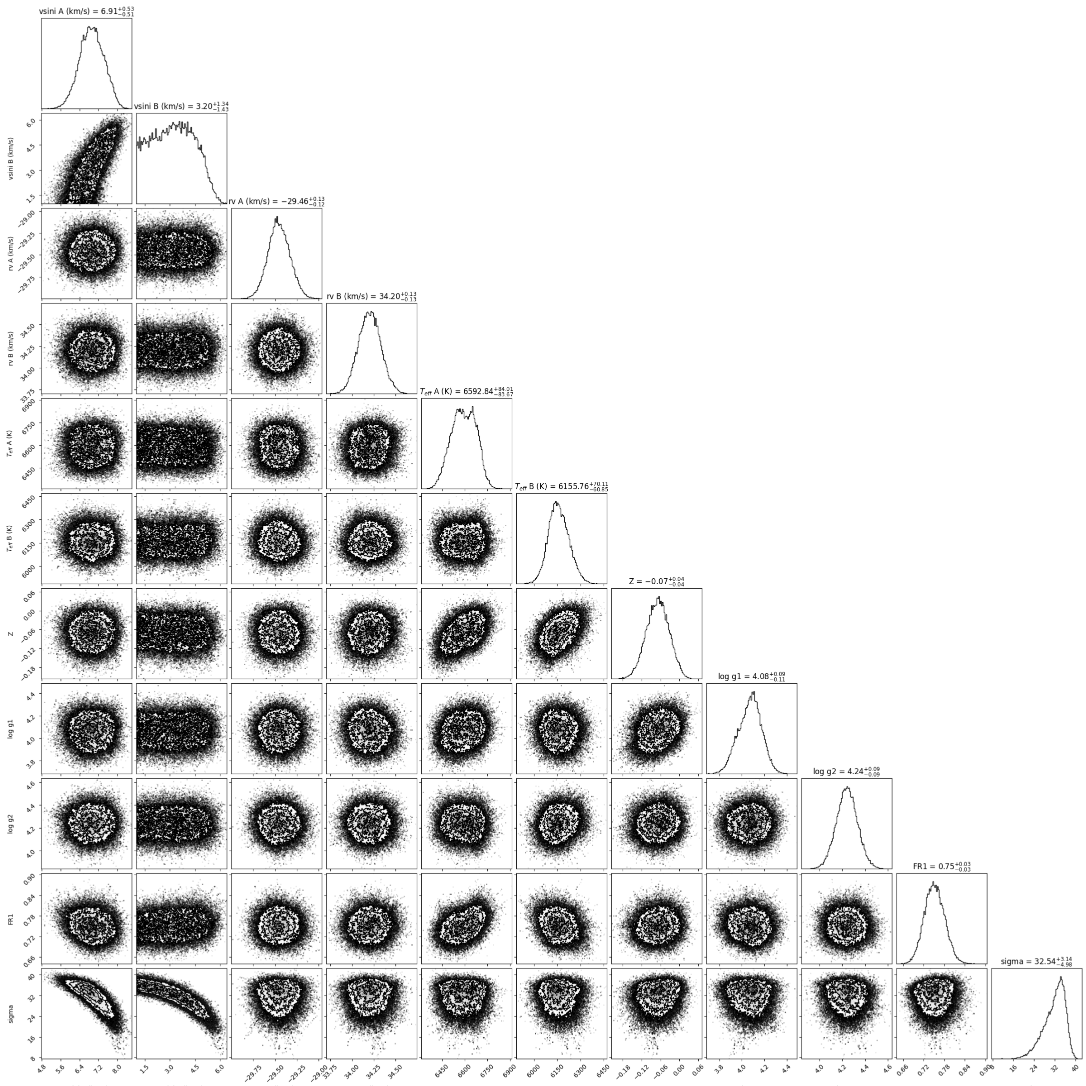

Fig. B.0.2.— The posterior distribution for the binary fit to the 2018-04-18 FEROS spectrum, fitting for stellar parameters and radial velocities of both components of the binary.

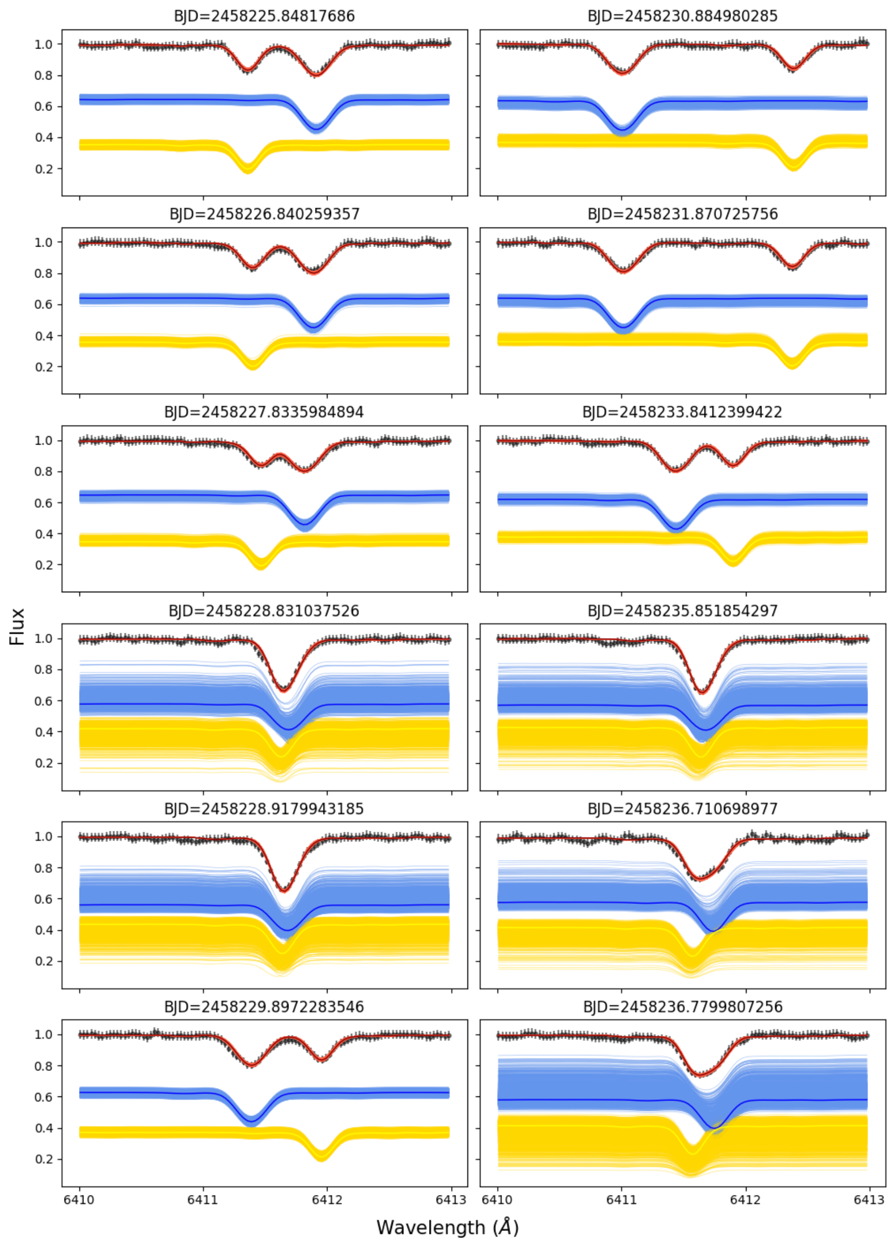

Fig. B.0.3.— A single wavelength region used for the spectral fitting for all 22 epochs. Black points with error bars are the data. Red tracks are random draws from the posterior for the two star model fit. Blue and yellow tracks are random draws from the posterior for single star models for the primary and secondary, respectively. Our method finds reliable RVs for both epochs with widely-separated lines and epochs with blended lines.

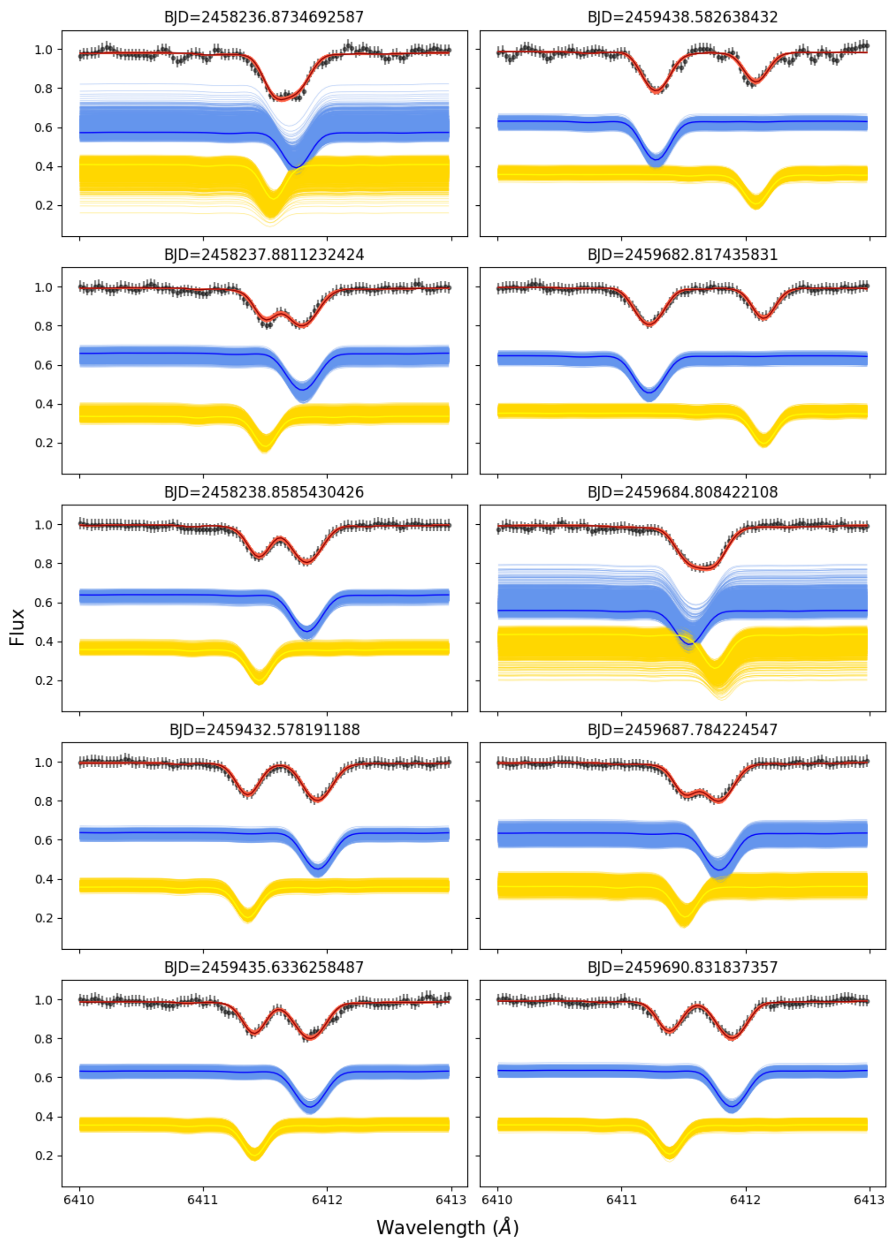

Fig. B.0.4.— Continued from Figure B.0.3.

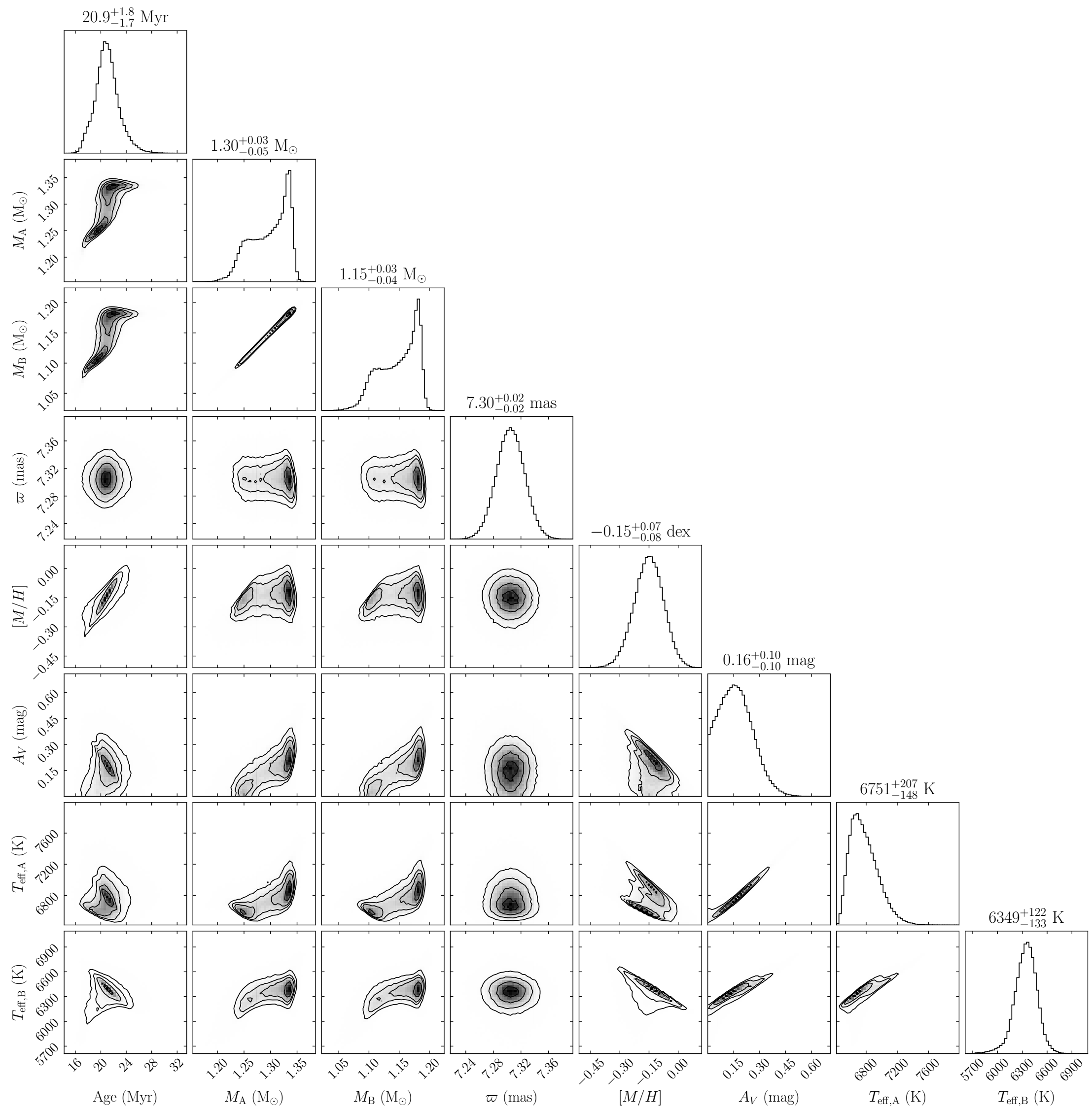

Fig. B.0.5.— The posterior distribution for the two-star fit to the unresolved photometric measurements, giving the six fitted parameters and their covariances. The effective temperature, a derived parameter, of both components is also shown.

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